Syllabus of the course

Categorical dualities in logic

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Introduction

John Baez and James Dolan [Baez and Dolan, 2001] remarked that

> "an equation is only interesting or useful to the extent that the two sides are different."

For instance, compare

$$2 = 2 \qquad \text{with} \qquad \sum_{n=0}^{\infty} \frac{1}{2^n} = 2.$$

The first equality is correct but uninformative: both sides express the same object in the same language. The second one is interesting precisely because it connects two apparently different descriptions, and allows us to switch freely between them depending on which is more convenient for a given computation.

In this course, we will study an analogous phenomenon, but at the level of *mathematical structures*. On one side, we will have algebraic structures that arise naturally in logic — most notably *Boolean algebras*, which provide an algebraic semantics for classical propositional logic. On the other side, we will encounter structures of a completely different nature — in this case, *Stone spaces*, which are certain topological spaces.

The connection between these two worlds is not a literal equality, but a *categorical duality* (a.k.a. categorical dual equivalence): a two-way translation that preserves all information. This means that we can translate a problem about Boolean algebras into a corresponding problem about Stone spaces, and vice versa. In practice, this is useful because, on the side of Stone spaces, many constructions are simpler, and one can use geometric intuition.

A slogan to keep in mind is:

> *Categorical dualities in logic relate "algebras of formulas" to "spaces of models".*

We begin with the simplest case: classical propositional logic, which is modeled by Boolean algebras and Stone spaces (Stone, 1936).

Later, we will see an analogous duality for classical *first-order* logic, where quantifiers enter the picture.

Chapter 1

Classical propositional logic

In the first part of the course, we will see *Stone duality*: a connection (in the form of a categorical duality) between *Boolean algebras* and *Stone spaces*. Informally, Boolean algebras encode the *syntax* of classical propositional logic (algebras of formulas), while Stone spaces encode its *semantics* (spaces of models).

1.1 From classical propositional logic towards Boolean algebras

1.1.1 Syntax: propositional languages, formulas

Definition 1.1.1 (Propositional language, propositional symbols)**.** A ***(propositional) language*** $\mathcal{L}$ is a set; its elements are called ***propositional symbols*** (or also *propositional variables*).

We use the fancy name "(propositional) language" for a plain set just to declare the usage we want to make out of it.

Propositional symbols are typically denoted by $p, q, r, \ldots$.

The connectives of classical propositional logic are

$$\vee, \quad \wedge, \quad \neg, \quad 0, \quad 1,$$

where $\vee$ is a binary operation denoting *or* (sometimes called *join*), $\wedge$ is a binary operation denoting *and* (sometimes called *meet*), $\neg$ is a unary operation denoting *not* (i.e. negation, or complement), 0 is a constant symbol denoting *false* (bottom) and 1 is a constant symbol denoting *true* (top).

Definition 1.1.2 (Formula)**.** The set of ***formulas*** $\mathrm{Form}(\mathcal{L})$ is defined inductively as follows:

- every propositional symbol $p \in \mathcal{L}$ is a formula;
- if $\varphi, \psi \in \mathrm{Form}(\mathcal{L})$, then $(\varphi \vee \psi) \in \mathrm{Form}(\mathcal{L})$ and $(\varphi \wedge \psi) \in \mathrm{Form}(\mathcal{L})$;
- if $\varphi \in \mathrm{Form}(\mathcal{L})$, then $(\neg\varphi) \in \mathrm{Form}(\mathcal{L})$;
- 0 and 1 are formulas.

In other words, $\mathrm{Form}(\mathcal{L})$ is the smallest set containing $\mathcal{L}$ and closed under $\vee$, $\wedge$, $\neg$, 0 and 1.

We use the standard abbreviations for implications and bi-implication:

$$\varphi \to \psi \;\coloneqq\; \neg\varphi \vee \psi, \qquad \varphi \leftrightarrow \psi \;\coloneqq\; (\varphi \to \psi) \wedge (\psi \to \varphi).$$

While propositional symbols are usually denoted by $p, q, r, \ldots$, formulas are typically denoted by $\varphi, \psi, \sigma, \ldots$.

1.1.2 Semantics: valuations, models, semantic equivalence

Valuations

Set $2 := \{0, 1\}$ (the "set of truth values").

Definition 1.1.3 (Valuation, interpretation of formulas)**.** A ***valuation*** on $\mathcal{L}$ is a function $v\colon \mathcal{L} \to 2$. Given a valuation v, its unique extension[a] to all formulas is a map

$$\bar{v}\colon \mathrm{Form}(\mathcal{L}) \longrightarrow 2$$

defined by recursion on the complexity of formulas:

$$\bar{v}(p) = v(p) \quad (p \in \mathcal{L}), \qquad \bar{v}(0) = 0, \qquad \bar{v}(1) = 1,$$

$$\bar{v}(\neg\varphi) = \neg\bar{v}(\varphi), \qquad \bar{v}(\varphi \vee \psi) = \bar{v}(\varphi) \vee \bar{v}(\psi), \qquad \bar{v}(\varphi \wedge \psi) = \bar{v}(\varphi) \wedge \bar{v}(\psi),$$

where on the right-hand side we use the usual Boolean operations on $2 = \{0, 1\}$, which you can find in Remark 1.1.4 below.

[a] This is a special case of a standard universal-algebraic fact: for any algebraic signature Σ, the forgetful functor $\mathsf{Alg}(\Sigma) \to \mathsf{Set}$ has a left adjoint sending a set X to the *free* Σ-*algebra* on X, whose underlying set can be described as the set of Σ-terms over X. For the signature $\{\vee, \wedge, \neg, 0, 1\}$, the left adjoint maps a set $\mathcal{L}$ to $\mathrm{Form}(\mathcal{L})$, the unit at $\mathcal{L}$ is the inclusion $\mathcal{L} \hookrightarrow \mathrm{Form}(\mathcal{L})$, and the extension $v \mapsto \bar{v}$ is given by the universal property of the unit.

For a bibliographic reference for terms algebras and their universal properties as free algebras, see [Burris and Sankappanavar, 2012, §10], and in particular their Def. 10.4 (definition of term algebra) and Thm. 10.8 (universal property of term algebras). In these sources, you can ignore the hypotheses $T(X) \neq \varnothing$ in Def. 10.4 and "where $X \neq \varnothing$ if $\mathcal{F}_0 = \varnothing$", which are due to the convention in [Burris and Sankappanavar, 2012] that the empty algebra is not considered as an algebra.

For a bibliographic reference for adjunctions, see [Mac Lane, 1998, Ch. IV].

Remark 1.1.4 (Boolean operations on $2 = \{0, 1\}$)**.** By convention,

$$0 \wedge 0 = 0, \;\; 0 \wedge 1 = 0, \;\; 1 \wedge 0 = 0, \;\; 1 \wedge 1 = 1,$$

$$0 \vee 0 = 0, \;\; 0 \vee 1 = 1, \;\; 1 \vee 0 = 1, \;\; 1 \vee 1 = 1,$$

$$\neg 0 = 1, \;\; \neg 1 = 0.$$

Semantic equivalence

Definition 1.1.5 (Satisfaction: "valuation $\models$ formula")**.** Let $\mathcal{L}$ be a propositional language. Given a valuation $v\colon \mathcal{L} \to 2$ and a formula $\varphi \in \mathrm{Form}(\mathcal{L})$, we write

$$v \models \varphi$$

and we say that v ***satisfies*** φ if

$$\bar{v}(\varphi) = 1.$$

Definition 1.1.6 (Semantic equivalence)**.** Let $\varphi, \psi \in \mathrm{Form}(\mathcal{L})$. We write $\varphi \equiv \psi$ and say that φ and ψ are ***semantically equivalent***[a] if, for every valuation $v\colon \mathcal{L} \to 2$,

$$v \models \varphi \iff v \models \psi$$

(i.e., $\bar{v}(\varphi) = \bar{v}(\psi)$).

[a] In Definition 1.1.6, "semantic" is in opposition to "syntactic": two formulas are *syntactically equivalent* if they are interprovable in a certain proof system, which we do not have the time to see here. Let me just mention that Stone's

Representation Theorem, which will be seen later and which is the core of this chapter, can be seen as an algebraic way to affirm that the syntactic and semantic notions of equivalence coincide.

In other words, φ and ψ are equivalent if and only if they have the same truth table (a function from $\mathcal{L}$ to 2 corresponds to a row of a truth table). For standard background on propositional semantics, truth tables, and semantic consequence, see [Enderton, 2001, Ch. 1].

Example 1.1.7. Let $\mathcal{L} = \{p, q\}$. Then $p \vee q \equiv q \vee p$, as can be checked via the truth table

p	q	$p \vee q$	$q \vee p$
0	0	0	0
0	1	1	1
1	0	1	1
1	1	1	1

Example 1.1.8. If $\mathcal{L} = \{p\}$ has one variable, then every formula is equivalent to exactly one of

$$0, \quad p, \quad \neg p, \quad 1.$$

(For instance, $\neg\neg p \equiv p$, $p \vee \neg p \equiv 1$, and $p \wedge \neg p \equiv 0$.) Thus, $\mathrm{Form}(\mathcal{L})/\equiv$ has 4 elements.

One may prove that, in a language $\mathcal{L} = \{p, q\}$ with two propositional symbols, there are 16 equivalence classes of formulas.[1]

Adding assumptions: theories and semantic equivalence modulo a theory

We can incorporate semantic assumptions by restricting the class of admissible valuations.

Definition 1.1.9 (Propositional theory)**.** A ***(propositional) theory*** $\mathcal{T}$ in a propositional language $\mathcal{L}$ is a subset $\mathcal{T} \subseteq \mathrm{Form}(\mathcal{L})$.

Definition 1.1.10 (Model)**.** Let $\mathcal{T} \subseteq \mathrm{Form}(\mathcal{L})$ be a propositional theory in a propositional language. A ***model*** of $\mathcal{T}$ is a function $v\colon \mathcal{L} \to 2$ that satisfies every element of $\mathcal{T}$. We denote by $\mathrm{Mod}(\mathcal{T})$ the set of models of $\mathcal{T}$.

Example 1.1.11. Let $\mathcal{L} = \{p, q\}$ and let $\mathcal{T} = \{p \vee q\}$. Then $\mathrm{Mod}(\mathcal{T})$ consists of the three valuations

$$(p, q) = (1, 0),\ (1, 1),\ (0, 1).$$

Definition 1.1.12 (Semantic equivalence modulo a theory)**.** Let $\mathcal{T} \subseteq \mathrm{Form}(\mathcal{L})$ be a propositional theory and let $\varphi, \psi \in \mathrm{Form}(\mathcal{L})$. We say that φ and ψ are ***semantically equivalent modulo the theory*** $\mathcal{T}$ (or ***relative to*** $\mathcal{T}$, or ***semantically*** $\mathcal{T}$***-equivalent***), and write $\varphi \equiv_{\mathcal{T}} \psi$, if for every $v \in \mathrm{Mod}(\mathcal{T})$ we have

$$v \models \varphi \iff v \models \psi$$

Example 1.1.13. Let $\mathcal{L} = \{p, q\}$ and let $\mathcal{T} = \{p \vee q\}$. One can check that $p \vee q \equiv_{\mathcal{T}} 1$, $\neg p \wedge \neg q \equiv_{\mathcal{T}} 0$ and $p \vee \neg q \equiv_{\mathcal{T}} p$.

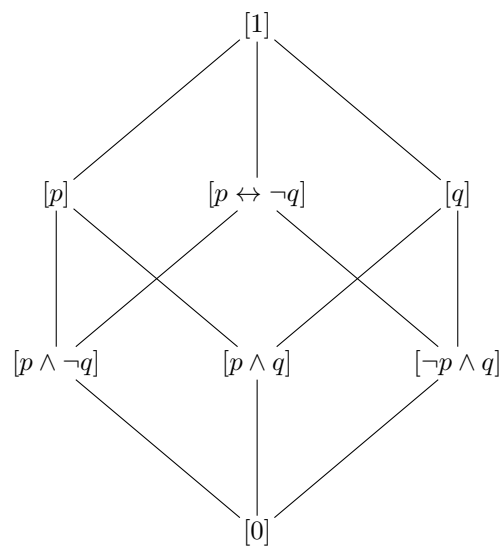
[1] More generally, if $\mathcal{L}$ is finite of cardinality n, then $|\mathrm{Form}(\mathcal{L})/\equiv| = 2^{2^n}$. Indeed, every formula determines a truth function $2^{\mathcal{L}} \to 2$, and vice versa one can prove that every function $2^{\mathcal{L}} \to 2$ is the truth function of a formula (this is called "functional completeness", see, e.g., [Enderton, 2001, Ch. 1, Cor. 15C, p. 49] or [Crama and Hammer, 2011, Thm. 1.4]), so that $|\mathrm{Form}(\mathcal{L})/\equiv| = 2^{2^{|\mathcal{L}|}}$. In particular, for $\mathcal{L} = \{p, q\}$ we have $|\mathrm{Form}(\mathcal{L})/\equiv| = 2^{2^2} = 16$.

The relation $\equiv_{\mathcal{T}}$ is again an equivalence relation, and we write $\mathrm{Form}(\mathcal{L})/\!\equiv_{\mathcal{T}}$ for the corresponding quotient set. It is convenient to partially order equivalence classes by *implication*:

$$\varphi \leq_{\mathcal{T}} \psi \quad :\Longleftrightarrow \quad \forall v \in \mathrm{Mod}(\mathcal{T}),\ \big((v \models \varphi) \Rightarrow (v \models \psi)\big).$$

(So "$\varphi \leq_{\mathcal{T}} \psi$" means that φ *implies* ψ on every model of $\mathcal{T}$.)

Example 1.1.14. Let $\mathcal{L} = \{p, q\}$ and $\mathcal{T} = \{p \vee q\}$. Then $\mathrm{Form}(\mathcal{L})/\!\equiv_{\mathcal{T}}$ has 8 elements and its Hasse diagram (with respect to $\leq_{\mathcal{T}}$) can be drawn as follows:



Boolean algebras (whose definition we will see soon) are meant to capture the algebraic structures of the form

$$\langle \mathrm{Form}(\mathcal{L})/\!\equiv_{\mathcal{T}}; \vee, \wedge, \neg, 0, 1\rangle,$$

for $\mathcal{L}$ a propositional language and $\mathcal{T}$ a propositional theory in $\mathcal{L}$; here, $\vee, \wedge, \neg, 0, 1$ are defined on $\mathrm{Form}(\mathcal{L})/\!\equiv_{\mathcal{T}}$ by setting

$$[\varphi] \vee [\psi] \ \coloneqq\ [\varphi \vee \psi]\,, \qquad [\varphi] \wedge [\psi] \ \coloneqq\ [\varphi \wedge \psi]\,, \qquad \neg\,[\varphi] \ \coloneqq\ [\neg\varphi]\,, \qquad 0 \ \coloneqq\ [0]\,, \qquad 1 \ \coloneqq\ [1]\,.$$

(These are well-defined operations.)

In other words, Boolean algebras are meant to capture the algebras of formulas modulo a theory.

1.2 Boolean algebras

The quotient posets in Example 1.1.14 already display a key phenomenon: the logical connectives $\wedge$ and $\vee$ behave like "infimum" and "supremum" with respect to the implication order.

To recall the definition of infimum and supremum, let P be a poset. For $x, y \in P$, an *infimum* (or *greatest lower bound*) of $\{x, y\}$ is an element $x \wedge y \in P$ such that

$$x \wedge y \leq x, \qquad x \wedge y \leq y,$$

and, for every $z \in P$, if $z \leq x$ and $z \leq y$ then $z \leq x \wedge y$. Similarly, a *supremum* (or *least upper bound*) of $\{x, y\}$ is an element $x \vee y \in P$ such that

$$x \leq x \vee y, \qquad y \leq x \vee y,$$

and, for every $z \in P$, if $x \leq z$ and $y \leq z$ then $x \vee y \leq z$.

More generally, for any subset S of a poset P, an ***infimum*** of S (relative to P) is a greatest lower bound of S in P and a ***supremum*** of S (relative to P) is a smallest upper bound of S in P. Infima and suprema, if they exist, are unique. Note that being the infimum of $\varnothing$ relative to P means being the maximum of P, and being the supremum of $\varnothing$ relative to P means being the minimum of P.

One can show that, in the poset $\mathrm{Form}(\mathcal{L})/\equiv_{\mathcal{T}}$ ordered by implication, $[\varphi \wedge \psi]$ is the infimum of $[\varphi]$ and $[\psi]$, and $[\varphi \vee \psi]$ is the supremum of $[\varphi]$ and $[\psi]$.[2]

1.2.1 Lattices

Order-theoretic definition of lattices

Definition 1.2.1 (Lattice: order theoretic presentation)**.** A ***lattice*** is a poset $(L, \leq)$ in which every pair of elements admits an infimum and a supremum.

Example 1.2.2 (Power-set lattice)**.** Let X be a set and consider the poset $(\mathcal{P}(X), \subseteq)$. Then $(\mathcal{P}(X), \subseteq)$ is a lattice, with

$$A \wedge B \coloneqq A \cap B \qquad \text{and} \qquad A \vee B \coloneqq A \cup B.$$

Equational definition of lattices

Lattices can also be presented *algebraically*, by taking $\wedge$ and $\vee$ as primitive operations and listing a small family of identities. This is useful because identities are stable under the kind of constructions we will use later (products, subalgebras, quotients).

Definition 1.2.3 (Lattice: equational presentation)**.** A ***lattice*** is a set L equipped with two binary operations

$$\wedge, \vee \colon L \times L \to L$$

such that:

1. (commutativity) for all $a, b \in L$, $a \wedge b = b \wedge a$ and $a \vee b = b \vee a$;
2. (associativity) for all $a, b, c \in L$, $(a \wedge b) \wedge c = a \wedge (b \wedge c)$ and $(a \vee b) \vee c = a \vee (b \vee c)$;
3. (absorption) for all $a, b \in L$, $a \vee (a \wedge b) = a$ and $a \wedge (a \vee b) = a$.

Remark 1.2.4. If L is a lattice in the equational sense, one can recover an order by setting

$$a \leq b \quad :\Longleftrightarrow \quad a \wedge b = a,$$

equivalently $a \vee b = b$. With this order, the operations $\wedge$ and $\vee$ are precisely infimum and supremum, so Definitions 1.2.1 and 1.2.3 are equivalent viewpoints.[3]

1.2.2 Bounded lattices

Definition 1.2.5 (Bounded lattice)**.** A lattice is ***bounded*** if it has a least element 0 and a greatest element 1, i.e. elements such that $0 \leq a \leq 1$ for all a. Equivalently (in the equational presentation), 0 and 1 satisfy

[2] Proof: We prove the statement for $\wedge$; the case of $\vee$ is analogous. For every model $v \in \mathrm{Mod}(\mathcal{T})$, if $\bar{v}(\varphi \wedge \psi) = 1$ then $\bar{v}(\varphi) = 1$ and $\bar{v}(\psi) = 1$, hence $[\varphi \wedge \psi] \leq_{\mathcal{T}} [\varphi]$ and $[\varphi \wedge \psi] \leq_{\mathcal{T}} [\psi]$. Now let $[\rho]$ be any lower bound of $[\varphi]$ and $[\psi]$. This means that, for every $v \in \mathrm{Mod}(\mathcal{T})$, $\bar{v}(\rho) = 1$ implies both $\bar{v}(\varphi) = 1$ and $\bar{v}(\psi) = 1$. Therefore $\bar{v}(\rho) = 1$ implies $\bar{v}(\varphi \wedge \psi) = 1$, i.e. $[\rho] \leq_{\mathcal{T}} [\varphi \wedge \psi]$.

[3] For more details on the equivalence between these viewpoints, see [Davey and Priestley, 2002, Sec. 2, pp. 33–41], and, in particular, Thm. 2.10 therein, or [Birkhoff, 1967, Ch. 1, Sec. 5], and, in particular, Thm. 8 (p. 10) therein.

1. For all a, $a \wedge 0 = 0$ (equivalently: $a \vee 0 = a$),
2. For all a, $a \vee 1 = 1$ (equivalently; $a \wedge 1 = a$).

1.2.3 Distributive lattices

Definition 1.2.6 (Distributive lattice)**.** A lattice L is ***distributive*** if it satisfies any (and hence both) of the following equivalent[a] conditions:

1. for all $a, b, c \in L$, $a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$.
2. for all $a, b, c \in L$, $a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$,

[a]For a proof of the equivalence between the two, see [Birkhoff, 1967, Sec. 6, Thm. 9, p. 11].

Example 1.2.7. The two-element chain is distributive:

Example 1.2.8. For every set X, the power-set lattice $\mathcal{P}(X)$ is distributive.

Two small lattices play a special role as "minimal" obstructions to distributivity. They are usually denoted by M_3 (the *diamond*) and N_5 (the *pentagon*).

M_3 $\qquad$ N_5 (1.1)

These are not distributive; one can easily verify that $a \vee (b \wedge c) \neq (a \vee b) \wedge (a \vee c)$.

In fact, one can prove that a lattice L is distributive if and only if it does not contain a sublattice isomorphic to M_3 or N_5, i.e., there is no injective map $M_3 \to L$ or $N_5 \to L$ preserving both $\wedge$ and $\vee$.[4]

1.2.4 Boolean algebras

Definition 1.2.9 (Boolean algebra)**.** A ***Boolean algebra*** is an algebraic structure

$$\langle B; \vee, \wedge, \neg, 0, 1 \rangle$$

(arities: 2, 2, 1, 0, 0, respectively) such that:

1. $\langle B; \vee, \wedge, 0, 1 \rangle$ is a bounded distributive lattice;
2. for every $a \in B$,
$$a \wedge \neg a = 0 \qquad \text{and} \qquad a \vee \neg a = 1.$$

The element $\neg a$ is called the *complement* of a.

[4]For a textbook reference, see, e.g., [Davey and Priestley, 2002, 4.10].

Remark 1.2.10. All axioms in the definition of Boolean algebras are *equational*, i.e. they are identities of the form

$$\forall x_1, \dots, x_n, \quad t_1(x_1, \dots, x_n) = t_2(x_1, \dots, x_n)$$

between terms.

Remark 1.2.11. A Boolean algebra is completely determined by its underlying partial order. This is because: $\vee, \wedge, 0, 1$ are the binary supremum, binary infimum, smallest element and greatest element, and, in every bounded distributive lattice L, if an element a has a complement (i.e., there is an element b such that $a \wedge b = 0$ and $a \vee b = 1$) it is unique.[5]

Example 1.2.12. The prototypical example of a Boolean algebra is $2 = \{0, 1\}$ with partial order $0 \leq 1$. The Boolean operations are those described in Remark 1.1.4.

Proposition 1.2.13 ($\mathrm{Form}(\mathcal{L})/\equiv_{\mathcal{T}}$ is a Boolean algebra)**.** *Let $\mathcal{L}$ be a propositional language and $\mathcal{T} \subseteq \mathrm{Form}(\mathcal{L})$ a theory. Then the quotient set $\mathrm{Form}(\mathcal{L})/\equiv_{\mathcal{T}}$ becomes a Boolean algebra by setting*

$$[\varphi] \vee [\psi] \coloneqq [\varphi \vee \psi]\,, \qquad [\varphi] \wedge [\psi] \coloneqq [\varphi \wedge \psi]\,, \qquad \neg\,[\varphi] \coloneqq [\neg\varphi]\,, \qquad 0 \coloneqq [0]\,, \qquad 1 \coloneqq [1]\,.$$

Proof sketch. Step 1: the operations are well defined. We show this for $\vee$; the cases of $\wedge$ and $\neg$ are analogous. Suppose $\varphi \equiv_{\mathcal{T}} \varphi'$ and $\psi \equiv_{\mathcal{T}} \psi'$. Then for every $v \in \mathrm{Mod}(\mathcal{T})$ we have $\bar{v}(\varphi) = \bar{v}(\varphi')$ and $\bar{v}(\psi) = \bar{v}(\psi')$. Using the truth table for $\vee$ in $2 = \{0, 1\}$ we compute:

$$\bar{v}(\varphi \vee \psi) = \bar{v}(\varphi) \vee \bar{v}(\psi) = \bar{v}(\varphi') \vee \bar{v}(\psi') = \bar{v}(\varphi' \vee \psi').$$

Hence, $\varphi \vee \psi \equiv_{\mathcal{T}} \varphi' \vee \psi'$, and so $[\varphi] \vee [\psi]$ does not depend on the choice of representatives. The constants $0 = [0]$ and $1 = [1]$ are trivially well defined.

Step 2: the Boolean algebra identities hold. As an example, we show that the axiom $a \wedge \neg a = 0$ holds. Fix $\varphi \in \mathrm{Form}(\mathcal{L})$. For every $v \in \mathrm{Mod}(\mathcal{T})$ we compute

$$\bar{v}(\varphi \wedge \neg\varphi) = \bar{v}(\varphi) \wedge \bar{v}(\neg\varphi) = \bar{v}(\varphi) \wedge \neg\bar{v}(\varphi) = 0 = \bar{v}(0),$$

where we used the recursive definition of $\bar{v}$ and the fact that $b \wedge \neg b = 0$ for all $b \in 2$. Hence, $\varphi \wedge \neg\varphi \equiv_{\mathcal{T}} 0$, and therefore

$$[\varphi] \wedge \neg\,[\varphi] = [0] = 0.$$

This is the general mechanism: any identity between Boolean terms can be checked pointwise in 2 under every valuation. To give more details, let $t_1(x_1, \dots, x_n) = t_2(x_1, \dots, x_n)$ be any Boolean algebra identity (an equation between terms built from $\vee, \wedge, \neg, 0, 1$). To check it in $\mathrm{Form}(\mathcal{L})/\equiv_{\mathcal{T}}$, pick arbitrary formulas $\varphi_1, \dots, \varphi_n \in \mathrm{Form}(\mathcal{L})$ and consider the two formulas $t_1(\varphi_1, \dots, \varphi_n)$ and $t_2(\varphi_1, \dots, \varphi_n)$. For every $v \in \mathrm{Mod}(\mathcal{T})$, the evaluation map $\bar{v}\colon \mathrm{Form}(\mathcal{L}) \to 2$ respects the connectives, hence

$$\bar{v}\big(t_i(\varphi_1, \dots, \varphi_n)\big) = t_i\big(\bar{v}(\varphi_1), \dots, \bar{v}(\varphi_n)\big) \qquad (i = 1, 2).$$

Since the identity $t_1 = t_2$ holds in the two-element Boolean algebra 2, the right-hand sides are equal for all $v \in \mathrm{Mod}(\mathcal{T})$, so $t_1(\varphi_1, \dots, \varphi_n) \equiv_{\mathcal{T}} t_2(\varphi_1, \dots, \varphi_n)$. Therefore, the induced operations on equivalence classes satisfy all Boolean algebra axioms. ∎

[5] This is not true for arbitrary bounded lattice: for example, in the bounded lattice M_3 in (1.1), both b and c are complements of a, and in N_5 both a and c are complements of b.

Definition 1.2.14 (Lindenbaum–Tarski algebra)**.** Let $\mathcal{L}$ be a propositional language and $\mathcal{T} \subseteq \mathrm{Form}(\mathcal{L})$ a theory. The Boolean algebra

$$\langle \mathrm{Form}(\mathcal{L})/{\equiv_{\mathcal{T}}}; \vee, \wedge, \neg, 0, 1 \rangle$$

with operations

$$[\varphi] \vee [\psi] \coloneqq [\varphi \vee \psi]\,, \qquad [\varphi] \wedge [\psi] \coloneqq [\varphi \wedge \psi]\,, \qquad \neg\,[\varphi] \coloneqq [\neg\varphi]\,, \qquad 0 \coloneqq [0]\,, \qquad 1 \coloneqq [1]$$

is called the ***Lindenbaum–Tarski algebra of*** $\mathcal{T}$.

Proposition 1.2.13 states that every Lindenbaum–Tarski algebra is a Boolean algebra. Later on (Theorem 1.4.4), we will see that also the converse holds: every Boolean algebra is isomorphic to $\mathrm{Form}(\mathcal{L})/{\equiv_{\mathcal{T}}}$ for some language $\mathcal{L}$ and theory $\mathcal{T}$. This will be a consequence of Stone's Representation Theorem (or of the crucial lemmas used in proving Stone's Representation Theorem). It will provide a guarantee that the definition of Boolean algebras is "the correct one".

Example 1.2.15 (Power set Boolean algebras)**.** Let X be a set. The power set $\mathcal{P}(X)$ with the inclusion order is a Boolean algebra, with operations

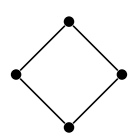
$$A \wedge B \coloneqq A \cap B, \qquad A \vee B \coloneqq A \cup B, \qquad \neg A \coloneqq X \setminus A, \qquad 0 \coloneqq \varnothing, \qquad 1 \coloneqq X.$$

See the footnote for a proof.[6]

Below, we draw the Hasse diagrams of the power sets of sets of cardinality 0, 1, 2, and 3. These are the smallest Boolean algebras, and probably the only ones that one can draw without losing one's sight. (The next one has 16 elements.)

The *trivial Boolean algebra*, of cardinality 1, isomorphic to $\mathcal{P}(\varnothing)$

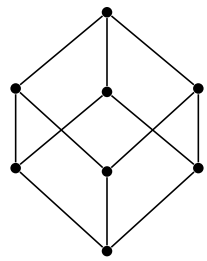
The Boolean algebra $2 = \{0, 1\}$, of cardinality 2, isomorphic to $\mathcal{P}(\{x\})$



A Boolean algebra of cardinality 4, isomorphic to $\mathcal{P}(\{x, y\})$

[6] One way to prove that it is a Boolean algebra is by observing it is isomorphic to the power $\prod_{x \in X} 2$ of 2, that 2 is a Boolean algebra, and that equationally defined classes of algebras are closed under products (by the easy direction in Birkhoff's theorem).

For Birkhoff's theorem, see, e.g., [Burris and Sankappanavar, 2012, Ch. II, Sec. 9–11], and, in particular, Thm. 11.9 therein (Birkhoff's theorem); please note that [Burris and Sankappanavar, 2012] has some minor conventional differences from this text: in [Burris and Sankappanavar, 2012], (1) products are assumed to be of nonempty families, and (2) algebras are assumed to be nonempty.



A Boolean algebra of cardinality 8, isomorphic to $\mathcal{P}(\{x, y, z\})$

There are examples of Boolean algebras that are not isomorphic to any power set. The following one is probably the simplest example.

Example 1.2.16 (The algebra of finite and cofinite subsets of $\mathbb{N}$)**.** Consider the collection

$$\mathrm{FC}(\mathbb{N}) \;:=\; \{\, A \subseteq \mathbb{N} \mid A \text{ is finite, or cofinite} \,\} \;\subseteq\; \mathcal{P}(\mathbb{N})$$

where "A *cofinite*" means that $\mathbb{N} \setminus A$ is finite. To show that this is indeed a Boolean algebra, the following is of help.

Remark 1.2.17 (Boolean subalgebras of power sets)**.** If $\mathcal{A} \subseteq \mathcal{P}(X)$ contains $\varnothing$ and X and is closed under $\cup$, $\cap$ and complement in X, then $\mathcal{A}$ is a Boolean algebra, as well.

This follows from the fact that Boolean algebras are equationally definable and hence closed under subalgebras.

Now one can prove that $\mathrm{FC}(\mathbb{N})$ is a Boolean algebra:

- $\varnothing$ and $\mathbb{N}$ belong to $\mathrm{FC}(\mathbb{N})$;
- $\mathrm{FC}(\mathbb{N})$ is closed under complement in $\mathbb{N}$: if A is finite then $\mathbb{N} \setminus A$ is cofinite, and if A is cofinite then $\mathbb{N} \setminus A$ is finite;
- $\mathrm{FC}(\mathbb{N})$ is closed under unions: if A and B are finite then $A \cup B$ is finite, while if at least one of A, B is cofinite then $A \cup B$ is cofinite.
- Analogously, $\mathrm{FC}(\mathbb{N})$ is closed under intersections.

The Boolean algebra $\mathrm{FC}(\mathbb{N})$ is infinite but much smaller than $\mathcal{P}(\mathbb{N})$: for instance, the set of even numbers is neither finite nor cofinite, hence it does not belong to $\mathrm{FC}(\mathbb{N})$. Moreover, $\mathrm{FC}(\mathbb{N})$ is *countably infinite* (i.e., in bijection with $\mathbb{N}$). This also shows that it is not isomorphic to any power set, since no power set is countably infinite: if X is finite then $\mathcal{P}(X)$ is finite, and if X is infinite then $\mathcal{P}(X)$ is uncountable.

Remark 1.2.18 (A different obstruction: completeness)**.** More generally, for any set X, one can consider the Boolean subalgebra

$$\mathrm{FC}(X) \coloneqq \{\, A \subseteq X \mid A \text{ is finite or cofinite} \,\} \subseteq \mathcal{P}(X).$$

If X is infinite, then $\mathrm{FC}(X)$ is *not complete* as a Boolean algebra[7]: there are families that do not admit a supremum. For instance, choose a subset $Y \subseteq X$ such that both Y and $X \setminus Y$ are infinite, and consider the family $\{\, \{y\} \mid y \in Y \,\} \subseteq \mathrm{FC}(X)$. Its union in $\mathcal{P}(X)$ is Y, which does not belong to $\mathrm{FC}(X)$. Moreover, there is no least element of $\mathrm{FC}(X)$ containing all singletons $\{y\}$ (one can always remove a point from any upper bound and still obtain a cofinite upper bound), hence the family has no supremum in $\mathrm{FC}(X)$. By contrast, every power set $\mathcal{P}(X)$ is complete (arbitrary unions exist), so $\mathrm{FC}(X)$ for X infinite cannot be isomorphic to any power set.

[7] A poset P is *complete* if every subset of P has a supremum; equivalently, if every subset has an infimum. For the equivalence between the two conditions, see [Burris and Sankappanavar, 2012, Thm. 4.2, p. 14].

1.3 Stone's Representation Theorem

Remark 1.2.17 gives a zoo of examples: every *Boolean subalgebra* of some power set is a Boolean algebra. Stone's Representation Theorem states that all Boolean algebras are of this form! We state it now, and we will prove it later Theorem 1.3.20.

Theorem (Stone's Representation Theorem for Boolean algebras)**.** *For every Boolean algebra B there is a set X such that B is isomorphic to a Boolean subalgebra of the power set $\mathcal{P}(X)$.*

This means that for every Boolean algebra B there are a set X and an injective map

$$\iota\colon B \hookrightarrow \mathcal{P}(X)$$

such that, under ι, the operations on B correspond to intersection, union, complement, empty set and whole set in X:

$$\iota(a \wedge b) = \iota(a) \cap \iota(b), \qquad \iota(a \vee b) = \iota(a) \cup \iota(b), \qquad \iota(\neg a) = X \setminus \iota(a), \qquad \iota(0) = \varnothing, \qquad \iota(1) = X.$$

The next goal is to prove Stone's Representation Theorem for Boolean algebras. We will first need to present auxiliary notions and lemmas.

Given a Boolean algebra B, how can we find a set X such that B embeds into the power set $\mathcal{P}(X)$ of X, as required by the statement of Stone's Representation Theorem?

Idea, from a logical perspective: In the special Boolean algebra $\mathrm{Form}(\mathcal{L})/{\equiv_{\mathcal{T}}}$ (for $\mathcal{L}$ a language and $\mathcal{T}$ a theory), an equivalence class $[\varphi]$ can be identified with the set of models $v \in \mathrm{Mod}(\mathcal{T})$ such that $\bar{v}(\varphi) = 1$; i.e., a formula can be identified with the models that satisfy it. This gives a very concrete embedding into a power set: the power set of models of $\mathcal{T}$. So, the idea for $B = \mathrm{Form}(\mathcal{L})/{\equiv_{\mathcal{T}}}$ is to take

$$X = \mathrm{Mod}(\mathcal{T}).$$

To translate this idea to a general Boolean algebra B, we note that a model

$$v\colon \mathcal{L} \longrightarrow 2$$

of $\mathcal{T}$ induces a function

$$\bar{v}\colon \mathrm{Form}(\mathcal{L}) \longrightarrow 2$$

which passes to the quotient

$$\begin{aligned} \mathrm{Form}(\mathcal{L})/{\equiv_{\mathcal{T}}} &\longrightarrow 2 \\ [\varphi] &\longmapsto \bar{v}(\varphi). \end{aligned}$$

In fact, one can prove that the models of $\mathcal{T}$ are in bijection with the Boolean homomorphisms (i.e., functions preserving all Boolean connectives, see Definition 1.3.1 below) from $\mathrm{Form}(\mathcal{L})/{\equiv_{\mathcal{T}}}$ to 2. We recall the definition of Boolean homomorphism, and then we state this bijection in Proposition 1.3.2.

Definition 1.3.1 (Boolean homomorphism)**.** Let A and B be Boolean algebras. A ***Boolean homomorphism*** (or, simply, a *homomorphism*) $f\colon A \to B$ is a function that preserves all Boolean operations, i.e., such that, for all $a, b \in A$,

$$f(a \vee b) = f(a) \vee f(b), \qquad f(a \wedge b) = f(a) \wedge f(b), \qquad f(\neg a) = \neg f(a), \qquad f(0) = 0, \qquad f(1) = 1.$$

(These conditions are redundant; for example, it is enough to ask the preservation of $\wedge, \neg, 1$.)

Proposition 1.3.2 (Models $\overset{1:1}{\leftrightarrow}$ homomorphisms to 2)**.** *Let $\mathcal{L}$ be a propositional language and $\mathcal{T} \subseteq \mathrm{Form}(\mathcal{L})$ a propositional theory. The following function defines a bijection*

$$\begin{aligned} \mathrm{Mod}(\mathcal{T}) &\longrightarrow \hom(\mathrm{Form}(\mathcal{L})/\equiv_{\mathcal{T}}, 2) \\ v &\longmapsto \big([\varphi] \mapsto \bar{v}(\varphi)\big). \end{aligned}$$

Proof. We skip the proof. ■

This suggests that, for a general Boolean algebra B, we shall take

$$X = \hom(B, 2),$$

the set of homomorphisms from B to 2.

Idea, from another perspective: If $B \subseteq \mathcal{P}(X)$ is a Boolean subalgebra of $\mathcal{P}(X)$, then every element $x \in X$ induces a function

$$\begin{aligned} B &\longrightarrow 2 \\ A &\longmapsto \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{otherwise,} \end{cases} \end{aligned}$$

which is a Boolean homomorphism. This suggests a close relationship between X and $\hom(B, 2)$. This suggests, from yet another perspective, that to prove Stone's Representation Theorem we shall take

$$X = \hom(B, 2).$$

1.3.1 Ultrafilters

A homomorphism $f\colon B \to 2$ can be encoded via $f^{-1}[\{1\}]$. The subsets of B arising as $f^{-1}[\{1\}]$ for some homomorphism $f\colon B \to 2$ can be characterized as the *ultrafilters*. To define this, we first define filters (of which ultrafilters are special instances) and ideals.

Think of a filter as a coherent notion of 'truth': a set of propositions closed under strengthening and finite conjunction.

Definition 1.3.3 (Filter)**.** A ***filter*** of a Boolean algebra B is a subset $F \subseteq B$ such that:

1. F is *upward closed*, i.e., if $x \in F$ and $x \leq y$, then $y \in F$;
2. F is *closed under finite meets*, i.e.,
   (a) if $x, y \in F$, then $x \wedge y \in F$;
   (b) $1 \in F$.

A filter F is *proper* if $0 \notin F$.

Turning the definition of a filter upside down, we get the following.

Definition 1.3.4 (Ideal)**.** An ***ideal*** of a Boolean algebra B is a subset $I \subseteq B$ such that:

1. I is *downward closed*, i.e., if $x \in I$ and $y \leq x$, then $y \in I$;
2. I is *closed under finite joins*, i.e.,
   (a) if $x, y \in I$, then $x \vee y \in I$;

(b) $0 \in I$.

An ultrafilter is what you get when you push consistency to the limit: every proposition gets a definitive yes/no answer.

Definition 1.3.5 (Ultrafilter)**.** An ***ultrafilter*** of a Boolean algebra is a filter whose complement is an ideal.

Spelling out the details, this means that an ultrafilter is a filter F such that

1. if $x \vee y \in F$, then $x \in F$ or $y \in F$;
2. $0 \notin F$.

There is also another useful characterization of ultrafilters.

Lemma 1.3.6 (Ultrafilters via negation)**.** *The ultrafilters of a Boolean algebra B are precisely the filters F such that, for every $x \in B$, exactly one between x and $\neg x$ belongs to F.*

Proof. ($\Rightarrow$). Suppose that F is an ultrafilter, and let $x \in B$. Since $1 = x \vee \neg x \in F$, we have $x \in F$ or $\neg x \in F$. Moreover, they cannot both lie in F because $x \wedge \neg x = 0 \notin F$.

($\Leftarrow$). Suppose that F is a filter such that, for every $x \in B$, exactly one between x and $\neg x$ belongs to F.

- Suppose $x \vee y \in F$, and let us prove that $x \in F$ or $y \in F$. If $x \in F$, we are done. Otherwise, from $x \notin F$ and from the hypothesis on F, we deduce $\neg x \in F$. Then, using distributivity,

$$F \ni (x \vee y) \wedge \neg x = (x \wedge \neg x) \vee (y \wedge \neg x) = y \wedge \neg x.$$

  By upward closure, this implies $y \in F$ (since $y \wedge \neg x \leq y$). $x \in F$ or $y \in F$.

- Since 1 belongs to F, its negation $\neg 1 = 0$ does not. ∎

Proposition 1.3.7 (Ultrafilters $\overset{1:1}{\leftrightarrow}$ homomorphisms to 2)**.** *For every Boolean algebra B, there is a bijection*

$$\begin{aligned} \hom(B, 2) &\longleftrightarrow \{\textit{ultrafilters of } B\}, \\ h &\longmapsto h^{-1}[\{1\}], \\ \left(\chi_U \colon x \mapsto \begin{cases} 1, & x \in U, \\ 0, & x \notin U, \end{cases}\right) &\longleftarrow\!\shortmid U. \end{aligned}$$

Proof sketch. If $h \colon B \to 2$ is a homomorphism, set $U := h^{-1}[\{1\}]$. Then $1 \in U$, U is upward closed and closed under meets, so it is a filter; it is proper since $h(0) = 0$. Moreover, for each $x \in B$, $h(\neg x) = \neg h(x)$, and hence exactly one of x and $\neg x$ lands in 1. Thus, U satisfies the conditions in Lemma 1.3.6 and so is an ultrafilter.

Conversely, if U is an ultrafilter, define $h_U \colon B \to 2$ by $h_U(x) = 1$ if and only if $x \in U$. The ultrafilter axioms ensure that h_U preserves $\wedge$, $\neg$, and 1, hence h_U is a homomorphism. ∎

Therefore, ultrafilters of B are encodings of homomorphisms from B to 2. Thus, we will use the set of ultrafilters of B as the set X such that B embeds into $\mathcal{P}(X)$.

Definition 1.3.8 (Principal filter)**.** The ***principal filter generated by an element*** a of a Boolean algebra B is

$$\uparrow a \;:=\; \{\, x \in B \mid a \leq x \,\}.$$

Therefore, a filter (and, in particular, an ultrafilter) is called ***principal*** if it has a minimum.

Remark 1.3.9. If B is finite, then every filter is principal.

We will soon see that $\uparrow a$ is an ultrafilter if and only if a satisfies the following simple condition.

Definition 1.3.10 (Atom)**.** An ***atom*** of a Boolean algebra B is a minimal element of $B \setminus \{0\}$.

Proposition 1.3.11 (Atoms and ultrafilters)**.**

1. *For every Boolean algebra B we have a bijection*

$$\begin{aligned} \{\textit{atoms of } B\} &\longleftrightarrow \{\textit{principal ultrafilters of } B\} \\ a &\longmapsto \uparrow a \\ \min U &\longleftarrow\!\shortmid U. \end{aligned} \tag{*}$$

2. *If B is <u>finite</u>, every ultrafilter is principal and so $*$ gives a bijection between atoms and ultrafilters.*

Proof. We only need to prove that the two functions are well defined; that they are mutually inverse will then be immediate.

To prove that the left-to-right function is well defined, let us assume that a is an atom, and let us prove that $U := \uparrow a$ is an ultrafilter. It is a filter (easy). $0 \notin U$ because $a > 0$. Suppose $x \vee y \in \uparrow a$, i.e., $x \vee y \geq a$. Then, by distributivity,

$$a = a \wedge (x \vee y) = (a \wedge x) \vee (a \wedge y).$$

Therefore, since the only elements below a are 0 and a, and the join of $(a \wedge x)$ and $(a \wedge y)$ is a, at least one of the two is a (otherwise they would both be 0 and so their join would be 0). If $a \wedge x = a$ then $x \in \uparrow a$, and if $a \wedge y = a$ then $y \in \uparrow a$. Thus, U is an ultrafilter.

To prove that the right-to-left function is well defined, let us assume that $\uparrow a$ is a principal ultrafilter, and let us prove that $\min U$ is an atom. We have $a \neq 0$ (otherwise 0 would belong to U). If $0 < x < a$, then $x \notin \uparrow a$, and hence $\neg x \in \uparrow a$ by Lemma 1.3.6. Thus $a \leq \neg x$, and since $x \leq a$ we get $x \leq \neg x$, hence $x = x \wedge \neg x = 0$, a contradiction. Therefore, there is no x with $0 < x < a$, so a is an atom. ∎

Example 1.3.12. The trivial Boolean algebra of cardinality 1 has no ultrafilters (and no homomorphisms to 2). The Boolean algebra $2 = \{0, 1\}$ has exactly one ultrafilter, namely $\{1\}$. The Boolean algebra of cardinality 4 has exactly two ultrafilters, corresponding to its two atoms. The Boolean algebra of cardinality 8 has exactly three ultrafilters, corresponding to its three atoms.

In an infinite Boolean algebra, an ultrafilter may fail to be principal, i.e., to have a minimum. The following gives an example.

Example 1.3.13 (Ultrafilters of $\mathrm{FC}(\mathbb{N})$)**.** For each $n \in \mathbb{N}$, the set

$$U_n \;:=\; \{\, A \in \mathrm{FC}(\mathbb{N}) \mid n \in A \,\}$$

is a (principal) ultrafilter on the finite-cofinite algebra $\mathrm{FC}(\mathbb{N})$ on $\mathbb{N}$.

In addition, the family of cofinite sets

$$U_\infty \;:=\; \{\, A \in \mathrm{FC}(\mathbb{N}) \mid A \text{ is cofinite} \,\}$$

is an ultrafilter on $\mathrm{FC}(\mathbb{N})$; it is not principal, since it has no minimum. As a simple exercise, you can prove that it has no other ultrafilters, i.e., that

$$\mathrm{Ult}(\mathrm{FC}(\mathbb{N})) = \{\, U_n \mid n \in \mathbb{N} \,\} \cup \{U_\infty\}.$$

A proof is in the footnote.[8]

Remark 1.3.14 (Ultrafilters on a power set)**.** What are the ultrafilters of $\mathcal{P}(X)$, for X a set? First of all, for every $x \in X$, we have an ultrafilter

$$U_x \coloneqq \{\, A \in \mathcal{P}(X) \mid x \in A \,\}.$$

These are precisely the *principal* ultrafilters.

- If X is finite, these are the only ultrafilters (see Proposition 1.3.11).
- For X infinite, does $\mathcal{P}(X)$ have nonprincipal ultrafilters? This is a nontrivial question; we will see that, if we assume the axiom of choice, we can prove the existence of nonprincipal ultrafilters; this will follow from the Boolean Prime Ideal Theorem (Theorem 1.3.17) proved below. Do not hope to find an explicit description of any of them, though.

1.3.2 Stone's Representation Theorem

Let us now take $X = \mathrm{Ult}(B)$ and define the embedding $B \hookrightarrow \mathcal{P}(X)$ required by Stone's Representation Theorem. Intuitively, this sends an element b to the set of "models" (ultrafilters) where b holds.

Definition 1.3.15 (Stone map)**.** For a Boolean algebra B, we define the ***Stone map***

$$\begin{aligned} \eta_B \colon B &\longrightarrow \mathcal{P}(\mathrm{Ult}(B)) \\ b &\longmapsto \{\, U \in \mathrm{Ult}(B) \mid b \in U \,\}. \end{aligned}$$

Logical intuition: the set $\eta_B(b)$ can be thought of as the set of "models" in which the "formula" b holds. To prove Stone's Representation Theorem it suffices to show that:

1. η_B is a Boolean homomorphism (easy),
2. η_B is injective (hard).

Lemma 1.3.16 (The Stone map is a homomorphism)**.** *The Stone map η_B is a Boolean homomorphism, i.e. for all $a, b \in B$,*

$$\eta_B(1) = \mathrm{Ult}(B), \qquad \eta_B(a \wedge b) = \eta_B(a) \cap \eta_B(b), \qquad \eta_B(\neg a) = \mathrm{Ult}(B) \setminus \eta_B(a),$$

(and hence also $\eta_B(a \vee b) = \eta_B(a) \cup \eta_B(b)$ and $\eta_B(0) = \varnothing$).

Proof. These are all straightforward computations. Details are in the footnote.[a] ∎

[a] The equality $\eta_B(1) = \mathrm{Ult}(B)$ is immediate: every ultrafilter contains 1.
For the binary meet, let $U \in \mathrm{Ult}(B)$. Then

$$U \in \eta_B(a \wedge b) \iff a \wedge b \in U \iff (a \in U \text{ and } b \in U) \iff U \in \eta_B(a) \cap \eta_B(b),$$

[8] Let U be an ultrafilter on $\mathrm{FC}(\mathbb{N})$. If U contains a finite set A, write A as a finite union of singletons: $A = \{n_1\} \cup \cdots \cup \{n_k\}$. Since U is an ultrafilter, the primeness condition in Definition 1.3.5 implies that some singleton $\{n_i\}$ belongs to U. Then U must coincide with U_{n_i}: if $B \in \mathrm{FC}(\mathbb{N})$ contains n_i, then $\{n_i\} \subseteq B$, hence $B \in U$ by upward closure; if $n_i \notin B$, then $n_i \in \mathbb{N} \setminus B$ and $\mathbb{N} \setminus B \in U$, hence $B \notin U$. If instead U contains no finite set, then for every finite $F \subseteq \mathbb{N}$ we have $F \notin U$, hence $\mathbb{N} \setminus F \in U$ by Lemma 1.3.6. Thus, U contains every cofinite set, i.e. $U_\infty \subseteq U$. Since U_∞ is itself an ultrafilter, maximality forces $U = U_\infty$.

where the middle equivalence uses upward closure and closure under binary meets of U.

For complements, using Lemma 1.3.6 we have

$$U \in \eta_B(\neg a) \iff \neg a \in U \iff a \notin U \iff U \notin \eta_B(a).$$

To conclude the proof of Stone's Representation Theorem (Theorem 1.3.20), it remains to show that η_B is injective. This is a separation statement: given two distinct elements of B, we must find an ultrafilter containing one but not the other. The required separation will follow from the Boolean Prime Ideal Theorem, proved next.

Let us recall Zorn's Lemma:

> Let $(P, \leq)$ be a partially ordered set. If
>
> 1. $P \neq \varnothing$,
> 2. and every nonempty chain[9] in P has an upper bound in P,
>
> then P has a maximal element.

Recall that Zorn's lemma is not provable in ZF, and is equivalent to the Axiom of Choice over ZF. We will use it in the following, which in turn we will use to conclude the proof of Stone's Representation Theorem.

Can every partial consistent choice of truth values be extended to a total one? The following can be thought of as a positive answer.

Theorem 1.3.17 (Boolean Prime Ideal Theorem)**.** *Let B be a Boolean algebra, let F be a filter, and let I be an ideal such that $F \cap I = \varnothing$. Then there is an ultrafilter U on B such that $F \subseteq U$ and $U \cap I = \varnothing$.*

Proof. Consider the poset

$$\mathcal{P} \coloneqq \{\, (G, J) \mid G \text{ is a filter},\ J \text{ is an ideal},\ F \subseteq G,\ I \subseteq J,\ G \cap J = \varnothing \,\},$$

ordered by componentwise inclusion: $(G, J) \leq (G', J')$ iff $G \subseteq G'$ and $J \subseteq J'$.

Nonemptiness. This poset is nonempty since $(F, I) \in \mathcal{P}$.

Nonempty chains have upper bounds. Let $\mathcal{C} \subseteq \mathcal{P}$ be a nonempty chain and set

$$G^* \coloneqq \bigcup_{(G,J)\in\mathcal{C}} G, \qquad J^* \coloneqq \bigcup_{(G,J)\in\mathcal{C}} J.$$

It is straightforward to prove that $(G^*, J*)$ belongs to $\mathcal{P}$: one checks that G^* is a filter, J^* is an ideal, $F \subseteq G^*$, $I \subseteq J^*$ and $G^* \cap J^* = \varnothing$. See the footnote for details. Thus, $(G^*, J^*) \in \mathcal{P}$ is an upper bound of $\mathcal{C}$.

By Zorn's lemma, $\mathcal{P}$ has a maximal element (U, K). At this point, the key step is to show that the maximal pair (U, K) "splits" B: once we know that $U \cup K = B$, the disjointness $U \cap K = \varnothing$ forces $K = B \setminus U$. Since K is an ideal, this will imply that $B \setminus U$ is an ideal, and hence U is an ultrafilter.

We claim that $U \cup K = B$. Suppose not, and pick $x \in B \setminus (U \cup K)$.

Let U' be the filter generated by $U \cup \{x\}$. Concretely,

$$U' = \{\, y \in B \mid \exists u \in U,\ (u \wedge x) \leq y \,\}.$$

If $U' \cap K = \varnothing$, then $(U', K) \in \mathcal{P}$ strictly extends (U, K), contradicting maximality. Hence, there are $u \in U$ and $k \in K$ such that $u \wedge x \leq k$, and thus $u \wedge x \in K$.

Similarly, let K' be the ideal generated by $K \cup \{x\}$. Concretely,

$$K' = \{\, y \in B \mid \exists k \in K,\ y \leq (k \vee x) \,\}.$$

[9] A chain is a totally ordered poset. Thus, a *chain in P* is a subset S of P such that, for all $x, y \in S$, either $x \leq y$ or $y \leq x$.

If $U \cap K' = \varnothing$, then $(U, K') \in \mathcal{P}$ strictly extends (U, K), again contradicting maximality. Hence, there are $u' \in U$ and $k' \in K$ such that $u' \leq k' \vee x$.

Set $w \coloneqq u \wedge u' \in U$. Then $w \wedge x \leq u \wedge x \in K$, hence $w \wedge x \in K$ (since K is downward closed), and also $w \leq u' \leq k' \vee x$. Using distributivity,

$$w = w \wedge (k' \vee x) = (w \wedge k') \vee (w \wedge x).$$

Now $w \wedge k' \leq k'$, hence $w \wedge k' \in K$, and we already know $w \wedge x \in K$. Since K is closed under finite joins, we conclude $w \in K$. This contradicts $w \in U$ and $U \cap K = \varnothing$. Therefore $U \cup K = B$. This proves that U is an ultrafilter. It is clear that $F \subseteq U$ and $U \cap I = \varnothing$. ■

Corollary 1.3.18 (Ultrafilters separate elements)**.** *If $a \not\leq b$ in B, then there is an ultrafilter $U \in \mathrm{Ult}(B)$ such that $a \in U$ and $b \notin U$.*

Proof. Let $F \coloneqq \uparrow a$ be the principal filter generated by a, and let

$$I \coloneqq \downarrow b \coloneqq \{\, x \in B \mid x \leq b \,\}$$

be the principal ideal generated by b. We notice that $F \cap I = \varnothing$, since otherwise there would be $x \in \uparrow a \cap \downarrow b$, which would imply $a \leq x \leq b$, which would contradict $a \not\leq b$. By Theorem 1.3.17 there is an ultrafilter U such that $F \subseteq U$ and $U \cap I = \varnothing$. In particular, $a \in U$ and $b \notin U$. ■

As a consequence:

Lemma 1.3.19 (The Stone map is injective)**.** *For every Boolean algebra B, the Stone map $\eta_B \colon B \to \mathcal{P}(\mathrm{Ult}(B))$ defined in Definition 1.3.15 is injective.*

Proof. Let $a \neq b$ in B. Then either $a \not\leq b$ or $b \not\leq a$. Assume $a \not\leq b$. (The proof in the other case is perfectly symmetrical.) By Corollary 1.3.18, there is $U \in \mathrm{Ult}(B)$ with $a \in U$ and $b \notin U$. Thus $U \in \eta_B(a)$ but $U \notin \eta_B(b)$, so $\eta_B(a) \neq \eta_B(b)$. ■

We are now ready to prove Stone's Representation Theorem for Boolean algebras, proved by Stone in [Stone, 1936].[10] Stone's Representation Theorem for Boolean algebras is a positive answer to the question: can every abstract Boolean algebra be made concrete by representing its elements as actual subsets of some set?

Theorem 1.3.20 (Stone's Representation Theorem for Boolean algebras)**.** *For every Boolean algebra B there is a set X such that B is isomorphic to a Boolean subalgebra of the power set $\mathcal{P}(X)$.*

Proof. This follows from the fact that the Stone map $\eta_B \colon B \to \mathcal{P}(\mathrm{Ult}(B))$ is a Boolean homomorphism (Lemma 1.3.16) and is injective (Lemma 1.3.19). ■

Remark 1.3.21 (Choice)**.** To prove Stone's Representation Theorem we used the axiom of choice (since we used Zorn's lemma to prove the Boolean Prime Ideal Theorem). In fact, Stone's Representation Theorem is not provable in ZF alone. Over ZF, Stone's Representation Theorem for Boolean algebras is equivalent to the Boolean Prime Ideal Theorem; see, for instance, [Jech, 1973, Sec. 2.6]. The Boolean Prime Ideal Theorem is weaker than the Axiom of Choice, in the sense that it follows from Choice (via Zorn's lemma), but it does not imply Choice [Halpern and Lévy, 1971].

[10] If you want a modern, succint and easily readable textbook reference, I recommend [Givant and Halmos, 2009, Ch. 22] (see, in particular, Cor. 1, p. 190).

1.4 Boolean algebras are precisely the Lindenbaum–Tarski algebras

In Section 1.2.4 we saw that every propositional theory $\mathcal{T}$ gives rise to a Boolean algebra $\mathrm{Form}(\mathcal{L})/{\equiv_{\mathcal{T}}}$, where $\equiv_{\mathcal{T}}$ denotes the semantic equivalence between formulas modulo $\mathcal{T}$: this Boolean algebra is called the Lindenbaum–Tarski algebra of $\mathcal{T}$.

Now we prove the converse statement announced earlier: *every Boolean algebra arises in this way.*

Warm-up examples

Example 1.4.1 (2 as a Lindenbaum–Tarski algebra)**.** Let $\mathcal{L} = \varnothing$ and $\mathcal{T} = \varnothing$. Then $\mathrm{Form}(\mathcal{L})$ only contains the constants $0, 1$ and the formulas built from them. Up to $\equiv_{\mathcal{T}}$ there are exactly two equivalence classes, so $\mathrm{Form}(\varnothing)/{\equiv_{\varnothing}} \cong 2$.

Example 1.4.2 (The 4-elements Boolean algebra as a Lindenbaum–Tarski algebra)**.** Let $\mathcal{L} = \{p\}$ and $\mathcal{T} = \varnothing$. Then $\mathrm{Form}(\mathcal{L})/{\equiv_{\mathcal{T}}}$ is a Boolean algebra with 4 elements (it is isomorphic to $\mathcal{P}(\{x, y\})$).

Example 1.4.3 (The trivial Boolean algebra as a Lindenbaum–Tarski algebra)**.** If a theory $\mathcal{T}$ has no models (i.e. $\mathrm{Mod}(\mathcal{T}) = \varnothing$), then by definition every two formulas are $\equiv_{\mathcal{T}}$-equivalent. Hence,Hence, $\mathrm{Form}(\mathcal{L})/{\equiv_{\mathcal{T}}}$ is the trivial Boolean algebra. For instance, one can take any $\mathcal{L}$ and $\mathcal{T} = \{0\}$.

Every Boolean algebra is a Lindenbaum–Tarski algebra

We now show that *every* Boolean algebra can be presented as $\mathrm{Form}(\mathcal{L})/{\equiv_{\mathcal{T}}}$ for a suitable language $\mathcal{L}$ and theory $\mathcal{T}$, i.e., that it is the algebra of propositions of a suitable language and theory.

Theorem 1.4.4 (Every Boolean algebra is a Lindenbaum–Tarski algebra)**.** *For every Boolean algebra B there is a propositional language $\mathcal{L}$ and a propositional theory $\mathcal{T}$ such that*

$$B \;\cong\; \mathrm{Form}(\mathcal{L})/{\equiv_{\mathcal{T}}}.$$

Proof. Let B be a Boolean algebra. We build a language whose propositional variables are *the elements of B themselves*:

$$\mathcal{L}_B \;\coloneqq\; B.$$

To avoid confusion, we will use $\wedge_B, \vee_B, \neg_B, 0_B, 1_B$ for the operations of the Boolean algebra B, while $\wedge, \vee, \neg, 0, 1$ are the connectives/constants in $\mathrm{Form}(\mathcal{L}_B)$.

Define a theory $\mathcal{T}_B$ that forces the propositional variables to behave like B:

$$\mathcal{T}_B \;\coloneqq\; \Big\{\, (\neg b) \leftrightarrow (\neg_B b) \;\Big|\; b \in B \,\Big\} \;\cup\; \Big\{\, (b \wedge c) \leftrightarrow (b \wedge_B c) \;\Big|\; b, c \in B \,\Big\} \;\cup\; \{1 \leftrightarrow 1_B\}.$$

(Here $(\neg_B b)$, $(b \wedge_B c)$ and 1_B are elements of B, and hence propositional variables in the language $\mathcal{L}_B$. Moreover, the axioms for $\vee$ and 0 are not necessary since $\vee$ and 0 are definable from $\wedge, \neg, 1$.)

Now consider the assignment $e\colon \mathcal{L}_B \to B$ given by the identity map $e(b) = b$. As for truth assignments into 2, the map e extends uniquely (by recursion on formulas)[a] to an evaluation map

$$\bar{e}\colon \mathrm{Form}(\mathcal{L}_B) \to B,$$

defined by interpreting $\vee, \wedge, \neg, 0, 1$ as $\vee_B, \wedge_B, \neg_B, 0_B, 1_B$. In particular, $\bar{e}$ is a $\{\vee, \wedge, \neg, 0, 1\}$-homomorphism, and it is surjective because every $b \in B$ is the value of the variable b.

We claim that for all formulas $\varphi, \psi \in \mathrm{Form}(\mathcal{L}_B)$ we have

$$\varphi \equiv_{\mathcal{T}_B} \psi \qquad \Longleftrightarrow \qquad \bar{e}(\varphi) = \bar{e}(\psi) \;\text{ in } B.$$

Granting this for a moment, the isomorphism $B \cong \mathrm{Form}(\mathcal{L}_B)/{\equiv_{\mathcal{T}_B}}$ follows immediately: define $\pi\colon \mathrm{Form}(\mathcal{L}_B) \twoheadrightarrow \mathrm{Form}(\mathcal{L}_B)/{\equiv_{\mathcal{T}_B}}$ and set

$$\Phi\colon B \to \mathrm{Form}(\mathcal{L}_B)/{\equiv_{\mathcal{T}_B}}, \qquad \Phi(b) \coloneqq [b]\,.$$

The claim says precisely that Φ is injective (two generators $[b]\,,[c]$ coincide iff $b = c$ in B), and surjectivity is clear because every class is represented by some formula and $\bar{e}$ hits all elements of B. Moreover, Φ is a Boolean homomorphism by construction. For instance, $\Phi(b \wedge_B c) = [b \wedge_B c] = [b \wedge c] = [b] \wedge [c] = \Phi(b) \wedge \Phi(c)$ by definition of $\mathcal{T}_B$, and similarly for $\neg$ and 1. Hence, Φ is an isomorphism.

It remains to prove the claim. First, observe the key fact (which is a direct unpacking of the axioms of $\mathcal{T}_B$):

Models of $\mathcal{T}_B$ are the same thing as Boolean homomorphisms $h\colon B \to 2$.

Indeed, if $h\colon B \to 2$ is a Boolean homomorphism, then the valuation $v_h\colon \mathcal{L}_B \to 2$ given by $v_h(b) = h(b)$ satisfies the axioms of $\mathcal{T}_B$, so $v_h \in \mathrm{Mod}(\mathcal{T}_B)$. Conversely, if $v \in \mathrm{Mod}(\mathcal{T}_B)$, then the map $h_v\colon B \to 2$ defined by $h_v(b) \coloneqq v(b)$ is a Boolean homomorphism, because the axioms in $\mathcal{T}_B$ exactly say that v respects $\neg_B$, $\wedge_B$, and 1_B.

Now suppose $\bar{e}(\varphi) = \bar{e}(\psi)$. Let $v \in \mathrm{Mod}(\mathcal{T}_B)$ and let $h_v\colon B \to 2$ be the corresponding homomorphism. By construction of evaluation-by-recursion, the interpretation of formulas agrees:

$$\bar{v} \;=\; h_v \circ \bar{e}.$$

Therefore $\bar{v}(\varphi) = h_v(\bar{e}(\varphi)) = h_v(\bar{e}(\psi)) = \bar{v}(\psi)$. Since this holds for all $v \in \mathrm{Mod}(\mathcal{T}_B)$, we get $\varphi \equiv_{\mathcal{T}_B} \psi$.

Conversely, suppose $\bar{e}(\varphi) \neq \bar{e}(\psi)$ in B. By the fact that ultrafilters separate distinct elements of B (i.e., that $\eta_B\colon B \to \mathcal{P}(\mathrm{Ult}(B))$ is injective, Lemma 1.3.19), there is a Boolean homomorphism $h\colon B \to 2$ such that

$$h(\bar{e}(\varphi)) \;\neq\; h(\bar{e}(\psi)).$$

Let $v_h \in \mathrm{Mod}(\mathcal{T}_B)$ be the corresponding model. Then again $\bar{v}_h = h \circ \bar{e}$, hence $\bar{v}_h(\varphi) \neq \bar{v}_h(\psi)$, so $\varphi \not\equiv_{\mathcal{T}_B} \psi$. This proves the claim and completes the proof. ■

[a] Universal-algebra viewpoint: for the algebraic signature $\Sigma = \{0, 1, \neg, \wedge, \vee\}$, the forgetful functor $\mathsf{Alg}(\Sigma) \to \mathsf{Set}$ has a left adjoint sending a set $\mathcal{L}$ to the Σ-algebra of terms on $\mathcal{L}$. Here, $\bar{e}$ is exactly the unique Σ-homomorphism extending e, whose existence is guaranteed by the universal property of the unit of the adjunction.

1.5 Bijection between Boolean algebras and Stone spaces

Stone's Representation Theorem says that every Boolean algebra embeds into a power set. However, such a representation is typically *not unique*. There are two properties—*separation* and *compactness*—that single out the canonical representation

$$\eta_B\colon B \hookrightarrow \mathcal{P}(\mathrm{Ult}(B)).$$

Example 1.5.1 (Separated vs non-separated)**.** For instance, the two-element Boolean algebra 2 has the following two representations:

$$\begin{aligned} \iota_1\colon 2 &\hookrightarrow \mathcal{P}(\{*\}) \\ 0 &\longmapsto \varnothing \\ 1 &\longmapsto \{*\} \end{aligned} \qquad\qquad \begin{aligned} \iota_2\colon 2 &\hookrightarrow \mathcal{P}(\{x, y\}) \\ 0 &\longmapsto \varnothing \\ 1 &\longmapsto \{x, y\}. \end{aligned}$$

The representation ι_2 is not "*separated*": the only sets in its image are $\varnothing$ and $\{x, y\}$, so the image cannot distinguish x from y.

Example 1.5.2 (Non-compact vs compact)**.** Let $\mathrm{FC}(\mathbb{N})$ be the Boolean algebra of finite and cofinite subsets of $\mathbb{N}$. It has the following two representations:

$$\jmath_1 \colon \mathrm{FC}(\mathbb{N}) \hookrightarrow \mathcal{P}(\mathbb{N}) \qquad\qquad \jmath_2 \colon \mathrm{FC}(\mathbb{N}) \hookrightarrow \mathcal{P}(\mathbb{N} \cup \{\infty\})$$

$$A \longmapsto A \qquad\qquad A \longmapsto \begin{cases} A, & \text{if } A \text{ is finite,} \\ A \cup \{\infty\}, & \text{if } A \text{ is cofinite.} \end{cases}$$

The representation $\jmath_1$ is not "*compact*": the family of singletons $\{\{n\} \mid n \in \mathbb{N}\}$ consists of elements of $\mathrm{FC}(\mathbb{N})$ and covers $\mathbb{N}$, but it has no finite subcover.

On the other hand, $\jmath_2$ is *compact*. Indeed, let $\{\jmath_2(A_i)\}_{i \in I}$ be a cover of $\mathbb{N} \cup \{\infty\}$. Since ∞ is covered, there is $i_0 \in I$ such that $\infty \in \jmath_2(A_{i_0})$. By definition of $\jmath_2$, this means that A_{i_0} is cofinite in $\mathbb{N}$. Then $\jmath_2(A_{i_0})$ already covers all but finitely many natural numbers; the remaining finitely many points of $\mathbb{N}$ can be covered by finitely many singleton sets $\jmath_2(\{n\}) = \{n\}$. Hence, the original cover admits a finite subcover.

The previous examples show that representations into power sets can fail to be *separated* or *compact*. Spoiler: for every Boolean algebra B, the canonical representation

$$\eta_B \colon B \hookrightarrow \mathcal{P}(\mathrm{Ult}(B))$$

is (up to renaming the points of the underlying set) the unique one that is *both* separated and compact.

Now, both separation and compactness are strongly "topology-like" properties. So rather than treating them as ad hoc conditions on set-theoretic representations, we move to the topological setting: we equip $\mathrm{Ult}(B)$ with a natural topology generated by the basic subsets $\eta_B(b)$. With this topology, $\mathrm{Ult}(B)$ becomes a *Stone space*, i.e. a space that is totally separated and compact (definitions to follow).

Then, we will get a bijection between isomorphism classes of Boolean algebras and isomorphism classes of Stone spaces.

1.5.1 Stone spaces and the Boolean algebra of clopens

Let X be a topological space.[11] A subset $C \subseteq X$ is called ***clopen*** if it is both closed and open in X. We write

$$\mathrm{Clop}(X)$$

for the set of clopen subsets of X. Notice that $\mathrm{Clop}(X)$ is a Boolean subalgebra of $\mathcal{P}(X)$.

Definition 1.5.3 (Stone space)**.** A ***Stone space***[a] is a topological space that is

1. ***totally separated***: for all $x, y \in X$ with $x \neq y$ there is a clopen subset C of X such that $x \in C$ and $y \notin C$;
2. ***compact***: every open cover of X has a finite subcover.

[a] Two further equivalent characterizations of Stone spaces are: (i) compact Hausdorff zero-dimensional spaces, and (ii) compact Hausdorff totally disconnected spaces.

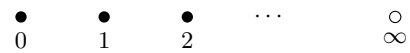
Notice that totally separated implies Hausdorff.

Example 1.5.4 (Examples of Stone spaces)**.**

1. (Finite discrete spaces) Every *finite discrete space* is a Stone space. In fact, these are the only finite Stone spaces.

[11] For a textbook on topology, see, e.g., [Munkres, 2000].

2. (One-point compactification of $\mathbb{N}$) The simplest example of an infinite Stone space is called the *one-point* (or *Alexandroff*) *compactification* of $\mathbb{N}$. It is denoted $\alpha\mathbb{N}$, and is obtained by adding to the discrete space $\mathbb{N}$ an accumulation point ∞.



More precisely, the opens of $\mathbb{N} \cup \{\infty\}$ are:

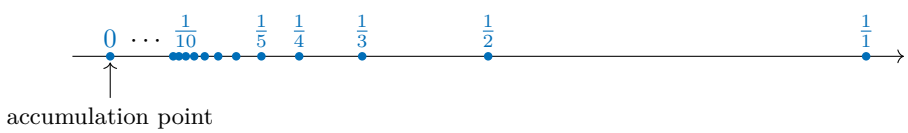
- every subset not containing ∞, and
- every cofinite subset containing ∞.

Then, the clopens are:

- every finite subset not containing ∞, and
- every cofinite subset containing ∞.

For a visual rendering, $\alpha\mathbb{N}$ is homeomorphic to the following subspace of $\mathbb{R}$ via the map $n \mapsto \frac{1}{n}$ and $\infty \mapsto 0$:

$$\{0\} \cup \left\{ \frac{1}{n} \mid n \in \mathbb{N} \setminus \{0\} \right\}.$$



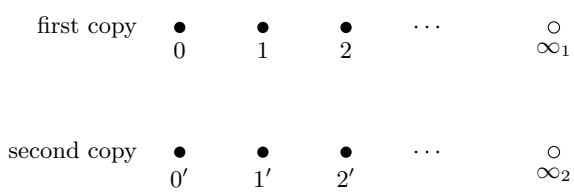
3. (One-point compactification of an infinite set X) Generalizing the previous example, for any infinite set X, we have a Stone space called the *one-point* (or *Alexandroff*) *compactification* of X. It is denoted αX, and is obtained by adding to the discrete space X an accumulation point ∞.

More precisely, it is the space obtained by equipping $X \cup \{\infty\}$ with the topology whose open sets are:

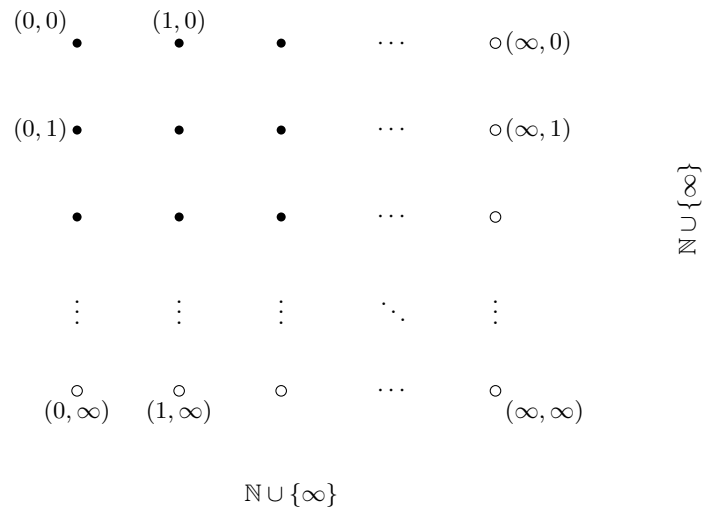
- every subset of X, and
- every cofinite subset U such that $\infty \in U$.

The clopens are the finite sets not containing ∞ and the cofinite sets containing ∞.

4. Any disjoint union of finitely many Stone spaces, equipped with the disjoint union topology, is a Stone space. So, for example, the union $\alpha\mathbb{N} + \alpha\mathbb{N}$ of two copies of the one-point compactification of $\mathbb{N}$ is a Stone space.



5. An arbitrary product of Stone spaces (with the product topology) is a Stone space (see Proposition 1.7.8(1) below for a proof). So, for example, the product of two copies of the one-point compactification $\alpha\mathbb{N}$ of the natural numbers is a Stone space; in contrast to the examples above, this has infinitely many accumulation points (the white ones).

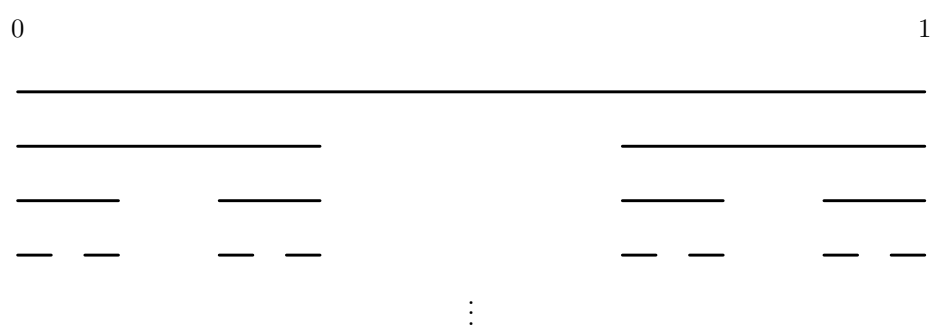


6. Taking products allows us also to provide an example of an infinite Stone space where all points are accumulation points: namely, the product

$$2^{\mathbb{N}} = \prod_{n\in\mathbb{N}} 2 = 2 \times 2 \times 2 \times \dots$$

of countably many copies of the two-element discrete space 2. In other words, this is the space of $\mathbb{N}$-indexed sequences of 0 and 1.

This space is called the *Cantor space* and is homeomorphic to the subset of $[0, 1]$ known as the *Cantor set*[12], equipped with the Euclidean topology.



Cantor set

Example 1.5.5 (*Non*-examples of Stone spaces)**.**

1. Any infinite discrete space is *not* a Stone space, because it is not compact.
2. $[0, 1]$ is *not* a Stone space, because it is not totally separated.

[12]For the definition of the Cantor set, see, e.g., [Munkres, 2000, Ch. 3, Sec. 27, Exercise 6, p. 178].

1.5.2 The Stone topology on the set of ultrafilters

Definition 1.5.6 (Stone topology on the set of ultrafilters)**.** Let B be a Boolean algebra. The ***Stone topology*** on $\mathrm{Ult}(B)$ is the topology *generated* by the subsets

$$\eta_B(b) \qquad (b \in B).$$

That is, the open sets are the arbitrary unions of elements of the form[a]

$$\eta_B(b) \qquad (b \in B).$$

[a] In general, the topology generated by a set $\mathcal{A}$ of subsets of a set X is the set of arbitrary unions of finite intersections of elements in $\mathcal{A}$. Here, there is no need to take finite intersections because the family $\{\, \eta_B(b) \mid b \in B \,\}$ is closed under finite intersections.

Thus, the family $\{\, \eta_B(b) \mid b \in B \,\}$ forms a basis of the Stone topology on $\mathrm{Ult}(B)$, and hence the elements of the form $\eta_B(b)$ are called *basic open* sets of $\mathrm{Ult}(B)$.

Theorem 1.5.7 (The space of ultrafilters is a Stone space)**.** *For every Boolean algebra B, the space $\mathrm{Ult}(B)$ of ultrafilters equipped with the Stone topology is a Stone space.*

Proof. Totally separated. Let $U, V \in \mathrm{Ult}(B)$ with $U \neq V$. Then, there is b which belongs to exactly one of the two, say, for example, U (the other case is perfectly symmetrical). Then $U \in \eta_B(b)$ and $V \notin \eta_B(b)$. Moreover, $\eta_B(b)$ is clopen, since its complement is $\eta_B(\neg b)$, which is open. Hence, $\eta_B(b)$ is a clopen separating U from V. Thus, $\mathrm{Ult}(B)$ is totally separated.

Compact. To check the compactness of a space, it is enough to check that any cover by opens in a given basis has a finite subcover. Suppose that $\mathrm{Ult}(B)$ is covered by basic opens:

$$\mathrm{Ult}(B) \;=\; \bigcup_{j \in J} \eta_B(b_j). \tag{1.2}$$

Consider the ideal $I \subseteq B$ generated by the set $\{\, b_j \mid j \in J \,\}$, i.e.

$$I = {\downarrow}\{\, b_{j_1} \vee \cdots \vee b_{j_n} \mid n \in \mathbb{N}, j_1, \ldots, j_n \in J \,\}.$$

We claim that $1 \in I$. Indeed, if, by way of contradiction, we had $1 \notin I$, then the filter $\{1\}$ would be disjoint from I; then, by the Boolean Prime Ideal Theorem (Theorem 1.3.17), there would be an ultrafilter U extending $\{1\}$ and disjoint from I. But by hypothesis ((1.2)), we deduce that there is $j \in J$ such that $U \in \eta_B(b_j)$, i.e., $b_j \in U$. This contradicts the facts that $b_j \in I$ and that I and U are disjoint. This proves the claim $1 \in I$.

Since $1 \in I$, by definition of I there are $j_1, \ldots, j_n \in J$ such that

$$1 \;\leq\; b_{j_1} \vee \cdots \vee b_{j_n},$$

and hence

$$1 = b_{j_1} \vee \cdots \vee b_{j_n}.$$

Applying η_B on both sides, and using the fact that it η_B a Boolean homomorphism (Lemma 1.3.16), we get

$$\mathrm{Ult}(B) = \eta_B(1) = \eta_B(b_{j_1} \vee \cdots \vee b_{j_n}) = \eta_B(b_{j_1}) \cup \cdots \cup \eta_B(b_{j_n}),$$

and so we have found a finite subcover. ∎

In the proof of Theorem 1.5.7 we used the (immediate) fact that the subsets of $\mathrm{Ult}(B)$ of the form $\eta_B(b)$

are clopen; in fact, the sets of the form $\eta_B(b)$ are all and the only clopens, as shown in the following.

Lemma 1.5.8 (In the space of ultrafilters: clopen = basic open)**.** *Let B be a Boolean algebra. The clopens of* $\mathrm{Ult}(B)$ *are precisely the subsets of the form $\eta_B(b)$, for some $b \in B$.*

Proof. ($\Leftarrow$). For every $b \in B$, $\eta_B(b)$ is clopen because it is open and its complement is $\eta_B(\neg b)$ (since η_B is a Boolean homomorphism by Lemma 1.3.16), which is open, too.

($\Rightarrow$). Let C be a clopen of $\mathrm{Ult}(B)$. Since C is open, we can write

$$C = \bigcup_{i \in I} \eta_B(b_i)$$

for some family $(b_i)_{i \in I}$ in B, and automatically each $\eta_B(b_i) \subseteq C$. Because C is closed in the space $\mathrm{Ult}(B)$, which is compact by Theorem 1.5.7, the subspace C is compact. Thus, the open cover $\{\, \eta_B(b_i) \mid i \in I \,\}$ of C has a finite subcover: there is a finite $I_0 \subseteq I$ such that

$$C = \bigcup_{i \in I_0} \eta_B(b_i).$$

Since η_B is a homomorphism,

$$C = \eta_B\left(\bigvee_{i \in I_0} b_i\right),$$

so C lies in the image of η_B. ∎

1.5.3 Bijection between Boolean algebras and Stone spaces

Theorem 1.5.9 ($B \cong \mathrm{Clop}(\mathrm{Ult}(B))$)**.** *For every Boolean algebra B, the map*

$$\begin{aligned} \eta_B \colon B &\longrightarrow \mathrm{Clop}(\mathrm{Ult}(B)), \\ b &\longmapsto \{\, U \in \mathrm{Ult}(B) \mid b \in U \,\} \end{aligned}$$

is an isomorphism of Boolean algebras.

Proof. By Lemma 1.5.8, for every b the set $\eta_B = \{\, U \in \mathrm{Ult}(B) \mid b \in U \,\}$ is indeed clopen, and the map $\eta_B \colon B \longrightarrow \mathrm{Clop}(\mathrm{Ult}(B))$ is surjective. By Lemmas 1.3.16 and 1.3.19, $\eta_B \colon B \longrightarrow \mathrm{Clop}(\mathrm{Ult}(B))$ is an injective homomorphism. ∎

It follows immediately:

Corollary 1.5.10 (Finite Boolean algebras are power sets)**.**

1. *Every <u>finite</u> Boolean algebra is isomorphic to the power set of some finite set.*
2. *Every <u>finite</u> Boolean algebra has cardinality 2^n for some $n \in \mathbb{N}$.*
3. *Any two <u>finite</u> Boolean algebras with the same cardinality are isomorphic.*

Proof. (1). Let B be finite. Then $\mathrm{Ult}(B)$ is finite as well (indeed $\mathrm{Ult}(B) \subseteq \mathcal{P}(B)$). Since $\mathrm{Ult}(B)$ is a finite Stone space, it is discrete[a], so every subset is clopen and hence

$$\mathrm{Clop}(\mathrm{Ult}(B)) = \mathcal{P}(\mathrm{Ult}(B)).$$

By the isomorphism $B \cong \mathrm{Clop}(\mathrm{Ult}(B))$ we conclude that $B \cong \mathcal{P}(X)$ for the finite set $X \coloneqq \mathrm{Ult}(B)$.

(2). This follows immediately from (1).

(3). Let B and B' be finite Boolean algebras and assume $|B| = |B'|$. By (1), B and B' are isomorphic to $\mathcal{P}(X)$ and $\mathcal{P}(X')$, respectively, for some finite sets X and X'. We have

$$2^{|X|} = |B| = |B'| = 2^{|X'|},$$

and so $|X| = |X'|$. It is immediate that two sets with the same cardinality have isomorphic power set Boolean algebras; hence, $B \cong \mathcal{P}(X) \cong \mathcal{P}(X') \cong B'$. ■

[a] It is easily seen that every T_1 finite space (and so, in particular, every Stone space) is discrete.

Theorem 1.5.11 ($X \cong \mathrm{Ult}(\mathrm{Clop}(X))$)**.** *For every Stone space X, the map*

$$\begin{aligned} \varepsilon_X \colon X &\longrightarrow \mathrm{Ult}(\mathrm{Clop}(X)) \\ x &\longmapsto \{\, C \in \mathrm{Clop}(X) \mid x \in C \,\} \end{aligned}$$

is a homeomorphism (where $\mathrm{Ult}(\mathrm{Clop}(X))$ *carries the Stone topology).*

Proof. First of all, the map is well-defined because, for every $x \in X$, the set $\varepsilon_X(x) = \{\, C \in \mathrm{Clop}(X) \mid x \in C \,\}$ is easily seen[a] to be an ultrafilter of $\mathrm{Clop}(X)$.

Since $\varepsilon_X \colon X \to \mathrm{Ult}(\mathrm{Clop}(X))$ is a function between compact Hausdorff spaces, to prove that it is a homeomorphism it is enough[b] to prove that it is

1. continuous (this is a straightforward computation),
2. injective (this will follow from <u>total separation</u>),
3. surjective (this will follow from <u>compactness</u>).

Continuity. This is a straightforward computation.[c]

Injectivity. If $x \neq y$, since X is <u>totally separated</u> there is a clopen $C \subseteq X$ such that $x \in C$ and $y \notin C$. Then $C \in \varepsilon_X(x)$ but $C \notin \varepsilon_X(y)$, and hence $\varepsilon_X(x) \neq \varepsilon_X(y)$.

Surjectivity. Let $U \in \mathrm{Ult}(\mathrm{Clop}(X))$. We claim that

$$\bigcap_{C \in U} C \neq \varnothing. \tag{1.3}$$

Since U is closed under finite intersections and does not contain $\varnothing$, the family of closed sets $\{\, C \mid C \in U \,\}$ has the finite intersection property (i.e., every intersection of finitely many of its elements is nonempty). Recall that every family of closed sets with the finite intersection property has nonempty intersection. This shows (1.3).

Having established that $\bigcap_{C \in U} C \neq \varnothing$, pick x in this intersection. We claim $U = \varepsilon_X(x)$. The inclusion $U \subseteq \varepsilon_X(x)$ is obvious because for all $C \in U$ we have $x \in C$. To prove the reverse inclusion ($\varepsilon_X(x) \subseteq U$), let $D \in \mathrm{Clop}(X)$. Reasoning by contraposition, suppose $D \notin U$ and let us prove $D \notin \varepsilon_X(x)$. Since U is an ultrafilter, exactly one of D and $X \setminus D$ belongs to U; hence, $X \setminus D \in U$. Thus, $x \in X \setminus D$ (since x belongs to the intersection of all clopens in U), and so $x \notin D$, i.e. $D \notin \varepsilon_X(x)$. Thus, $U = \varepsilon_X(x)$. Therefore, ε_X is surjective. ■

[a] Indeed, it is easily seen that $\varepsilon_X(x)$ is a filter. It is also clear that $\varnothing \notin \varepsilon_X(x)$. Finally, if $C \cup D \in \varepsilon_X(x)$ then $x \in C \cup D$, hence $x \in C$ or $x \in D$, i.e. $C \in \varepsilon_X(x)$ or $D \in \varepsilon_X(x)$. So $\varepsilon_X(x)$ is an ultrafilter.

[b] Indeed, every continuous bijection between compact Hausdorff spaces is a homeomorphism. This follows from the fact that every continuous map between compact Hausdorff spaces is closed, i.e. the image of a closed subset is closed. This latter

fact, in turn, follows from the facts that (i) a subset of a compact Hausdorff space is closed if and only if it is compact, and (ii) every continuous map maps a compact set to a compact set.

[c]Indeed, the Stone topology on $\mathrm{Ult}(\mathrm{Clop}(X))$ has a basis of opens of the form

$$\eta_{\mathrm{Clop}(X)}(C) \;=\; \{\, U \in \mathrm{Ult}(\mathrm{Clop}(X)) \mid C \in U \,\}, \qquad (C \in \mathrm{Clop}(X)).$$

For such a basic open we have

$$\varepsilon_X^{-1}\left[\eta_{\mathrm{Clop}(X)}(C)\right] \;=\; \{\, x \in X \mid \varepsilon_X(x) \in \eta_{\mathrm{Clop}(X)}(C) \,\} \;=\; \{\, x \in X \mid C \in \varepsilon_X(x) \,\} \;=\; \{\, x \in X \mid x \in C \,\} \;=\; C,$$

which is open. Hence, ε_X is continuous.

1.6 Stone duality

We finally prove Stone duality.[13]

Definition 1.6.1 (The categories BA and Stone)**.**

1. The category BA has Boolean algebras as objects and Boolean homomorphisms as morphisms.
2. The category Stone has Stone spaces as objects and continuous maps as morphisms.

Definition 1.6.2 (The functors Clop and Ult between BA and Stone)**.**

1. (Clop) We define a functor
$$\mathrm{Clop}\colon \mathsf{Stone}^{\mathrm{op}} \longrightarrow \mathsf{BA}$$
as follows:
   - (On objects:) it maps a Stone space X to the Boolean algebra $\mathrm{Clop}(X)$ of clopens of X.
   - (On morphisms:) it maps a continuous map $f\colon X \to Y$ to the preimage function
$$\begin{aligned} f^{-1}[-]\colon \mathrm{Clop}(Y) &\longrightarrow \mathrm{Clop}(X) \\ A &\longmapsto f^{-1}[A]. \end{aligned}$$
2. (Ult) We define a functor
$$\mathrm{Ult}\colon \mathsf{BA} \longrightarrow \mathsf{Stone}^{\mathrm{op}}$$
as follows:
   - (On objects:) it maps a Boolean algebra A to the Stone space $\mathrm{Ult}(A)$ of ultrafilters of A.
   - (On morphisms:) it maps a Boolean homomorphism $f\colon A \to B$ to the preimage function
$$\begin{aligned} f^{-1}[-]\colon \mathrm{Ult}(B) &\longrightarrow \mathrm{Ult}(A) \\ U &\longmapsto f^{-1}[U]. \end{aligned}$$

Theorem 1.6.3 (Stone duality for Boolean algebras)**.** *The categories* BA *of Boolean algebras and* Stone *of Stone spaces are dually equivalent*

$$\mathsf{BA} \simeq \mathsf{Stone}^{\mathrm{op}},$$

as witnessed by the quasi-inverse functors $\mathrm{Clop}\colon \mathsf{Stone}^{\mathrm{op}} \to \mathsf{BA}$ *and* $\mathrm{Ult}\colon \mathsf{BA} \to \mathsf{Stone}^{\mathrm{op}}$ *and the natural isomorphisms* $\Big(\eta_B\colon B \to \mathrm{Clop}(\mathrm{Ult}(B))\Big)_{B \in \mathsf{BA}}$ *and* $\Big(\varepsilon_X\colon X \to \mathrm{Ult}(\mathrm{Clop}(X))\Big)_{X \in \mathsf{Stone}}$.

[13]For categorical background, see, e.g., [Mac Lane, 1998].

Proof. Consider the families of maps

$$\begin{aligned} \eta_B \colon B &\longrightarrow \mathrm{Clop}(\mathrm{Ult}(B)) \\ b &\longmapsto \{\, U \in \mathrm{Ult}(B) \mid b \in U \,\} \end{aligned} \qquad \begin{aligned} \varepsilon_X \colon X &\longrightarrow \mathrm{Ult}(\mathrm{Clop}(X)) \\ x &\longmapsto \{\, C \in \mathrm{Clop}(X) \mid x \in C \,\}. \end{aligned}$$

By the theorems proved above, each η_B is an isomorphism of Boolean algebras (Theorem 1.5.9), and each ε_X is a homeomorphism (Theorem 1.5.11).

They are natural: for every Boolean homomorphism $h \colon A \to B$ and every continuous map $f \colon X \to Y$, the diagrams

$$\begin{tikzcd} A \arrow[r, "\eta_A"] \arrow[d, "h"'] & \mathrm{Clop}(\mathrm{Ult}(A)) \arrow[d, "\mathrm{Clop}(\mathrm{Ult}(h))"] \\ B \arrow[r, "\eta_B"'] & \mathrm{Clop}(\mathrm{Ult}(B)) \end{tikzcd} \qquad \begin{tikzcd} X \arrow[r, "\varepsilon_X"] \arrow[d, "f"'] & \mathrm{Ult}(\mathrm{Clop}(X)) \arrow[d, "\mathrm{Ult}(\mathrm{Clop}(f))"] \\ Y \arrow[r, "\varepsilon_Y"'] & \mathrm{Ult}(\mathrm{Clop}(Y)) \end{tikzcd}$$

commute (both are straightforward "unwind the definitions" computations, done in the footnote[a]).

Hence, Clop and Ult are quasi-inverse up to these natural isomorphisms, i.e. they define a dual equivalence.

In fact, this is an ***adjoint dual equivalence***: Clop is right adjoint[b] to Ult

$$\begin{tikzcd} \mathsf{BA} \arrow[r, shift left, "\mathrm{Ult}"] \arrow[r, phantom, "\perp"] & \mathsf{Stone}^{\mathrm{op}}, \arrow[l, shift left, "\mathrm{Clop}"] \end{tikzcd}$$

with unit $(\eta_B)_B$ and counit $(\varepsilon_X)_X$ (viewed in the opposite category); the triangle identities that one needs to verify are the following ones (for $B \in \mathsf{BA}$ and $X \in \mathsf{Stone}$, with the diagram on the left-hand side in Stone, and the diagram on the right-hand side in BA):

$$\begin{tikzcd} \mathrm{Ult}(B) \arrow[r, "\epsilon_{\mathrm{Ult}(B)}"] \arrow[dr, "\mathrm{id}_{\mathrm{Ult}(B)}"'] & \mathrm{Ult}(\mathrm{Clop}(\mathrm{Ult}(B))) \arrow[d, "\mathrm{Ult}(\eta_B)"] \\ & \mathrm{Ult}(B), \end{tikzcd} \qquad \begin{tikzcd} \mathrm{Clop}(X) \arrow[r, "\eta_{\mathrm{Clop}(X)}"] \arrow[dr, "\mathrm{id}_{\mathrm{Clop}(X)}"'] & \mathrm{Clop}(\mathrm{Ult}(\mathrm{Clop}(X))) \arrow[d, "\mathrm{Clop}(\epsilon_X)"] \\ & \mathrm{Clop}(X). \end{tikzcd}$$

The verification of the commutativity is a tedious unwinding of the definitions, done in the footnote[c]. ■

[a] For the η-square, let $a \in A$. We compute both composites as clopen subsets of $\mathrm{Ult}(B)$. For the η-square, let $a \in A$. We compute both composites as clopen subsets of $\mathrm{Ult}(B)$. First,

$$(\mathrm{Clop}(\mathrm{Ult}(h)) \circ \eta_A)(a) = \mathrm{Clop}(\mathrm{Ult}(h))\big(\eta_A(a)\big) = \mathrm{Ult}(h)^{-1}\big[\eta_A(a)\big],$$

since $\mathrm{Clop}(\mathrm{Ult}(h))$ is defined as inverse image along $\mathrm{Ult}(h) \colon \mathrm{Ult}(B) \to \mathrm{Ult}(A)$. Now let $U \in \mathrm{Ult}(B)$. Then

$$\begin{aligned} U \in \mathrm{Ult}(h)^{-1}\big[\eta_A(a)\big] &\iff \mathrm{Ult}(h)(U) \in \eta_A(a) \\ &\iff h^{-1}[U] \in \eta_A(a) \\ &\iff a \in h^{-1}[U] \\ &\iff h(a) \in U \\ &\iff U \in \eta_B(h(a)). \end{aligned}$$

Therefore $\mathrm{Ult}(h)^{-1}[\eta_A(a)] = \eta_B(h(a))$, i.e.

$$\mathrm{Clop}(\mathrm{Ult}(h))(\eta_A(a)) = \eta_B(h(a)),$$

which is exactly $(\mathrm{Clop}(\mathrm{Ult}(h)) \circ \eta_A)(a) = (\eta_B \circ h)(a)$.

For the ε-square, let $x \in X$. We compute both composites as ultrafilters on $\mathrm{Clop}(Y)$. We have

$$(\mathrm{Ult}(\mathrm{Clop}(f)) \circ \varepsilon_X)(x) = \mathrm{Ult}(\mathrm{Clop}(f))\big(\varepsilon_X(x)\big) = \mathrm{Clop}(f)^{-1}\big[\varepsilon_X(x)\big],$$

since $\mathrm{Ult}(\mathrm{Clop}(f))$ is defined by preimage along $\mathrm{Clop}(f)\colon \mathrm{Clop}(Y) \to \mathrm{Clop}(X)$. Now let $C \in \mathrm{Clop}(Y)$. Then

$$\begin{aligned} C \in \mathrm{Clop}(f)^{-1}\big[\varepsilon_X(x)\big] &\iff \mathrm{Clop}(f)(C) \in \varepsilon_X(x) \\ &\iff f^{-1}[C] \in \varepsilon_X(x) \\ &\iff x \in f^{-1}[C] \\ &\iff f(x) \in C \\ &\iff C \in \varepsilon_Y(f(x)). \end{aligned}$$

Therefore $\mathrm{Clop}(f)^{-1}[\varepsilon_X(x)] = \varepsilon_Y(f(x))$, i.e.

$$\mathrm{Ult}(\mathrm{Clop}(f))(\varepsilon_X(x)) = \varepsilon_Y(f(x)),$$

which is exactly $(\mathrm{Ult}(\mathrm{Clop}(f)) \circ \varepsilon_X)(x) = (\varepsilon_Y \circ f)(x)$.

[b]This is just because we put the "op" on Stone rather than on BA; otherwise it would have been Ult to be right adjoint to Clop.

[c]For the left triangle (in Stone), let $U \in \mathrm{Ult}(B)$. Recall that

$$\varepsilon_{\mathrm{Ult}(B)}(U) = \{\, C \in \mathrm{Clop}(\mathrm{Ult}(B)) \mid U \in C \,\} \in \mathrm{Ult}(\mathrm{Clop}(\mathrm{Ult}(B))),$$

and that $\mathrm{Ult}(\eta_B)\colon \mathrm{Ult}(\mathrm{Clop}(\mathrm{Ult}(B))) \to \mathrm{Ult}(B)$ is given by preimage:

$$\mathrm{Ult}(\eta_B)(V) = \eta_B^{-1}[V] \qquad (V \in \mathrm{Ult}(\mathrm{Clop}(\mathrm{Ult}(B)))).$$

We show $\mathrm{Ult}(\eta_B)(\varepsilon_{\mathrm{Ult}(B)}(U)) = U$ by element-chasing. For $b \in B$,

$$\begin{aligned} b \in \mathrm{Ult}(\eta_B)\big(\varepsilon_{\mathrm{Ult}(B)}(U)\big) &\iff b \in \eta_B^{-1}\big[\varepsilon_{\mathrm{Ult}(B)}(U)\big] \\ &\iff \eta_B(b) \in \varepsilon_{\mathrm{Ult}(B)}(U) \\ &\iff U \in \eta_B(b) \\ &\iff b \in U, \end{aligned}$$

hence $\mathrm{Ult}(\eta_B) \circ \varepsilon_{\mathrm{Ult}(B)} = \mathrm{id}_{\mathrm{Ult}(B)}$.

For the right triangle (in BA), let $C \in \mathrm{Clop}(X)$. Recall that

$$\eta_{\mathrm{Clop}(X)}(C) = \{\mathcal{U} \in \mathrm{Ult}(\mathrm{Clop}(X)) \mid C \in \mathcal{U}\,\} \in \mathrm{Clop}(\mathrm{Ult}(\mathrm{Clop}(X))),$$

and that $\mathrm{Clop}(\varepsilon_X)\colon \mathrm{Clop}(\mathrm{Ult}(\mathrm{Clop}(X))) \to \mathrm{Clop}(X)$ is inverse image along ε_X:

$$\mathrm{Clop}(\varepsilon_X)(D) = \varepsilon_X^{-1}[D] \qquad (D \in \mathrm{Clop}(\mathrm{Ult}(\mathrm{Clop}(X)))).$$

Then for $x \in X$,

$$\begin{aligned} x \in \mathrm{Clop}(\varepsilon_X)\big(\eta_{\mathrm{Clop}(X)}(C)\big) &\iff \varepsilon_X(x) \in \eta_{\mathrm{Clop}(X)}(C) \\ &\iff C \in \varepsilon_X(x) \\ &\iff x \in C, \end{aligned}$$

so $\mathrm{Clop}(\varepsilon_X) \circ \eta_{\mathrm{Clop}(X)} = \mathrm{id}_{\mathrm{Clop}(X)}$.

1.7 Using Stone duality

In this section, we collect some simple examples of the use of Stone duality.

1.7.1 Spaces of models and the compactness theorem

Remark 1.7.1. In Proposition 1.3.2 we stated that, for every propositional theory $\mathcal{T}$ in a propositional language $\mathcal{L}$, there is a bijection

$$\mathrm{Mod}(\mathcal{T}) \longrightarrow \hom(\mathrm{Form}(\mathcal{L})/{\equiv_{\mathcal{T}}}, 2).$$

Composing this bijection with the bijection

$$\hom(\mathrm{Form}(\mathcal{L})/{\equiv_{\mathcal{T}}}, 2) \longrightarrow \mathrm{Ult}(\mathrm{Form}(\mathcal{L})/{\equiv_{\mathcal{T}}}),$$

we obtain the following bijection:

$$\begin{aligned} \mathrm{Mod}(\mathcal{T}) &\longrightarrow \mathrm{Ult}(\mathrm{Form}(\mathcal{L})/{\equiv_{\mathcal{T}}}) \\ v &\longmapsto \{\, [\varphi] \in \mathrm{Form}(\mathcal{L})/{\equiv_{\mathcal{T}}} \mid v \models \varphi \,\}. \end{aligned} \tag{1.4}$$

Via this bijection, $\mathrm{Mod}(\mathcal{T})$ inherits a Stone topology from $\mathrm{Ult}(\mathrm{Form}(\mathcal{L})/{\equiv_{\mathcal{T}}})$. This can be defined directly as follows.

Definition 1.7.2 (The space of models)**.** Let $\mathcal{T}$ be a propositional theory in a propositional language $\mathcal{L}$. We equip $\mathrm{Mod}(\mathcal{T})$ with the topology that has as a basis the sets of the form

$$[\![\varphi]\!]_{\mathcal{T}} \;:=\; \{\, v \in \mathrm{Mod}(\mathcal{T}) \mid v \models \varphi \,\}, \qquad\qquad \varphi \in \mathrm{Form}(\mathcal{L}).$$

We now characterize the clopens.

Proposition 1.7.3 (The space of models is a Stone space)**.** $\mathrm{Mod}(\mathcal{T})$ *is a Stone space, whose clopens are precisely the sets of the form*

$$[\![\varphi]\!]_{\mathcal{T}} \;=\; \{\, v \in \mathrm{Mod}(\mathcal{T}) \mid v \models \varphi \,\}, \qquad\qquad \varphi \in \mathrm{Form}(\mathcal{L}).$$

Proof. It is enough to observe that, under the bijection (1.4) in Remark 1.7.1, the subset $[\![\varphi]\!]$ corresponds to the basic open set $\eta_{\mathrm{Form}(\mathcal{L})/\equiv_{\mathcal{T}}}([\varphi])$ of $\mathrm{Ult}(\mathrm{Form}(\mathcal{L})/{\equiv_{\mathcal{T}}})$, and that $\mathrm{Ult}(\mathrm{Form}(\mathcal{L})/{\equiv_{\mathcal{T}}})$ is a Stone space with clopens exactly the sets of the form $\eta_{\mathrm{Form}(\mathcal{L})/\equiv_{\mathcal{T}}}([\varphi])$ for $\varphi \in \mathrm{Form}(\mathcal{L})$. ■

Proposition 1.7.4 (No propositional theory captures "exactly one among countably many")**.** *Let* $\mathcal{L} = \{p_0, p_1, p_2, \dots\}$. *There is no propositional theory* $\mathcal{T} \subseteq \mathrm{Form}(\mathcal{L})$ *such that* $\mathrm{Mod}(\mathcal{T})$ *consists exactly of those valuations* $v\colon \mathcal{L} \to 2$ *for which there is* exactly one $n \in \mathbb{N}$ *with* $v(p_n) = 1$.

Proof. Suppose, for a contradiction, that such a theory $\mathcal{T}$ exists. For each $n \in \mathbb{N}$ consider the set

$$U_n \;:=\; [\![p_n]\!]_{\mathcal{T}} \;=\; \{\, v \in \mathrm{Mod}(\mathcal{T}) \mid v \models p_n \,\}.$$

Each U_n is clopen in $\mathrm{Mod}(\mathcal{T})$ by Proposition 1.7.3, and by assumption we have a cover

$$\mathrm{Mod}(\mathcal{T}) = \bigcup_{n \in \mathbb{N}} U_n.$$

Since $\mathrm{Mod}(\mathcal{T})$ is compact, there is a finite $I \subseteq \mathbb{N}$ such that $\mathrm{Mod}(\mathcal{T}) = \bigcup_{n \in I} U_n$.

Choose $m \in \mathbb{N} \setminus I$, and define the valuation $v_m\colon \mathcal{L} \to 2$ by $v_m(p_m) = 1$ and $v_m(p_n) = 0$ for $n \neq m$. By the defining property of $\mathcal{T}$, the valuation v_m belongs to $\mathrm{Mod}(\mathcal{T})$. But $v_m \notin U_n$ for every $n \in I$ (since $v_m(p_n) = 0$), contradicting $\mathrm{Mod}(\mathcal{T}) = \bigcup_{n \in I} U_n$. ■

Definition 1.7.5 (Semantic consequence: "theory $\models$ formula")**.** Let $\mathcal{T}$ be a propositional theory in a propositional language $\mathcal{L}$, and let $\varphi \in \mathrm{Form}(\mathcal{L})$. We write

$$\mathcal{T} \models \varphi$$

and say that φ ***is a semantic consequence of*** $\mathcal{T}$ if every model of $\mathcal{T}$ satisfies φ.

(Yes, we use the same symbol used for the relation of satisfiability between valuations and formulas.)

The compactness of the space of ultrafilters has the following logical flavour.[14]

[14] For a textbook reference for the compactness theorem of classical propositional logic, see, e.g., [Enderton, 2001, Sec. 1.7].

Theorem 1.7.6 (Compactness theorem of classical propositional logic)**.** *Let $\mathcal{L}$ be a propositional language, let $\mathcal{T} \subseteq \mathrm{Form}(\mathcal{L})$ be a propositional theory, and let $\varphi \in \mathrm{Form}(\mathcal{L})$. If $\mathcal{T} \models \varphi$, then there is a finite $\mathcal{T}_0 \subseteq \mathcal{T}$ such that $\mathcal{T}_0 \models \varphi$.*

Proof. Consider the Stone space $\mathrm{Mod}(\varnothing)$ (the space of all valuations on $\mathcal{L}$). For each formula $\psi \in \mathrm{Form}(\mathcal{L})$ set

$$[\![\psi]\!] \coloneqq \{\, v \in \mathrm{Mod}(\varnothing) \mid v \models \psi \,\}.$$

By Proposition 1.7.3 (with $\mathcal{T} = \varnothing$), each $[\![\psi]\!]$ is clopen, and hence closed.

The assumption says exactly that

$$\bigcap_{\sigma \in \mathcal{T}} [\![\sigma]\!] \subseteq [\![\varphi]\!].$$

From the definition of compactness, it follows immediately that, if the intersection of a family of closed subsets is contained in an open set, then there is a *finite* subfamily whose intersection is contained in the open set. Since every $[\![\sigma]\!]$ (for $\sigma \in \mathcal{T}$) is clopen and hence closed, and $[\![\varphi]\!]$ is clopen and hence open, there are $\sigma_1, \ldots, \sigma_n \in \mathcal{T}$ such that

$$[\![\sigma_1]\!] \cap \cdots \cap [\![\sigma_n]\!] \subseteq [\![\varphi]\!].$$

Let $\mathcal{T}_0 \coloneqq \{\sigma_1, \ldots, \sigma_n\}$. This is exactly the desired conclusion. ■

1.7.2 Question: are coproduct maps injective?

Question (Guiding question)**.** *Let A and B be Boolean algebras. Is the coproduct morphism*

$$i_A \colon A \longrightarrow A + B$$

injective?

In logical terms, suppose A and B are Lindenbaum–Tarski algebras of theories $\mathcal{T}_A$ and $\mathcal{T}_B$ in disjoint languages $\mathcal{L}_A$ and $\mathcal{L}_B$. One can easily show that the injectivity of a Boolean homomorphism between Boolean algebras is equivalent to the condition that a non-1 element is mapped to a non-1 element. The question then translates to the following:

is it true that, for every formula φ in the language $\mathcal{L}_A$,
if φ does not follow from $\mathcal{T}_A$ (i.e., $\mathcal{T}_A \not\models \varphi$),
then φ does not follow from $\mathcal{T}_A \sqcup \mathcal{T}_B$ (i.e., $\mathcal{T}_A \sqcup \mathcal{T}_B \not\models \varphi$)?

We answer this by translating the question to Stone via Stone duality. To do so, we make a brief detour in which

- we show that binary products of Stone spaces (and more generally, all limits) are easy: they are computed as in Set,
- we show that the injectivity (resp. surjectivity) of a Boolean homomorphism is equivalent to the surjectivity (resp. injectivity) of the dual continuous map.

1.7.3 Limits and colimits in BA and Stone

Limits and colimits in BA

The category of Boolean algebras is complete and cocomplete, as any equationally defined class of algebras[15]. However, there is a fundamental difference in the behaviour of limits and colimits:

[15] See, e.g., [Bergman, 2015, Prop. 9.4.8] for limits, and [Bergman, 2015, Thm. 9.4.14.] for colimits.

- (Limits ✓) Limits of Boolean algebras are great: they are computed as in Set, i.e., the forgetful functor $\mathsf{BA} \to \mathsf{Set}$ preserves limits (as the forgetful functor of any equationally defined class of algebras[16]).
- (Colimits ✗) However, colimits are crazy, meaning that they are not at all computed as in Set. For example:
  - the initial object of BA is 2 (while in Set the initial object is the empty set).
  - In BA, $2 + 2 = 2$ (since 2 is the initial object), while in Set the coproduct of a 2-element set with itself has 4 elements.

Limits and colimits in Stone

Since Stone is dually equivalent to BA, which is complete and cocomplete since BA is an equationally defined class of algebras, Stone is also complete and cocomplete.

In Stone, limits are nice (= computed as in Set):

Lemma 1.7.7 (Closed in Stone is Stone)**.** *Every closed subspace of a Stone space is a Stone space (with the subspace topology).*

Proof. Let X be a Stone space and let $F \subseteq X$ be closed. Since F is closed in the compact space X, F is compact, as well. To see that F is totally separated, let $x \neq y$ in F. Since X is totally separated, there is $C \in \mathrm{Clop}(X)$ such that $x \in C$ and $y \notin C$. Then $C \cap F$ is clopen in F and separates x and y. ■

Proposition 1.7.8 (Limits in Stone)**.** *The category* Stone *is complete and the forgetful functor*

$$\mathsf{Stone} \to \mathsf{Set}$$

preserves limits.
In fact, limits in Stone *are computed as in* Top*:*

1. *(Products) the product in* Stone *of a family* $(X_i)_{i \in I}$ *of objects of* Stone *is the Cartesian product* $\prod_{i \in I} X_i$ *with the product topology (and the projection as limit morphisms);*
2. *(Equalizer) the equalizer in* Stone *of a pair of morphisms* $f, g \colon X \to Y$ *in* Stone *is the space*

$$\{\, x \in X \mid f(x) = g(x) \,\},$$

with the subspace topology (and the inclusion as limit morphism).

Proof. It is enough to prove that Stone is closed in Top under limits. (We recall that the forgetful functor $\mathsf{Top} \to \mathsf{Set}$ preserves limits.)

(1). We prove that the topological product $\prod_{i \in I} X_i$ of Stone spaces is a Stone space.

- (Compactness) We recall that Tychonoff's theorem[a] states that an arbitrary product of compact spaces is compact. Therefore, an arbitrary product of Stone spaces is compact.[b]
- (Total separation) Let us prove that $\prod_{i \in I} X_i$ is totally separated. Let $(x_j)_j$ and $(y_j)_j$ be distinct elements. Then, they differ at some component, i.e., there is $i \in I$ such that $x_i \neq y_i$. Then, there is a clopen C of X_i such that $x_i \in C$ and $y_i \notin C$. Then, letting $\pi_i \colon \prod_j X_j \to X_i$ denote the i-th projection, the set $\pi_i^{-1}[C]$ is a clopen subset of $\prod_j X_j$ in the product topology, and $(x_j)_j \in \pi_i^{-1}[C]$ and $(y_j)_j \notin \pi_i^{-1}[C]$. Thus, $\prod_{i \in I} X_i$ is totally separated.

[16] See, e.g., [Bergman, 2015, Prop. 9.4.8].

(2). We prove that $\{ x \in X \mid f(x) = g(x) \}$ is a Stone space. Consider the map $\langle f, g \rangle \colon X \to Y \times Y$, $\langle f, g \rangle(x) = (f(x), g(x))$. Then $\mathsf{Eq}(f, g) = \langle f, g \rangle^{-1}[\Delta_Y]$, where $\Delta_Y = \{(y, y) \mid y \in Y\}$ is the diagonal. Since Y is Hausdorff, Δ_Y is closed in $Y \times Y$, and hence $\{ x \in X \mid f(x) = g(x) \}$ is closed in X. Since every closed subspace of a Stone space is a Stone space (Lemma 1.7.7), $\{ x \in X \mid f(x) = g(x) \}$ is a Stone space, as desired. ■

[a]See, e.g., [Munkres, 2000, Thm. 37.3].

[b]We recall that Tychonoff's theorem, stated in its full generality (for arbitrary compact spaces), is equivalent to the Axiom of Choice.

Remark 1.7.9 (Colimits in Stone are not as in Set)**.** While Stone is cocomplete (since BA is complete), colimits are, in general, not computed as in Set. A simple example of a colimit that is *not* preserved by the forgetful functor $\mathsf{Stone} \to \mathsf{Set}$ is the coproduct of countably many copies of a singleton. A proof is in the footnote; the idea is that one would like to equip $\mathbb{N}$ with the discrete topology, which, however, is not compact.[17] By Stone duality, the coproduct $\sum_{n \in \mathbb{N}} 1$ in Stone exists and is the Stone space $\mathrm{Ult}(\prod_{n \in \mathbb{N}} \mathrm{Clop}(1)) = \mathrm{Ult}(\prod_{n \in \mathbb{N}} 2) = \mathrm{Ult}(\mathcal{P}(X))$.

In general, infinite coproducts, coequalizers and pushouts are not preserved by the forgetful functor to Set. See the footnote for the case of coequalizers and pushouts.[18]

Finite coproducts, instead, *are* computed as in Set, and in fact as in Top: a finite coproduct of Stone

[17]We prove that the forgetful functor $U \colon \mathsf{Stone} \to \mathsf{Set}$ does *not* preserve the coproduct of countably many copies of the one-point Stone space.

Suppose it does, and let us reach a contradiction. Let $(\iota_n \colon 1_n \to C)_{n \in \mathbb{N}}$ be a coproduct cocone in Stone. Then $(U(\iota_n))_{n \in \mathbb{N}}$ is a coproduct cocone in Set, hence it induces a canonical bijection

$$\mathbb{N} \cong U(C),$$

under which $U(\iota_n)$ picks out the element $n \in U(C)$. Transporting the topology of C along this bijection, we may assume $C = (\mathbb{N}, \tau)$ for some topology τ.

We claim that τ is discrete. Let $A \subseteq \mathbb{N}$ be arbitrary, and consider the discrete two-point Stone space $2 = \{0, 1\}$. Define, for each $n \in \mathbb{N}$, a (necessarily continuous) map

$$G_n \colon 1_n \to 2, \qquad G_n(*) = \begin{cases} 1 & \text{if } n \in A, \\ 0 & \text{if } n \notin A. \end{cases}$$

By the universal property of the coproduct $C = \sum_{n \in \mathbb{N}} 1_n$ in Stone, there is a unique continuous map $f_A \colon (\mathbb{N}, \tau) \to 2$ such that $f_A \circ \iota_n = G_n$ for all $n \in \mathbb{N}$. Since $U(\iota_n)$ picks out $n \in \mathbb{N}$, this forces $f_A(n) = G_n(*)$ for all n, hence $f_A^{-1}[\{1\}] = A$. As $\{1\}$ is clopen in 2 and f_A is continuous, it follows that A is clopen in $(\mathbb{N}, \tau)$. Since A was arbitrary, every subset of $\mathbb{N}$ is clopen, so τ is discrete. But an infinite discrete space is not compact, whereas every Stone space is compact, a contradiction.

[18]Let $U \colon \mathsf{Stone} \to \mathsf{Set}$ be the forgetful functor. We sketch a proof of the fact that U does *not* preserve coequalizers. (The analogous statement for pushouts follows, since U preserves the initial object.)

Let $f \colon B \to [0, 1]$ be a continuous surjection, where B is a Stone space and $[0, 1]$ carries the usual Euclidean topology. (Such an f exists: for instance, take $B = \beta([0, 1]_{\mathrm{disc}}) = \mathrm{Ult}(\mathcal{P}([0, 1]))$ and extend the identity map $[0, 1]_{\mathrm{disc}} \to [0, 1]$ along the Stone–Čech compactification; by the universal property of β, the extension exists and is surjective.)

Consider the *kernel pair* of f in Top (equivalently in $\mathsf{CompHaus}$): $R := \{(x, y) \in B \times B \mid f(x) = f(y)\}$. Since $[0, 1]$ is Hausdorff, the diagonal $\Delta \subseteq [0, 1] \times [0, 1]$ is closed, and $R = (f \times f)^{-1}[\Delta]$ is a closed subset of $B \times B$. As B is Stone, so is $B \times B$, and every closed subspace of a Stone space is Stone. Hence, R is a Stone space. Let $r_1, r_2 \colon R \rightrightarrows B$ be the two projections.

Claim. The coequalizer of r_1, r_2 in Stone is the one-point Stone space 1.

Indeed, the constant map $q \colon B \to 1$ clearly coequalizes r_1, r_2. Now let $g \colon B \to X$ be a continuous map into a Stone space such that $g \circ r_1 = g \circ r_2$. Then $g(x) = g(y)$ whenever $f(x) = f(y)$, i.e. g is constant on each fiber of f. Define

$$h \colon [0, 1] \to X, \qquad h(t) := g(x) \ \text{ for any } x \in f^{-1}(t).$$

This is well-defined by the fiber-constancy of g, and it satisfies $g = h \circ f$.

Because B is compact and $[0, 1]$ is Hausdorff, the map f is a quotient map; hence from the continuity of $g = h \circ f$ it follows that h is continuous. Since $[0, 1]$ is connected, the image $h([0, 1])$ is connected. But every connected subset of a Stone space is a singleton (Stone spaces are totally disconnected). Therefore, h is constant, and so is $g = h \circ f$. Consequently, g factors uniquely through $q \colon B \to 1$. This proves that $\mathrm{coeq}_{\mathsf{Stone}}(r_1, r_2) \cong 1$.

Conclusion. Applying U, we have $U(\mathrm{coeq}_{\mathsf{Stone}}(r_1, r_2)) = U(1)$, a singleton. On the other hand, in Set the coequalizer of $U(r_1)$ and $U(r_2)$ is the quotient of the set $U(B)$ by the equivalence relation generated by $x \sim y \iff f(x) = f(y)$; since f is surjective, this quotient is (canonically) in bijection with $[0, 1]$, hence it is not a singleton. Therefore, U does not preserve this coequalizer, and thus the forgetful functor $\mathsf{Stone} \to \mathsf{Set}$ does not preserve coequalizers.

spaces is the disjoint union equipped with the disjoint union topology.

In summary:

	BA	Stone
Has all limits / colimits?	✓/ ✓	✓/ ✓
Are limits computed as in Set?	✓	✓
Are colimits computed as in Set?	✗	✗ (exception: *finite coproducts* are as in Set)

1.7.4 Injective is dual to surjective, and vice versa

In Stone spaces, open sets are approximated from below by clopens, in the following sense.

Lemma 1.7.10 (Open sets are unions of clopens)**.** *Every open set in a Stone space is the union of the clopens contained in it.*

Proof. Let X be a Stone space, U an open of X, and $x \in U$. We shall prove that there is a clopen C such that $x \in C \subseteq U$. Set $F \coloneqq X \setminus U$, which is closed. For each $y \in F$ we have $y \neq x$, so by total separation there is a clopen $C_y \subseteq X$ with

$$x \in C_y \quad \text{and} \quad y \notin C_y.$$

Equivalently, $y \in X \setminus C_y$, and the latter is open. Thus, the family $\{ X \setminus C_y \mid y \in F \}$ is an open cover of F.

Since F is a closed subspace of the compact space X, it is compact. Hence, there are $y_1, \dots, y_n \in F$ such that

$$F \subseteq (X \setminus C_{y_1}) \cup \dots \cup (X \setminus C_{y_n}).$$

Let

$$C \coloneqq C_{y_1} \cap \dots \cap C_{y_n}.$$

Then C is clopen, $x \in C$, and moreover $C \cap F = \varnothing$ (because F is contained in the union of the complements). Therefore, $C \subseteq X \setminus F = U$, as desired. ■

Lemma 1.7.10 says that every Stone space has a basis of clopens, i.e., it is 0-***dimensional***. In fact, Stone spaces can be equivalently defined as compact Hausdorff 0-dimensional spaces[19].

In Stone spaces, just like *open* sets are approximated from *below* by clopen sets (Lemma 1.7.10), *closed* sets are approximated from *above* by clopen sets.

Lemma 1.7.11 (Closed sets are intersections of clopens)**.** *Every closed set in a Stone space is the intersection of the clopens in which it is contained.*

Proof. This follows from Lemma 1.7.10 by taking complements. ■

Lemma 1.7.12 (Clopens separate disjoint closed sets)**.** *Let X be a Stone space and let $A, B \subseteq X$ be closed and disjoint. There is a clopen C such that $A \subseteq C$ and $B \cap C = \varnothing$.*

Proof. For each $a \in A$, the set $X \setminus B$ is an open neighbourhood of a. Since X has a basis of clopens (Lemma 1.7.10), there is $C_a \in \mathrm{Clop}(X)$ with $a \in C_a \subseteq X \setminus B$. Then $(C_a)_{a \in A}$ is an open cover of A. As A is closed in the compact space X, it is compact. Therefore, there are $a_1, \dots, a_n \in A$ such that $A \subseteq C_{a_1} \cup \dots \cup C_{a_n}$. Set $C \coloneqq C_{a_1} \cup \dots \cup C_{a_n}$. This is clopen, contains A, and is disjoint from B. ■

[19] Yet another alternative equivalent definition is: compact Hausdorff totally-disconnected spaces.

Theorem 1.7.13 (Injective is dual to surjective, and vice versa)**.** *Let $f\colon X \to Y$ be a continuous map of Stone spaces. Then:*

1. *f is injective iff $f^{-1}[-]\colon \mathrm{Clop}(Y) \to \mathrm{Clop}(X)$ is surjective;*
2. *f is surjective iff $f^{-1}[-]\colon \mathrm{Clop}(Y) \to \mathrm{Clop}(X)$ is injective.*

Proof. (1). ($\Rightarrow$). Suppose f is injective and let $A \in \mathrm{Clop}(X)$. Since X is compact and Y is Hausdorff, f is a homeomorphism onto the compact subset $f[X] \subseteq Y$, which is therefore closed. The sets $f[A]$ and $f[X \setminus A]$ are compact, hence closed in Y, and they are disjoint. By Lemma 1.7.12, there is $C \in \mathrm{Clop}(Y)$ such that $f[A] \subseteq C$ and $f[X \setminus A] \cap C = \varnothing$. Then $f^{-1}[C] = A$, so $f^{-1}[-]$ is surjective.

($\Leftarrow$). Conversely, suppose f is not injective and let us prove that $f^{-1}[-]$ is not surjective. Pick $x \neq y$ in X with $f(x) = f(y)$. Since X is totally separated, choose $A \in \mathrm{Clop}(X)$ with $x \in A$ and $y \notin A$. There is no clopen C of Y such that $f^{-1}[C] = A$, because for every clopen C of Y we have $x \in f^{-1}[C] \Leftrightarrow y \in f^{-1}[C]$; therefore, $f^{-1}[-]$ is not surjective. Hence, (2). ($\Rightarrow$). Suppose that f is surjective. To prove injectivity of $f^{-1}[-]$, suppose $f^{-1}[C] = f^{-1}[D]$ for clopens $C, D \subseteq Y$, and let us prove that $C = D$. Let $y \in Y$ and let us prove that $y \in C \iff y \in D$. By surjectivity, there is x with $f(x) = y$. Thus, $y \in C \Leftrightarrow x \in f^{-1}[C] = f^{-1}[D] \Leftrightarrow y \in D$. Hence, $C = D$, so $f^{-1}[-]$ is injective.

($\Leftarrow$). Conversely, suppose that f is not surjective, i.e., there is $y_0 \in Y \setminus f[X]$. Since $f[X]$ and $\{y_0\}$ are disjoint closed sets, by Lemma 1.7.12 there is $C \in \mathrm{Clop}(Y)$ such that $f[X] \subseteq C$ and $y_0 \notin C$. Then $f^{-1}[C] = X = f^{-1}[Y]$ but $C \neq Y$, so $f^{-1}[-]$ is not injective. ■

1.7.5 Answer to the question "are coproduct maps injective?"

We are now ready to answer our guiding question:

Question 1.7.14. (Guiding question) Let A and B be Boolean algebras. Is the coproduct morphism

$$i_A\colon A \longrightarrow A + B$$

injective?

Answer 1.7.15 (Answer to Question 1.7.14)**.** The coproduct morphism

$$i_A\colon A \longrightarrow A + B$$

is injective if and only if either A is trivial (i.e., the one-element Boolean algebra) or B is nontrivial. Equivalently, i_A fails to be injective if and only if A is nontrivial and B is trivial.

Proof. Let

$$X := \mathrm{Ult}(A) \qquad \text{and} \qquad Y := \mathrm{Ult}(B).$$

Under Stone duality, coproducts in BA correspond to products in Stone, and the morphism

$$i_A\colon A \longrightarrow A + B$$

is dual to the first projection

$$\pi_X\colon X \times Y \longrightarrow X.$$

By Theorem 1.7.13, i_A is injective if and only if π_X is surjective.

Now, π_X is surjective if and only if either $Y \neq \varnothing$ or $X = \varnothing$. Indeed, if $Y \neq \varnothing$, choose $y_0 \in Y$. Then for every $x \in X$ we have $\pi_X(x, y_0) = x$, so π_X is surjective. On the other hand, if $Y = \varnothing$, then $X \times Y = \varnothing$, so π_X is the unique map

$$\varnothing \longrightarrow X,$$

which is surjective if and only if $X = \varnothing$.

Finally, $X = \varnothing$ if and only if A is trivial, and $Y = \varnothing$ if and only if B is trivial. Therefore, i_A is injective if and only if either A is trivial or B is nontrivial. ■

In logical terms, suppose A and B are Lindenbaum–Tarski algebras of theories $\mathcal{T}_A$ and $\mathcal{T}_B$ in disjoint languages $\mathcal{L}_A$ and $\mathcal{L}_B$. One can easily show that the injectivity of a Boolean homomorphism between Boolean algebras is equivalent to the condition that a non-1 element is mapped to a non-1 element. The question then translates to the following:

is it true that, for every formula φ in the language $\mathcal{L}_A$,
if φ does not follow from $\mathcal{T}_A$ (i.e., $\mathcal{T}_A \not\models \varphi$),
then φ does not follow from $\mathcal{T}_A \sqcup \mathcal{T}_B$ (i.e., $\mathcal{T}_A \sqcup \mathcal{T}_B \not\models \varphi$)?

Answer 1.7.15 tells us that this is always true, except when $\mathcal{T}_B$ is inconsistent (i.e., $\mathcal{T}_B \models 0$) and $\mathcal{T}_A$ is not (i.e., $\mathcal{T}_A \not\models 0$). Intuitively, if $\mathcal{T}_B$ is inconsistent and $\mathcal{T}_A$ is not, then the theory $\mathcal{T}_A \sqcup \mathcal{T}_B$ proves the falsum 0. Since 0 is also a formula of $\mathcal{L}_A$, this gives a formula of the language $\mathcal{L}_A$ that was not provable from $\mathcal{T}_A$ but becomes provable after adding $\mathcal{T}_B$.

1.7.6 Monomorphisms and epimorphisms in BA and Stone

We now summarize how the categorical notions of monomorphism, epimorphism, regular monomorphism and regular epimorphism are characterized in BA and in Stone. To prove some of these characterizations, we use Stone duality: we characterize monos and regular epis in each category, and then use Stone duality to characterize epis and regular monos on the dual category.

Theorem 1.7.16 (Monos, regular monos, epis and regular epis in BA and Stone)**.**

1. *In* BA*, monomorphisms and regular monomorphisms are precisely the injective homomorphisms.*
2. *In* BA*, epimorphisms and regular epimorphisms are precisely the surjective homomorphisms.*
3. *In* Stone*, monomorphisms and regular monomorphisms are precisely the injective continuous maps.*
4. *In* Stone*, epimorphisms and regular epimorphisms are precisely the surjective continuous maps.*

Proof. (Monos and regular epis in BA, epis and regular monos in Stone:) Recall that BA is a variety of algebras, i.e., an equationally defined class of algebras. In any variety of algebras, the monomorphisms are precisely the injective homomorphisms, and the regular epimorphisms are precisely the surjective homomorphisms. Therefore, the monomorphisms in BA are precisely the injective homomorphisms, and the regular epimorphisms in BA are precisely the surjective homomorphisms. By Stone duality and Theorem 1.7.13, we immediately obtain that, in Stone, the epimorphisms are precisely the surjective continuous maps, and the regular monomorphisms are precisely the injective continuous maps.

(Epis in BA, monos in Stone:) Proving directly that epimorphisms in BA are surjective is not immediate. To simplify the argument, we use Stone duality. Let us prove that the monomorphisms in Stone are precisely the injective continuous maps.

- First, the easiest direction: every injective morphism in Stone is a monomorphism, since this is true in Set and faithful functors reflect monomorphisms.
- For the converse direction, let $f\colon X \to Y$ be a continuous map that is not injective, and let us prove that it is not a monomorphism. Pick $x_1 \neq x_2$ in X such that $f(x_1) = f(x_2)$. Let 1 be the one-point space, which is a Stone space, and define

$$g_1, g_2 \colon 1 \longrightarrow X$$

to be the constant maps with values x_1 and x_2, respectively. Then $g_1 \neq g_2$, but

$$f \circ g_1 = f \circ g_2,$$

so f is not a monomorphism.

Therefore, monomorphisms in Stone are precisely the injective continuous maps. By Stone duality and Theorem 1.7.13 (injective is dual to surjective, and vice versa), it follows that the epimorphisms in BA are precisely the surjective homomorphisms.

(Regular monos in BA, regular epis in Stone:) Also proving that injective homomorphisms in BA are regular monomorphisms is not immediate. The obvious idea would be to take the equalizer of the *cokernel pair* of one such injective morphism, but the cokernel pair in BA is not computed as in Set, and is thus difficult to manage. So, we use Stone duality: we characterize regular epimorphisms in Stone as precisely the surjective continuous maps.

- First, the easiest direction: since every regular epimorphism is an epimorphism, and epimorphisms in Stone are surjective, it follows that regular epimorphisms in Stone are surjective continuous maps.
- Let us prove the converse implication: let $f\colon X \to Y$ be a surjective continuous map of Stone spaces, and let us prove that it is a regular epimorphism. Consider its kernel pair

  $$R \coloneqq \{(x_1, x_2) \in X \times X \mid f(x_1) = f(x_2)\},$$

  with the two projections

  $$r_1, r_2\colon R \rightrightarrows X.$$

  Since Y is Hausdorff, the diagonal Δ_Y of $Y \times Y$ is closed, and hence the set $R = (f \times f)^{-1}[\Delta_Y]$ is a closed subspace of $X \times X$. As $X \times X$ is a Stone space, so is R.

  Now let $g\colon X \to Z$ be a continuous map such that

  $$g \circ r_1 = g \circ r_2.$$

  Since surjections are coequalizers of their kernel pairs in Set, there is a unique set-theoretic map $\bar{g}\colon Y \to Z$ making the triangle in the following diagram commute.

  $$\begin{tikzcd} R \arrow[r, shift left, "r_1"] \arrow[r, shift right, "r_2"'] & X \arrow[r, "f"] \arrow[dr, "g"'] & Y \arrow[d, dashed, "\bar{g}"] \\ & & Z \end{tikzcd}$$

  We claim that $\bar{g}$ is continuous. To prove continuity of $\bar{g}$, it is enough to show that the preimage of every closed subset of Z is closed in Y. Let $C \subseteq Z$ be closed. Then

  $$\bar{g}^{-1}[C] = f[g^{-1}[C]].$$

  Indeed, if $y \in Y$, choose $x \in X$ such that $f(x) = y$. Then

  $$y \in \bar{g}^{-1}[C] \;\Leftrightarrow\; \bar{g}(y) \in C \;\Leftrightarrow\; g(x) \in C \;\Leftrightarrow\; x \in g^{-1}[C] \;\Leftrightarrow\; y \in f[g^{-1}[C]].$$

  Now $g^{-1}[C]$ is closed in the compact space X, hence compact. Its image under the continuous map f is compact, and therefore closed in the Hausdorff space Y. So $\bar{g}^{-1}[C]$ is closed, which proves that $\bar{g}$ is continuous. Hence, f is the coequalizer of r_1, r_2 in Stone, and therefore f is a regular epimorphism.

Therefore, regular epimorphisms in Stone are precisely the surjective continuous maps. By Stone duality and Theorem 1.7.13 (injective is dual to surjective, and vice versa), it follows that the regular monomorphisms in BA are precisely the injective homomorphisms. ■

Remark 1.7.17. The fact that epimorphisms in BA are surjective has a logical counterpart: it is related to Beth definability.

1.7.7 Amalgamation

We conclude this part with another useful categorical property of BA.

Theorem 1.7.18 (Amalgamation property of Boolean algebras)**.** *Given injective Boolean homomorphisms*

$$\begin{tikzcd} A \arrow[r, hook] \arrow[d, hook] & B \\ C & \end{tikzcd}$$

there are a Boolean algebra D and injective homomorphisms

$$\begin{tikzcd} A \arrow[r, hook] \arrow[d, hook] & B \arrow[d, hook] \\ C \arrow[r, hook] & D \end{tikzcd}$$

such that the square commutes.

Proof. Let

$$i\colon A \hookrightarrow B \qquad \text{and} \qquad j\colon A \hookrightarrow C$$

be injective homomorphisms. By Stone duality and Theorem 1.7.13 ("Injective is dual to surjective, and vice versa"), they correspond to surjective continuous maps

$$\mathrm{Ult}(B) \twoheadrightarrow \mathrm{Ult}(A) \qquad \text{and} \qquad \mathrm{Ult}(C) \twoheadrightarrow \mathrm{Ult}(A).$$

Let

$$P \coloneqq \mathrm{Ult}(B) \times_{\mathrm{Ult}(A)} \mathrm{Ult}(C)$$

be their pullback in Stone. By Proposition 1.7.8, it is computed as in Set, so

$$P = \{(x, y) \in \mathrm{Ult}(B) \times \mathrm{Ult}(C) \mid f(x) = g(y)\},$$

where

$$f\colon \mathrm{Ult}(B) \to \mathrm{Ult}(A) \qquad \text{and} \qquad g\colon \mathrm{Ult}(C) \to \mathrm{Ult}(A)$$

are the two surjective maps. Thus, we have a commutative diagram

$$\begin{tikzcd} P \arrow[r, "\pi_2"] \arrow[d, "\pi_1"'] & \mathrm{Ult}(C) \arrow[d, "g"] \\ \mathrm{Ult}(B) \arrow[r, "f"'] & \mathrm{Ult}(A). \end{tikzcd}$$

We claim that both projections π_1 and π_2 are surjective. We prove this for π_1; the proof for π_2 is symmetric. Let $x \in \mathrm{Ult}(B)$. Since g is surjective, there is $y \in \mathrm{Ult}(C)$ such that

$$g(y) = f(x).$$

Then $(x, y) \in P$, and therefore

$$\pi_1(x, y) = x.$$

So π_1 is surjective.

Now dualize the above pullback square. Since surjective maps in Stone correspond to injective homomorphisms in BA, we obtain a commutative square

$$\begin{tikzcd} A \arrow[r, hook] \arrow[d, hook] & B \arrow[d, hook] \\ C \arrow[r, hook] & \mathrm{Clop}(P), \end{tikzcd}$$

which proves the claim. ■

Remark 1.7.19. The amalgamation property of BA has a logical counterpart: it corresponds to a property called interpolation.

1.7.8 Free Boolean algebras and counting formulas modulo equivalence

We now settle the question: given a finite language $\mathcal{L}$ of cardinality n, how many formulas (up to semantic equivalence) are there?

Theorem 1.7.20 (Formulas modulo equivalence are free Boolean algebras)**.** *For every propositional language $\mathcal{L}$, the Boolean algebra $\mathrm{Form}(\mathcal{L})/\equiv$ is the free Boolean algebra on $\mathcal{L}$.*

Proof. Let B be a Boolean algebra, and let

$$v\colon \mathcal{L} \longrightarrow B$$

be a function.

Since $\mathrm{Form}(\mathcal{L})$ is the algebra of formulas built from the generators in $\mathcal{L}$ using the Boolean connectives, there is a unique homomorphism

$$\bar{v}\colon \mathrm{Form}(\mathcal{L}) \longrightarrow B$$

such that the following triangle commutes:

$$\begin{tikzcd} \mathcal{L} \arrow[r, hook] \arrow[dr, "v"'] & \mathrm{Form}(\mathcal{L}) \arrow[d, "\bar{v}"] \\ & B. \end{tikzcd}$$

Now let

$$\pi\colon \mathrm{Form}(\mathcal{L}) \twoheadrightarrow \mathrm{Form}(\mathcal{L})/\equiv$$

be the quotient map. To prove that $\mathrm{Form}(\mathcal{L})/\equiv$ is the free Boolean algebra on $\mathcal{L}$, it is enough to show that $\bar{v}$ factors uniquely through π, i.e., that there is a unique Boolean homomorphism

$$\widetilde{v}\colon \mathrm{Form}(\mathcal{L})/\equiv \longrightarrow B$$

making the following diagram commute:

$$\begin{tikzcd} \mathcal{L} \arrow[r] \arrow[dr, "v"'] & \mathrm{Form}(\mathcal{L}) \arrow[r, two heads, "\pi"] \arrow[d, "\bar{v}"] & \mathrm{Form}(\mathcal{L})/\equiv \arrow[dl, dashed, "\widetilde{v}"] \\ & B & \end{tikzcd}.$$

So it remains to show that $\bar{v}$ is constant on $\equiv$-equivalence classes. In other words, whenever $\varphi \equiv \psi$, we must prove that

$$\bar{v}(\varphi) = \bar{v}(\psi).$$

Suppose not. Then $\bar{v}(\varphi) \neq \bar{v}(\psi)$ in B. By Corollary 1.3.18 ("Ultrafilters separate elements"), distinct elements of a Boolean algebra can be separated by a homomorphism into 2. So there is a Boolean homomorphism

$$h\colon B \longrightarrow 2$$

such that $h(\bar{v}(\varphi)) \neq h(\bar{v}(\psi))$.

$$\begin{tikzcd}
\mathcal{L} \arrow[r] \arrow[dr, "v"'] & \mathrm{Form}(\mathcal{L}) \arrow[r, two heads, "\pi"] \arrow[d, "\bar{v}"] & \mathrm{Form}(\mathcal{L})/\equiv \arrow[dl, dashed, "\widetilde{v}"] \\
& B \arrow[d, "h"] & \\
& 2 &
\end{tikzcd}$$

Consider now the composite

$$h \circ v\colon \mathcal{L} \longrightarrow 2.$$

This is a valuation on $\mathcal{L}$, and its unique extension to a homomorphism

$$\mathrm{Form}(\mathcal{L}) \longrightarrow 2$$

is precisely $h \circ \bar{v}$. So we have another commutative triangle:

$$\begin{tikzcd}
\mathcal{L} \arrow[r, hook] \arrow[dr, "h\circ v"'] & \mathrm{Form}(\mathcal{L}) \arrow[d, "h\circ\bar{v}"] \\
& 2.
\end{tikzcd}$$

Thus

$$(h \circ \bar{v})(\varphi) \neq (h \circ \bar{v})(\psi),$$

which means that the valuation $h \circ v$ distinguishes φ and ψ. But this contradicts the assumption that $\varphi \equiv \psi$.

Therefore $\bar{v}$ is constant on $\equiv$-equivalence classes, so it factors uniquely through $\mathrm{Form}(\mathcal{L})/\equiv$. Hence, $\mathrm{Form}(\mathcal{L})/\equiv$ is the free Boolean algebra on $\mathcal{L}$. ■

Proposition 1.7.21 (The Stone dual of the free Boolean algebra on $\mathcal{L}$)**.** *For every set $\mathcal{L}$, the Stone dual of the free Boolean algebra $\mathrm{Free}_{\mathsf{BA}}(\mathcal{L})$ on $\mathcal{L}$ is homeomorphic to $2^{\mathcal{L}}$ equipped with the product topology.*

Proof. Since the free Boolean algebra functor

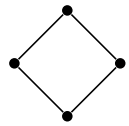
$$\mathrm{Free}_{\mathsf{BA}}\colon \mathsf{Set} \longrightarrow \mathsf{BA}$$

is left adjoint to the forgetful functor, it preserves coproducts. Hence

$$\mathrm{Free}_{\mathsf{BA}}(\mathcal{L}) \cong \mathrm{Free}_{\mathsf{BA}}\left(\sum_{p\in\mathcal{L}} 1\right) \cong \sum_{p\in\mathcal{L}} \mathrm{Free}_{\mathsf{BA}}(1),$$

where 1 denotes a singleton. By Stone duality, coproducts in BA correspond to products in Stone. Therefore the Stone dual of $\mathrm{Free}_{\mathsf{BA}}(\mathcal{L})$ is homeomorphic to the product of $|\mathcal{L}|$ copies of the Stone dual of $\mathrm{Free}_{\mathsf{BA}}(1)$.

It remains to identify the Stone dual of $\mathrm{Free}_{\mathsf{BA}}(1)$. The free Boolean algebra on one generator is easily seen to be the four-element Boolean algebra,



and so it is isomorphic to $\mathcal{P}(2)$. Hence, its Stone dual is the two-point discrete space (which we denote by 2).[a]

Therefore, the Stone dual of $\mathrm{Free}_{\mathsf{BA}}(\mathcal{L})$ is homeomorphic to $2^{\mathcal{L}}$ with the product topology. ■

[a] Alternatively, without explicitly identifying the free Boolean algebra on one generator, one can use the fact that $\mathrm{Free}_{\mathsf{BA}}$ is left adjoint to the forgetful functor, and hence the points of the Stone dual of $\mathrm{Free}_{\mathsf{BA}}(1)$ are in bijection with

$$\hom_{\mathsf{BA}}(\mathrm{Free}_{\mathsf{BA}}(1), 2) \cong \hom_{\mathsf{Set}}(1, 2) \cong 2,$$

to obtain that the Stone dual of $\mathrm{Free}_{\mathsf{BA}}(1)$ is the two-point discrete space.

Corollary 1.7.22 (How many formulas are there up to semantic equivalence?)**.** *If $\mathcal{L}$ is finite of cardinality n, then*

$$\mathrm{Form}(\mathcal{L})/{\equiv} \cong \mathcal{P}(2^{\mathcal{L}}),$$

and therefore

$$|\mathrm{Form}(\mathcal{L})/{\equiv}| = 2^{2^n}.$$

So, there are 2^{2^n} many formulas in n propositional symbols up to semantic equivalence.

Proof. If $|\mathcal{L}| = n < \infty$, then $2^{\mathcal{L}}$ is a finite discrete space. Hence, every subset of $2^{\mathcal{L}}$ is clopen, so

$$\mathrm{Clop}(2^{\mathcal{L}}) = \mathcal{P}(2^{\mathcal{L}}).$$

By Proposition 1.7.21 and Stone duality,

$$\mathrm{Form}(\mathcal{L})/{\equiv} \cong \mathrm{Clop}(2^{\mathcal{L}}) = \mathcal{P}(2^{\mathcal{L}}).$$

Since $|2^{\mathcal{L}}| = 2^n$, it follows that

$$|\mathrm{Form}(\mathcal{L})/{\equiv}| = |\mathcal{P}(2^{\mathcal{L}})| = 2^{2^n}.$$

■

Example 1.7.23. If $\mathcal{L} = \varnothing$, then there are 2 formulas up to equivalence ($[0], [1]$); if $\mathcal{L} = \{p\}$, then there are 4 ($[0], [1], [p], [\neg p]$); if $\mathcal{L} = \{p, q\}$, then there are 16.

1.7.9 Exercises on Stone duality

The following exercises are meant to reinforce the use of Stone duality. I would solve them (or get the intuition for them) via Stone duality (and that's the reason why they are here), as it is easier for me to think via Stone spaces, although for you it may be different. Try to discover your own preferred point of view.

Exercise 1.7.24. How many endomorphisms does an 8-element Boolean algebra have? How many automorphisms does it have?

Solution. Every Boolean algebra with 8 elements is isomorphic to $\mathcal{P}(X)$ with $|X| = 3$. Indeed, every finite Boolean algebra of cardinality 2^n is isomorphic to the powerset algebra of an n-element set, and here $8 = 2^3$.

By Stone duality, endomorphisms of $\mathcal{P}(X)$ correspond to continuous maps from the discrete Stone space X on 3 points to itself. Since every map between finite discrete spaces is continuous, the number of endomorphisms is

just the number of functions from 3 to 3, which is 27.

Similarly, automorphisms of $\mathcal{P}(X)$ correspond to homeomorphisms between X and itself, i.e., to permutations of a 3-element set. Therefore, the number of automorphisms is $|S_3| = 3! = 6$. ■

Exercise 1.7.25. How many elements does the coproduct in BA of an 8-element Boolean algebra and a 16-element Boolean algebra have?

Solution. If B_1 has $8 = 2^3$ elements and B_2 has $16 = 2^4$ elements, then $B_1 \cong \mathcal{P}(X)$ and $B_2 \cong \mathcal{P}(Y)$ with $|X| = 3$ and $|Y| = 4$. By Stone duality,

$$B_1 + B_2 \;\cong\; \mathrm{Clop}(X \times Y) = \mathcal{P}(X \times Y),$$

and so

$$|B_1 + B_2| = |\mathcal{P}(X \times Y)| = 2^{|X \times Y|} = 2^{3 \cdot 4} = 2^{12}.$$

Therefore, the solution is $2^{12} = 4096$. ■

Exercise 1.7.26. Let A be an infinite Boolean algebra and B a finite nontrivial Boolean algebra. Show that B is isomorphic to a subalgebra of A.

Solution. Under Stone duality, this corresponds to the following:

Let X be an infinite Stone space and F a finite nonempty discrete space.
Show that there is a surjective continuous map from X to F.

It is enough to partition X into $n := |F|$ pairwise disjoint nonempty clopen subsets.

We prove by induction on $n \geq 1$ that every infinite Stone space admits such a partition. The case $n = 1$ is trivial. Assume we have a partition

$$X = U_1 \sqcup \cdots \sqcup U_k$$

into k pairwise disjoint nonempty clopen subsets. Since X is infinite, at least one of the U_i's must be infinite; let it be U_j. Now U_j, being a clopen subspace of a Stone space, is again a Stone space. Choose two distinct points $x, y \in U_j$, which exist because U_j is infinite. Since U_j is totally separated, there is a clopen subset

$$V \subseteq U_j$$

such that $x \in V$ and $y \notin V$. Then both V and $U_j \setminus V$ are nonempty clopen subsets of the clopen U_j (and hence of X). Replacing U_j by these two pieces, we obtain a partition of X into $k+1$ pairwise disjoint nonempty clopen subsets. Thus, by induction, X admits a partition

$$X = U_1 \sqcup \cdots \sqcup U_n$$

into n pairwise disjoint nonempty clopen subsets.

Let $z_1, \ldots, z_n$ denote the elements of F. Define the function $q\colon X \twoheadrightarrow F$ by setting $q(x) = z_i$ whenever $x \in U_i$. It is surjective because each U_i is nonempty. Moreover, since F is discrete and each fiber $q^{-1}[\{z_i\}] = U_i$ is clopen, the map q is continuous. ■

The assertion in Exercise 1.7.26 can be strenghtened as follows.

Exercise 1.7.27. Let A be an infinite Boolean algebra and B a finite nontrivial Boolean algebra. Show that B is a retract of A, i.e., that there are Boolean homomorphisms $\iota\colon B \hookrightarrow A$ and $\pi\colon A \twoheadrightarrow B$ such that the composite $B \overset{\iota}{\hookrightarrow} A \overset{\pi}{\twoheadrightarrow} B$ is the identity on B.

(By $\pi \circ \iota = \mathrm{id}_B$, one necessarily has that ι is injective and π is surjective.)

Solution. Under Stone duality, this corresponds to the following:

Let X be an infinite Stone space and F a finite nonempty discrete space.
Show that there are continuous maps $q\colon X \twoheadrightarrow F$ and $j\colon F \hookrightarrow X$
such that the composite $F \overset{j}{\hookrightarrow} X \overset{q}{\twoheadrightarrow} F$ is the identity on F.

So, let X be an infinite Stone space and F a finite nonempty discrete space. By the argument in Exercise 1.7.26, there is a continuous surjective function $q\colon X \twoheadrightarrow F$. Since q is surjective, for every $z \in F$ we can choose a point $x_z \in q^{-1}[\{z\}]$. Define $j\colon F \hookrightarrow X$ by setting $j(z) \coloneqq x_z$ for every $z \in F$. Then, by construction, $q(j(z)) = z$ for every $z \in F$; i.e., the composite $F \overset{j}{\hookrightarrow} X \overset{q}{\twoheadrightarrow} F$ is the identity on F. Moreover, since F is discrete, every function from F to X is continuous, and hence j is continuous. ■

Exercise 1.7.28. Is the free Boolean algebra $\mathrm{Free}_{\mathsf{BA}}(\omega)$ on countably many generators complete?

Hint: you can use the following facts:

1. A Boolean algebra is complete if and only if its dual Stone space is *extremally disconnected* (i.e. the closure of every open set is clopen; equivalently, the closures of any two disjoint open sets are disjoint)[20].
2. The Stone dual of $\mathrm{Free}_{\mathsf{BA}}(\omega)$ is homeomorphic to $2^{\mathbb{N}}$ with the product topology, which in turn is homeomorphic to the subset of $[0,1]$ known as the *Cantor set*[21], equipped with the Euclidean topology.

Solution. No.

By the hint, it is enough to show that the Cantor space $C \subseteq [0,1]$ is not extremally disconnected. To do so, we find two disjoint open subsets of the Cantor space whose closures intersect. Consider the open subsets of C

$$U \coloneqq C \cap (-\infty, 1/4), \qquad V \coloneqq C \cap (1/4, +\infty).$$

They are disjoint and open in the subspace topology on C. However, their closures in C intersect, since $1/4 \in \overline{U} \cap \overline{V}$. Therefore C is not extremally disconnected, and hence $\mathrm{Free}_{\mathsf{BA}}(\omega)$ is not complete.[a] ■

[a](Logical rephrasing: completeness would amount to saying that for any set Γ of formulas there is a single formula ψ such that for every formula φ one has $\Gamma \models \varphi$ if and only if $\psi \models \varphi$, which fails in classical propositional logic.)

Exercise 1.7.29. Let $\mathrm{FC}(\mathbb{N})$ be the algebra of finite and cofinite subsets of $\mathbb{N}$ (Example 1.2.16). Is the coproduct $\mathrm{FC}(\mathbb{N}) + \mathrm{FC}(\mathbb{N})$ isomorphic to $\mathrm{FC}(\mathbb{N})$?

Solution. No.

Let $X \coloneqq \mathrm{Ult}(\mathrm{FC}(\mathbb{N}))$. As discussed in Example 1.2.16, the Stone space X is homeomorphic to the one-point compactification $\mathbb{N} \cup \{\infty\}$ of the discrete space $\mathbb{N}$: the opens are the sets not containing ∞ and the cofinite sets containing ∞.

Now, by Stone duality, coproducts in BA correspond to products in Stone. Therefore, if one had $\mathrm{FC}(\mathbb{N}) + \mathrm{FC}(\mathbb{N}) \cong \mathrm{FC}(\mathbb{N})$, then one would also have a homeomorphism $X \times X \cong X$.

We show that this is impossible. In X, there is exactly one non-isolated point, namely ∞. On the other hand, in $X \times X$, the isolated points are exactly the pairs (n, m) for $n, m \in \mathbb{N}$. Thus, X has exactly one non-isolated point—namely, ∞—, whereas $X \times X$ has infinitely many non-isolated points—namely, (∞, ∞), (n, ∞) for $n \in \mathbb{N}$, and (∞, n) for $n \in \mathbb{N}$. So, $X \times X$ is not homeomorphic to X. Therefore, $\mathrm{FC}(\mathbb{N}) + \mathrm{FC}(\mathbb{N})$ is not isomorphic to $\mathrm{FC}(\mathbb{N})$. ■

Remark 1.7.30. Every Boolean algebra embeds into a complete Boolean algebra. (This is simply by Stone's Representation Theorem.)

[20]For a reference for the fact that a Boolean algebra is complete if and only if its dual Stone space is extremally disconnected, see [Givant and Halmos, 2009, Ch. 38, Thm. 39, p. 369] (where "extremally disconnected" is called "complete").

[21]For the definition of the Cantor set, see, e.g., [Munkres, 2000, Ch. 3, Sec. 27, Exercise 6, p. 178].

Chapter 2

Classical first-order logic

In the second part of the course, we move from classical propositional logic to classical first-order logic, where quantifiers enter the picture. As in the propositional case, we will relate *algebras of formulas* to *spaces of models* via a categorical duality.

2.1 From classical first-order logic towards Boolean doctrines

2.1.1 Syntax: first-order languages, formulas, free variables, sentences, substitutions

Fix once and for all an infinite supply of variables, typically denoted by

$$x, y, z, \dots$$

Definition 2.1.1 (Relational first-order language, relation symbol, arity)**.** A ***(one-sorted relational first-order) language*** $\mathcal{L}$ consists of a set, whose elements are called ***relation symbols***, together with an assigned natural number for each relation symbol. The number assigned to a relation symbol R is called the ***arity*** of R, and is denoted $\mathrm{ar}(R)$.
In this chapter, we consider only the relational one-sorted case, and we do not include equality.

Relation symbols are typically denoted by $R, S, \dots$

Example 2.1.2 (Graphs and preorders)**.** A language with one binary relation symbol can be used to describe directed graphs. Indeed, let $\mathcal{L} = \{R\}$, where R has arity 2. Then, we can form the expression

$$R(x, y),$$

which one may read informally as "there is an edge from x to y".

The same syntactic language can also be used to describe preorders. For this, one may write the unique binary relation symbol as $\leq$, so that the atomic formulas have the form

$$x \leq y.$$

The difference between graphs and preorders is not in the language, but in the axioms that we impose later on the structures interpreting that language.

Definition 2.1.3 (Formula)**.** The set of ***formulas*** $\mathrm{Form}(\mathcal{L})$ is defined inductively as follows:

- if $R \in \mathcal{L}$ has arity n, and $x_1, \dots, x_n$ are (not necessarily distinct) variables, then $R(x_1, \dots, x_n) \in$

Form($\mathcal{L}$) (the formulas of this form are called ***atomic formulas***);

- if $\varphi, \psi \in \mathrm{Form}(\mathcal{L})$, then $(\varphi \vee \psi) \in \mathrm{Form}(\mathcal{L})$ and $(\varphi \wedge \psi) \in \mathrm{Form}(\mathcal{L})$;
- if $\varphi \in \mathrm{Form}(\mathcal{L})$, then $(\neg\varphi) \in \mathrm{Form}(\mathcal{L})$;
- $0, 1 \in \mathrm{Form}(\mathcal{L})$;
- if x is a variable and $\varphi \in \mathrm{Form}(\mathcal{L})$, then $(\exists x\, \varphi) \in \mathrm{Form}(\mathcal{L})$ and $(\forall x\, \varphi) \in \mathrm{Form}(\mathcal{L})$.

In other words, Form($\mathcal{L}$) is the smallest set containing all atomic formulas and closed under the Boolean connectives and the quantifiers.

Formulas are typically denoted by $\varphi, \psi, \sigma, \ldots$.

Remark 2.1.4. In principle, quantifiers may be nested using the same variable symbol. For example, the expression

$$\forall x\, (R(x) \wedge \forall x\, P(x))$$

is a well-formed formula. However, it should be thought of as a needlessly confusing variant of

$$\forall x\, (R(x) \wedge \forall y\, P(y)).$$

Similarly, a formula of the form

$$R(x) \wedge \forall x\, P(x)$$

should be thought of as a needlessly confusing variant of

$$R(x) \wedge \forall y\, P(y).$$

Definition 2.1.5 (Free and bound occurrences, free variable, sentence)**.** Let $\mathcal{L}$ be a one-sorted relational first-order language, and let $\varphi \in \mathrm{Form}(\mathcal{L})$.
An occurrence of a variable x in φ is said to be ***bound*** if either it is the occurrence of x in a quantifier $\exists x$ or $\forall x$, or it lies within the scope of a quantifier $\exists x$ or $\forall x$. Otherwise, the occurrence is said to be ***free***.
A ***free variable*** of φ is a variable having a free occurrence in φ. We denote by

$$\mathrm{FV}(\varphi)$$

the set of free variables of φ.
A formula with no free variables is called a ***sentence***.

Example 2.1.6 (Free and bound occurrences)**.** In the formula

$$R(\underset{\substack{\uparrow \\ \text{free}}}{x}, y) \wedge \forall \underset{\substack{\uparrow \\ \text{bound}}}{x}\, S(\underset{\substack{\uparrow \\ \text{bound}}}{x}, z),$$

the first occurrence of x is free, while the second and the third occurrences of x are bound. Since the formula has a free occurrence of x, the variable x is a free variable of this formula.

Example 2.1.7 (Free variables)**.** Consider a language with one binary relation symbol R. Then:

$$\mathrm{FV}(R(x, y)) = \{x, y\},$$
$$\mathrm{FV}(\forall x\, R(x, y)) = \{y\},$$
$$\mathrm{FV}(\exists y\, \forall x\, R(x, y)) = \varnothing.$$

So $R(x, y)$ and $\forall x\, R(x, y)$ are not sentences, whereas $\exists y\, \forall x\, R(x, y)$ is a sentence.

Before defining the semantics of formulas, it is useful to isolate one technical point that will later play an important role: substitution.

Definition 2.1.8 (Substitution of a variable by a variable)**.** Let $\mathcal{L}$ be a relational first-order language, let $\varphi \in \mathrm{Form}(\mathcal{L})$, and let x, y be variables. The formula obtained by ***substituting*** y ***for*** x ***in*** φ is denoted by

$$\varphi[y/x].$$

It is obtained by replacing the *free* occurrences of x in φ by y. (Bound occurrences are not replaced.) Moreover, when passing through a quantifier, one must avoid ***variable capture***: if replacing a free occurrence of x by y would make it fall under a quantifier binding y, then one first renames that bound variable to a fresh variable.

For example, if one starts from the formula

$$\forall y\, R(x, y),$$

then the substitution of y for x is *not*

$$\forall y\, R(y, y),$$

because, in this formula, the new occurrence of y would become bound. Instead, one first renames the bound variable, for instance from y to z, and obtains

$$\forall z\, R(x, z),$$

so that

$$(\forall y\, R(x, y))[y/x] = \forall z\, R(y, z).$$

2.1.2 Semantics: structures, assignments, validity of formulas, models, semantic equivalence

We now pass from syntax to semantics.

Definition 2.1.9 (Structure)**.** Let $\mathcal{L}$ be a relational first-order language. An ***$\mathcal{L}$-structure*** is a set[a] A equipped, for each relation symbol $R \in \mathcal{L}$, with a subset (denoted by R^A and called the *interpretation of R in A*) of

$$\underbrace{A \times \cdots \times A}_{\mathrm{ar}(R)\text{ times}}.$$

In other words, each relation symbol R is interpreted in A as an $(\mathrm{ar}(R))$-ary relation on the set A.

[a] Unlike many standard treatments of first-order logic, we allow the underlying set of an $\mathcal{L}$-structure to be empty.

For example, if $\mathcal{L} = \{\leq\}$ and $\mathrm{ar}(\leq) = 2$, then an $\mathcal{L}$-structure is just a set equipped with a binary relation.

Remark 2.1.10 (Nullary relation symbols)**.** A relation symbol of arity 0 behaves like a propositional symbol. Indeed, if $R \in \mathcal{L}$ and $\mathrm{ar}(R) = 0$, then in every $\mathcal{L}$-structure A its interpretation is a subset

$$R^A \subseteq A^0.$$

But A^0 is a singleton, namely the set consisting of the unique function from the empty set to A. Therefore, R^A is either empty or all of A^0.

So one may think of the interpretation of R as simply *false* in the first case, and *true* in the second. In this sense, nullary relation symbols play the role of the propositional symbols of classical propositional logic.

Definition 2.1.11 (Assignment)**.** Let A be an $\mathcal{L}$-structure, and let X be a finite set of variables. An ***assignment on*** X ***in*** A is a function

$$\nu \colon X \to A.$$

Definition 2.1.12 (Validity of a formula in a structure under an assignment ($\models$))**.** Let $\mathcal{L}$ be a relational first-order language and let A be an $\mathcal{L}$-structure. By induction on the complexity of a formula φ in the language $\mathcal{L}$, we define, for every finite set X of variables with $\mathrm{FV}(\varphi) \subseteq X$ and every function $\nu\colon X \to A$, the expression

$$A, \nu \models \varphi,$$

read as "φ holds in A under the assignment ν", as follows.

- $A, \nu \models R(x_1, \dots, x_n) \iff (\nu(x_1), \dots, \nu(x_n)) \in R^A$, for $R \in \mathcal{L}$ of arity n and $x_1, \dots, x_n \in X$;
- $A, \nu \models \varphi \vee \psi \iff A, \nu \models \varphi$ or $A, \nu \models \psi$;
- $A, \nu \models \varphi \wedge \psi \iff A, \nu \models \varphi$ and $A, \nu \models \psi$;
- $A, \nu \models \neg\varphi \iff$ not $A, \nu \models \varphi$;
- $A, \nu \models 0 \iff$ false (i.e., $A, \nu \models 0$ never holds);
- $A, \nu \models 1 \iff$ true (i.e., $A, \nu \models 1$ always holds);
- $A, \nu \models \exists x\, \varphi$ if and only if, picking any variable[a] $z \notin X$ (such as x itself if $x \notin X$), there is $a \in A$ such that
$$A, \nu' \models \varphi[z/x],$$
where
$$\nu'\colon X \cup \{z\} \to A$$
is given by
$$\nu'(y) = \begin{cases} \nu(y) & \text{if } y \in X, \\ a & \text{if } y = z; \end{cases}$$
- $A, \nu \models \forall x\, \varphi$ if and only if, picking any variable $z \notin X$ (such as x itself if $x \notin X$), for every $a \in A$ we have
$$A, \nu' \models \varphi[z/x],$$
where
$$\nu'\colon X \cup \{z\} \to A$$
is given by
$$\nu'(y) = \begin{cases} \nu(y) & \text{if } y \in X, \\ a & \text{if } y = z. \end{cases}$$

[a] One can prove that the ensuing condition is independent of the choice of the variable z (as well as of any possible additional choice of variable needed to avoid variable capture in the substitution $\varphi[z/x]$). The same holds for the case $\forall$.

Definition 2.1.13 (Validity of a sentence in a structure ($\models$))**.** Let $\mathcal{L}$ be a relational first-order language, let A be an $\mathcal{L}$-structure, and let φ be a sentence in the language $\mathcal{L}$. We write

$$A \models \varphi$$

if

$$A, \varnothing \models \varphi,$$

where $\varnothing\colon \varnothing \to A$ denotes the unique function from the empty set to A.

Example 2.1.14. Let $\mathcal{L} = \{R\}$, where R is a binary relation symbol, and let A be an $\mathcal{L}$-structure. We want

to understand what the sentence

$$\exists x\, \forall y\, R(x, y)$$

means in A.

By definition,

$$A \models \exists x\, \forall y\, R(x, y)$$

means

$$A, \varnothing \models \exists x\, \forall y\, R(x, y), \tag{2.1}$$

where $\varnothing \colon \varnothing \to A$ is the unique function from the empty set to A.

Now pick a variable z not in the empty set. By the case $\exists$ in the definition of $\models$ (Definition 2.1.12), the condition (2.1) means that there is $a \in A$ such that

$$A, \nu \models \forall y\, R(z, y),$$

where

$$\nu \colon \{z\} \to A$$

is given by $\nu(z) = a$.

Next, by the case $\forall$ in the definition of $\models$ (Definition 2.1.12), this means that, picking a variable $w \notin \{z\}$, for every $b \in A$ we have

$$A, \nu' \models R(z, w),$$

where

$$\nu' \colon \{z, w\} \to A$$

is given by

$$\nu'(z) = a, \qquad \nu'(w) = b.$$

By the definition of validity for atomic formulas, this is equivalent to saying that

$$(a, b) \in R^A$$

for every $b \in A$.

Therefore,

$$A \models \exists x\, \forall y\, R(x, y)$$

holds if and only if there is $a \in A$ such that for every $b \in A$ one has $(a, b) \in R^A$.

We can now incorporate semantic assumptions, just as we did in the propositional case.

Definition 2.1.15 (Theory)**.** Let $\mathcal{L}$ be a relational first-order language. A ***theory*** in the language $\mathcal{L}$ is a set of sentences in the language $\mathcal{L}$.

Definition 2.1.16 (Model of a theory)**.** Let $\mathcal{T}$ be a theory in a relational first-order language $\mathcal{L}$. A ***model*** of $\mathcal{T}$ is an $\mathcal{L}$-structure A such that, for every sentence $\varphi \in \mathcal{T}$, $A \models \varphi$.

Definition 2.1.17 (Semantic equivalence modulo a theory, in a context)**.** Let $\mathcal{T}$ be a theory in the language $\mathcal{L}$, let X be a finite set of variables, and let $\varphi, \psi \in \mathrm{Form}(\mathcal{L})$ be such that

$$\mathrm{FV}(\varphi) \subseteq X \qquad \text{and} \qquad \mathrm{FV}(\psi) \subseteq X.$$

We say that φ and ψ are ***semantically equivalent modulo $\mathcal{T}$ in the context*** X if, for every model A of $\mathcal{T}$ and every assignment

$$\nu \colon X \to A,$$

one has

$$A, \nu \models \varphi \qquad \Longleftrightarrow \qquad A, \nu \models \psi.$$

In other words, once the assumptions in $\mathcal{T}$ are fixed, two formulas are identified in the context X when they have the same truth value in every model of $\mathcal{T}$ under every assignment of values to the variables in X.

The next example shows a case in which two sentences are not equivalent in the empty context, but become equivalent in any nonempty context.

Example 2.1.18 (Dependence on the context)**.** Let $\mathcal{L}$ be the empty language, let $\mathcal{T}$ be the empty theory, and consider the formula

$$\exists x\, 1.$$

We first look at the empty context $\varnothing$. In this context, the formula $\exists x\, 1$ is *not* semantically equivalent modulo $\mathcal{T}$ to 1. Indeed, if $A = \varnothing$, then A is an $\mathcal{L}$-structure and hence a model of $\mathcal{T}$. Now

$$A \models 1,$$

but

$$A \not\models \exists x\, 1,$$

since there is no element of A that can witness the existential quantifier.

We now look at the context $\{y\}$. In this context, the formula $\exists x\, 1$ *is* semantically equivalent modulo $\mathcal{T}$ to 1. Indeed, let A be any model of $\mathcal{T}$, and let

$$\nu \colon \{y\} \to A$$

be any assignment. Then A must be nonempty, since otherwise no such assignment could exist. Hence there is some $a \in A$, and therefore

$$A, \nu \models \exists x\, 1.$$

On the other hand,

$$A, \nu \models 1$$

always holds. So, for every model A of $\mathcal{T}$ and every assignment $\nu \colon \{y\} \to A$, we have

$$A, \nu \models \exists x\, 1 \qquad \Longleftrightarrow \qquad A, \nu \models 1.$$

Thus, $\exists x\, 1$ is not equivalent to 1 in the empty context, but it is equivalent to 1 in the context $\{y\}$. A similar reasoning shows that $\exists x\, 1$ is equivalent to 1 in any nonempty context.

The point is that choosing a context amounts to declaring which variables are available as *generic elements* of the structure. If the context is nonempty, then only nonempty structures admit assignments on that context.

2.2 First-order Boolean doctrines

The previous discussion suggests that, in first-order logic, one should not look at a single Boolean algebra of formulas, but rather at a family of Boolean algebras: one for each finite set of variables. Here we only model one-sorted purely relational first-order logic without equality.

Let FinSet denote the category of finite sets and functions, and recall that BA is the category of Boolean algebras and Boolean homomorphisms. The following notion—of first-order Boolean doctrine—is essentially due to Lawvere's notion of hyperdoctrine ([Lawvere, 1969])[1]. To understand the definition, I suggest comparing each passage of the definition with its meaning in the intended example put right after: the Lindenbaum–Tarski first-order Boolean doctrine of a theory (Definition 2.2.3).

[1] See also [Lawvere, 1970] for the treatment of equality and comprehension, which we do not discuss here.

Definition 2.2.1 (First-order Boolean doctrine)**.** A ***first-order Boolean doctrine*** is a functor

$$P\colon \mathsf{FinSet} \to \mathsf{BA}$$

with the following properties.

1. (Quantifiers as adjoints to adding a dummy variable) For every finite set X and every $y \notin X$, the Boolean homomorphism[a]

$$P(X \hookrightarrow X \cup \{y\})\colon P(X) \longrightarrow P(X \cup \{y\})$$

has both[b] a left adjoint and a right adjoint (as order-preserving functions between posets), denoted respectively by

$$(\exists y)_X\colon P(X \cup \{y\}) \longrightarrow P(X) \qquad \text{and} \qquad (\forall y)_X\colon P(X \cup \{y\}) \longrightarrow P(X).$$

This means that, for every $\varphi \in P(X \cup \{y\})$, there are (necessarily unique) elements $(\exists y)_X\, \varphi$ and $(\forall y)_X\, \varphi$ such that, for every $\psi \in P(X)$,

$$(\exists y)_X\, \varphi \leq \psi \iff \varphi \leq P(X \hookrightarrow X \cup \{y\})(\psi) \qquad \text{and} \qquad P(X \hookrightarrow X \cup \{y\})(\psi) \leq \varphi \iff \psi \leq (\forall y)_X\, \varphi.$$

2. (Beck–Chevalley property: substitutions commute with quantifiers) For every function $\sigma\colon X \to X'$ between finite sets, every $y \notin X$ and every $y' \notin X'$, calling $\widetilde{\sigma}\colon X \cup \{y\} \to X' \cup \{y'\}$ the extension of σ that maps y to y', the following diagrams commute.[c]

$$\begin{tikzcd} P(X \cup \{y\}) \arrow[r, "(\exists y)_X"] \arrow[d, "P(\widetilde{\sigma})"'] & P(X) \arrow[d, "P(\sigma)"] \\ P(X' \cup \{y'\}) \arrow[r, "(\exists y')_{X'}"'] & P(X') \end{tikzcd} \qquad \begin{tikzcd} P(X \cup \{y\}) \arrow[r, "(\forall y)_X"] \arrow[d, "P(\widetilde{\sigma})"'] & P(X) \arrow[d, "P(\sigma)"] \\ P(X' \cup \{y'\}) \arrow[r, "(\forall y')_{X'}"'] & P(X') \end{tikzcd}$$

[a] Intuitively, the Boolean homomorphism $P(X \hookrightarrow X \cup \{y\})\colon P(X) \to P(X \cup \{y\})$ corresponds to adding a dummy variable.

[b] For a Boolean homomorphism between Boolean algebras, the existence of a left adjoint is equivalent to the existence of a right adjoint. Indeed, one can define the right adjoint R from the left one L as $R(x) = \neg L(\neg x)$, and, vice versa, $L(x) = \neg R(\neg x)$.

[c] The commutativity of any of the two diagrams implies the commutativity of the other one.

For a morphism $f\colon X \to Y$ in FinSet, the Boolean homomorphism $P(f)$ is called the ***reindexing*** along f.

Warning 2.2.2 (Quantifiers are not Boolean homomorphisms)**.** The existential and universal quantifiers $(\exists y)_X\colon P(X \cup \{y\}) \to P(X)$ and $(\forall y)_X\colon P(X \cup \{y\}) \to P(X)$ are not, in general, Boolean homomorphisms.

What one can say in general is that they are order-preserving maps. Moreover, the existential preserves 0 and $\vee$ (since it is a left adjoint), and the universal preserves 1 and $\wedge$ (since it is a right adjoint).

Definition 2.2.3 (The Lindenbaum–Tarski first-order Boolean doctrine)**.** Let $\mathcal{T}$ be a theory in a one-sorted relational first-order language $\mathcal{L}$, with no equality. The ***Lindenbaum–Tarski first-order Boolean doctrine of*** $\mathcal{T}$ is the functor

$$\mathrm{LT}_{\mathcal{T}}\colon \mathsf{FinSet} \to \mathsf{BA}$$

defined as follows.
For every finite set X of variables, the Boolean algebra $\mathrm{LT}_{\mathcal{T}}(X)$ is the quotient of the set of formulas $\varphi \in \mathrm{Form}(\mathcal{L})$ such that

$$\mathrm{FV}(\varphi) \subseteq X$$

by semantic equivalence modulo $\mathcal{T}$ in the context X.
For every function $\sigma\colon X \to Y$ of finite sets, the Boolean homomorphism

$$\mathrm{LT}_{\mathcal{T}}(\sigma)\colon \mathrm{LT}_{\mathcal{T}}(X) \to \mathrm{LT}_{\mathcal{T}}(Y)$$

is induced by simultaneous substitution along σ: it sends the equivalence class of a formula φ to the equivalence class of the formula obtained by replacing each free occurrence of a variable $x \in X$ by the variable $\sigma(x) \in Y$, renaming bound variables when necessary in order to avoid variable capture.
Finally, for every finite set X and every variable $y \notin X$, the adjoints

$$(\exists y)_X, (\forall y)_X\colon \mathrm{LT}_{\mathcal{T}}(X \cup \{y\}) \to \mathrm{LT}_{\mathcal{T}}(X)$$

are induced by the usual existential and universal quantification:

$$(\exists y)_X[\varphi] = [\exists y\, \varphi], \qquad (\forall y)_X[\varphi] = [\forall y\, \varphi].$$

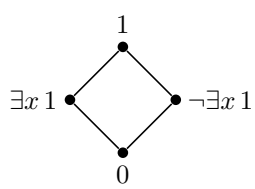
One checks that $\mathrm{LT}_{\mathcal{T}}$ is indeed a first-order Boolean doctrine.

We now examine some concrete examples of Lindenbaum–Tarski first-order Boolean doctrines.

Example 2.2.4 (The Lindenbaum–Tarski doctrine of the empty theory in the empty language)**.** Let $\mathcal{L} = \varnothing$ and $\mathcal{T} = \varnothing$. We describe the first-order Boolean doctrine

$$\mathrm{LT}_{\mathcal{T}}\colon \mathsf{FinSet} \to \mathsf{BA}.$$

A model of $\mathcal{T}$ is simply a set. Hence, in the empty context $\varnothing$, the only information one can express is whether the underlying set is empty or not. It follows that $\mathrm{LT}_{\mathcal{T}}(\varnothing)$ is the four-element Boolean algebra:



(With a little abuse of notation, we wrote formulas in place of their equivalence classes.)

Now let X be a nonempty finite set. Then $\mathrm{LT}_{\mathcal{T}}(X)$ is the two-element Boolean algebra:

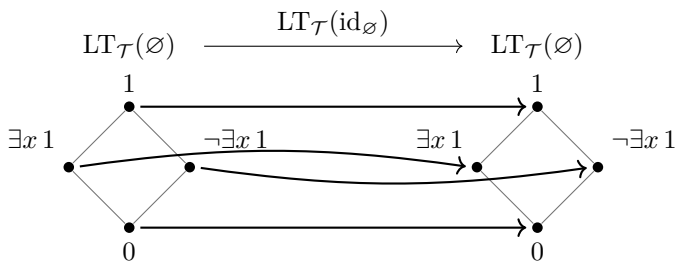
Indeed, if

$$\nu\colon X \to A$$

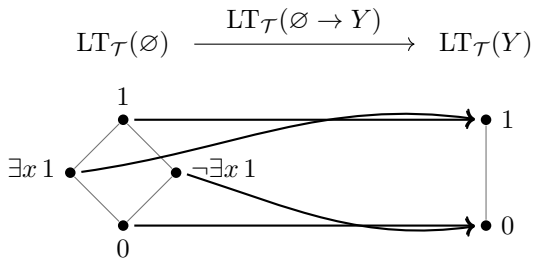
is an assignment, then A must be nonempty. So, in the context X, the formula $\exists x\, 1$ is semantically equivalent to 1, and $\neg\exists x\, 1$ is semantically equivalent to 0. Thus, only the two classes 0 and 1 remain.

Let us now describe the action on morphisms: for a morphism $f\colon X \to Y$ in FinSet, what is the Boolean homomorphism $\mathrm{LT}_{\mathcal{T}}(f)\colon \mathrm{LT}_{\mathcal{T}}(X) \to \mathrm{LT}_{\mathcal{T}}(Y)$? We divide the analysis into three cases.

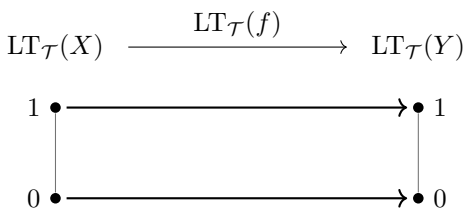
1. (Case $X = \varnothing$, $Y = \varnothing$) $\mathrm{LT}_{\mathcal{T}}(\mathrm{id}_{\varnothing} \colon \varnothing \to \varnothing)$ is the identity on $\mathrm{LT}_{\mathcal{T}}(\varnothing)$:



2. (Case $X = \varnothing$, $Y \neq \varnothing$) $\mathrm{LT}_{\mathcal{T}}(\varnothing \to Y)$ is the following:



3. (Case $X \neq \varnothing$, $Y \neq \varnothing$) In this case, $\mathrm{LT}_{\mathcal{T}}(f)$ is the identity on the two-element Boolean algebra:



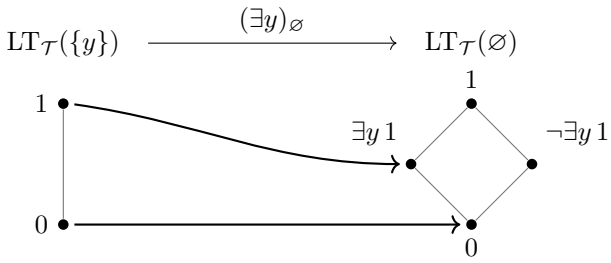
There is no fourth case, since no map $X \to \varnothing$ exists when $X \neq \varnothing$.

Let us also describe the existential quantification maps: given a finite set X and $y \notin X$, what is

$$(\exists y)_X \colon \mathrm{LT}_{\mathcal{T}}(X \cup \{y\}) \to \mathrm{LT}_{\mathcal{T}}(X)?$$

We divide the analysis into two cases.

1. (Case $X = \varnothing$) The map $(\exists y)_{\varnothing} \colon \mathrm{LT}_{\mathcal{T}}(\{y\}) \to \mathrm{LT}_{\mathcal{T}}(\varnothing)$ is the following:



2. (Case $X \neq \varnothing$) In this case, $(\exists y)_X \colon \mathrm{LT}_{\mathcal{T}}(X \cup \{y\}) \to \mathrm{LT}_{\mathcal{T}}(X)$ is the identity map on the two-element Boolean algebra:

$$\begin{tikzcd}
\mathrm{LT}_{\mathcal{T}}(X \cup \{y\}) \arrow[r, "(\exists y)_X"] & \mathrm{LT}_{\mathcal{T}}(X) \\
1 \arrow[r] \arrow[d, dash] & 1 \arrow[d, dash] \\
0 \arrow[r] & 0
\end{tikzcd}$$

Example 2.2.5 (The Lindenbaum–Tarski doctrine of the empty theory in the language with a unary relation symbol)**.** Let $\mathcal{L} = \{R\}$, where R is a unary relation symbol, and let $\mathcal{T} = \varnothing$. A model of $\mathcal{T}$ is a set A together with a subset

$$R^A \subseteq A.$$

We now describe the Boolean algebras $\mathrm{LT}_{\mathcal{T}}(X)$ in a few small contexts.

In the empty context, one can only express whether R^A is empty or nonempty, and whether its complement $A \setminus R^A$ is empty or nonempty. Accordingly, the following four formulas are pairwise incompatible, jointly exhaustive, and each is satisfiable:

$$\neg\exists x\, R(x) \;\wedge\; \neg\exists x\, \neg R(x),$$

$$\neg\exists x\, R(x) \;\wedge\; \exists x\, \neg R(x),$$

$$\exists x\, R(x) \;\wedge\; \neg\exists x\, \neg R(x),$$

$$\exists x\, R(x) \;\wedge\; \exists x\, \neg R(x).$$

In fact, they are the atoms of $\mathrm{LT}_{\mathcal{T}}(\varnothing)$, and $\mathrm{LT}_{\mathcal{T}}(\varnothing)$ has $2^4 = 16$ elements.

Let us now pass to the context $\{x\}$. Here one may ask whether the distinguished element x belongs to R^A or not, and whether the opposite region is empty or nonempty. Accordingly, the following four formulas are again pairwise incompatible, jointly exhaustive, and each is satisfiable:

$$R(x) \wedge \exists y\, \neg R(y),$$

$$R(x) \wedge \neg\exists y\, \neg R(y),$$

$$\neg R(x) \wedge \exists y\, R(y),$$

$$\neg R(x) \wedge \neg\exists y\, R(y).$$

In fact, these are the atoms of $\mathrm{LT}_{\mathcal{T}}(\{x\})$, and again $\mathrm{LT}_{\mathcal{T}}(\{x\})$ has 16 elements.

Note that, since the language has no equality, one cannot express cardinality statements such as "there are at least two distinct elements satisfying R".

In the two-variable context $\{x, y\}$, there is a useful way to organize all possibilities. The set A is partitioned into the two regions

$$R^A \qquad \text{and} \qquad A \setminus R^A,$$

and the variables x and y should be thought of as two distinguished elements of A. Since the language has no equality, one cannot ask whether x and y are equal or distinct. One can only ask:

- in which region x lies;
- in which region y lies;
- and, if x and y belong to the same region, whether the other region is empty or nonempty.

In fact, the atoms of $\mathrm{LT}_{\mathcal{T}}(\{x, y\})$ are exactly the following six formulas:

$$\begin{aligned}
&R(x) \wedge R(y) \wedge \exists z\, \neg R(z), && R(x) \wedge R(y) \wedge \neg\exists z\, \neg R(z),\\
&R(x) \wedge \neg R(y), && \neg R(x) \wedge R(y),\\
&\neg R(x) \wedge \neg R(y) \wedge \exists z\, R(z), && \neg R(x) \wedge \neg R(y) \wedge \neg\exists z\, R(z).
\end{aligned}$$

Therefore $\mathrm{LT}_{\mathcal{T}}(\{x, y\})$ has 6 atoms, and hence it has

$$2^6 = 64$$

elements.

So, the Boolean algebras in larger contexts quickly become richer, even in a very small language.

The next exercises continue the previous two examples. They are meant to help the reader get used to the shape of explicit Lindenbaum–Tarski doctrines. It is not essential to prove every claim in full detail; the main point is to understand why the resulting doctrine has the shape it has.

Exercise 2.2.6 (The empty language with the axiom $\exists x\, 1$)**.** Consider the language $\mathcal{L} = \varnothing$ and the theory $\mathcal{T} = \{\exists x\, 1\}$. What is the Lindenbaum–Tarski first-order Boolean doctrine of $\mathcal{T}$? Describe the functor on objects and morphisms, and describe the existential quantification maps.

Solution. For every finite set X, the Boolean algebra $\mathrm{LT}_{\mathcal{T}}(X)$ is the two-element Boolean algebra. Indeed, all models of $\mathcal{T}$ are nonempty, so in every context the formulas $\exists x\, 1$ and 1 are equivalent, while $\neg\exists x\, 1$ and 0 are equivalent. Hence $\mathrm{LT}_{\mathcal{T}}$ is the constant functor at 2. Every reindexing map is therefore the identity, and every existential quantification map is the identity as well. ■

Exercise 2.2.7 (The empty language with the axiom $\neg\exists x\, 1$)**.** Consider the language $\mathcal{L} = \varnothing$ and the theory $\mathcal{T} = \{\neg\exists x\, 1\}$. What is the Lindenbaum–Tarski first-order Boolean doctrine of $\mathcal{T}$? Describe the functor on objects and morphisms, and describe the existential quantification maps.

Solution. The only model of $\mathcal{T}$ is the empty set. Hence $\mathrm{LT}_{\mathcal{T}}(\varnothing)$ is the two-element Boolean algebra: in the empty context, formulas are identified by their truth value in the unique empty model.

For every nonempty finite set X, there are no assignments $X \to A$ into a model A, so all formulas in context X are equivalent; therefore, $\mathrm{LT}_{\mathcal{T}}(X)$ is the trivial (i.e., one-element) Boolean algebra.

If $f\colon \varnothing \to \varnothing$, then $\mathrm{LT}_{\mathcal{T}}(f)$ is the identity on $\mathrm{LT}_{\mathcal{T}}(\varnothing)$. If $f\colon X \to Y$ with $X \neq \varnothing$ and $Y \neq \varnothing$, then $\mathrm{LT}_{\mathcal{T}}(f)$ is the unique endomap of the trivial Boolean algebra. If $f\colon \varnothing \to Y$ with $Y \neq \varnothing$, then $\mathrm{LT}_{\mathcal{T}}(f)\colon \mathrm{LT}_{\mathcal{T}}(\varnothing) \to \mathrm{LT}_{\mathcal{T}}(Y)$ is the unique map from the two-element Boolean algebra to the trivial Boolean algebra.

If $X \neq \varnothing$ and $y \notin X$, then

$$(\exists y)_X\colon \mathrm{LT}_{\mathcal{T}}(X \cup \{y\}) \to \mathrm{LT}_{\mathcal{T}}(X)$$

is the unique endomap of the trivial Boolean algebra. Finally,

$$(\exists y)_{\varnothing}\colon \mathrm{LT}_{\mathcal{T}}(\{y\}) \to \mathrm{LT}_{\mathcal{T}}(\varnothing)$$

sends the unique element of the trivial Boolean algebra to 0, because in the unique model of $\mathcal{T}$ there is no element witnessing an existential statement. ■

Exercise 2.2.8 (An inconsistent theory in the empty language)**.** Consider the language $\mathcal{L} = \varnothing$ and the theory $\mathcal{T}$ consisting of the single element 0 (the falsum). What is the Lindenbaum–Tarski first-order Boolean doctrine of $\mathcal{T}$? Describe the functor on objects and morphisms, and describe the existential quantification maps.

Solution. The theory $\mathcal{T}$ has no models. Therefore, for every finite set X, every two formulas in context X are semantically equivalent modulo $\mathcal{T}$, and hence $\mathrm{LT}_{\mathcal{T}}(X)$ is the trivial Boolean algebra. Every reindexing map is the unique map between trivial Boolean algebras, so $\mathrm{LT}_{\mathcal{T}}$ is the constant functor at the trivial Boolean algebra. The

existential quantification maps are again the unique maps between trivial Boolean algebras. ■

The same argument works for any language $\mathcal{L}$ and any inconsistent theory $\mathcal{T}$: if $\mathcal{T}$ has no models, then $\mathrm{LT}_{\mathcal{T}}$ is the constant doctrine at the trivial Boolean algebra.

Exercise 2.2.9 (A nullary relation symbol over nonempty structures)**.** Consider the language $\mathcal{L} = \{R\}$, where $\mathrm{ar}(R) = 0$, and the theory $\mathcal{T} = \{\exists x\, 1\}$. What is the Lindenbaum–Tarski first-order Boolean doctrine of $\mathcal{T}$? Describe the functor on objects and morphisms, and describe the existential quantification maps.

Solution. Since all models of $\mathcal{T}$ are nonempty, quantification over variables adds no new information, and the only genuine atomic datum is the truth value of the nullary predicate R. Hence, for every finite set X, every formula is equivalent modulo $\mathcal{T}$ to one of

$$0, \qquad R, \qquad \neg R, \qquad 1.$$

Therefore, $\mathrm{LT}_{\mathcal{T}}(X)$ is the four-element Boolean algebra for every finite set X. Then, $\mathrm{LT}_{\mathcal{T}}$ is the constant functor at the four-element Boolean algebra. Every reindexing map and every existential quantification map is the identity. ■

Exercise 2.2.10 (A nullary relation symbol over empty structures)**.** Consider the language $\mathcal{L} = \{R\}$, where $\mathrm{ar}(R) = 0$, and the theory $\mathcal{T} = \{\neg\exists x\, 1\}$. What is the Lindenbaum–Tarski first-order Boolean doctrine of $\mathcal{T}$? Describe the functor on objects and morphisms, and describe the existential quantification maps.

Solution. The only models of $\mathcal{T}$ are empty structures, and in such a structure the only genuine datum is the truth value of the nullary predicate R. Hence $\mathrm{LT}_{\mathcal{T}}(\varnothing)$ is the four-element Boolean algebra, with representatives

$$0, \qquad R, \qquad \neg R, \qquad 1.$$

For every nonempty finite set X, there are no assignments $X \to A$ into a model $A \models \mathcal{T}$, so all formulas in context X are equivalent. Therefore $\mathrm{LT}_{\mathcal{T}}(X)$ is the trivial Boolean algebra.

If $f\colon \varnothing \to \varnothing$, then $\mathrm{LT}_{\mathcal{T}}(f)$ is the identity on $\mathrm{LT}_{\mathcal{T}}(\varnothing)$. If $f\colon X \to Y$ with $X \neq \varnothing$ and $Y \neq \varnothing$, then $\mathrm{LT}_{\mathcal{T}}(f)$ is the unique endomap of the trivial Boolean algebra. If $f\colon \varnothing \to Y$ with $Y \neq \varnothing$, then $\mathrm{LT}_{\mathcal{T}}(f)\colon \mathrm{LT}_{\mathcal{T}}(\varnothing) \to \mathrm{LT}_{\mathcal{T}}(Y)$ is the unique map from the four-element Boolean algebra to the trivial Boolean algebra.

If $X \neq \varnothing$ and $y \notin X$, then $(\exists y)_X$ is the unique endomap of the trivial Boolean algebra. Finally,

$$(\exists y)_{\varnothing}\colon \mathrm{LT}_{\mathcal{T}}(\{y\}) \to \mathrm{LT}_{\mathcal{T}}(\varnothing)$$

sends the unique element of the trivial Boolean algebra to 0, since in an empty model no existential statement can be true. ■

Remark 2.2.11 (Constant doctrines from propositional theories)**.** Let $\mathcal{L}$ be a propositional language, and regard it as a first-order relational language by declaring every symbol of $\mathcal{L}$ to have arity 0. Let $\mathcal{T}$ be a propositional theory in the language $\mathcal{L}$, and define

$$\mathcal{T}' \coloneqq \mathcal{T} \cup \{\exists x\, 1\}.$$

Let B be the Lindenbaum–Tarski algebra of the propositional theory $\mathcal{T}$. For every finite set X, there is a canonical isomorphism

$$\mathrm{LT}_{\mathcal{T}'}(X) \cong B.$$

Moreover, under these isomorphisms, every reindexing map and every existential quantification map is the identity. Thus, $\mathrm{LT}_{\mathcal{T}'}$ is the constant first-order Boolean doctrine at B. This provides a large supply of concrete examples of constant first-order Boolean doctrines.

The next exercises illustrate a general principle: if the theory forces a relation symbol to carry no genuine information, then adding that symbol does not change the resulting Lindenbaum–Tarski doctrine.

Exercise 2.2.12 (A nullary relation symbol forced to be true)**.** Let $\mathcal{L} = \{R\}$, where R is a nullary relation symbol, and let $\mathcal{T} = \{R\}$. What is the Lindenbaum–Tarski first-order Boolean doctrine of $\mathcal{T}$? Describe the functor on objects and morphisms, and describe the existential quantification maps.

Solution. For every formula ϕ in the language $\mathcal{L}$, let ϕ^1 be the formula in the empty language obtained by replacing each occurrence of R by 1. For every finite set X and every formula ϕ with $\mathrm{FV}(\phi) \subseteq X$, the formulas ϕ and ϕ^1 are semantically equivalent modulo $\mathcal{T}$ in the context X. Therefore, up to the translation $R \mapsto 1$, the doctrine $\mathrm{LT}_{\mathcal{T}}$ is isomorphic to the doctrine of Example 2.2.4. In particular, adding a nullary symbol and then imposing that it is always true gives no new information. In particular, this means that $\mathrm{LT}_{\mathcal{T}}(\varnothing)$ is the four-element Boolean algebra and that $\mathrm{LT}_{\mathcal{T}}(X)$ is the two-element Boolean algebra for every nonempty finite set X, exactly as in Example 2.2.4. ■

Arguments analogous to the one in the solution of Exercise 2.2.12 also apply to the theory $\{\neg R\}$ in the same language, where every occurrence of R is replaced by 0, and to the theory $\{\forall x\, S(x)\}$ in the language $\{S\}$ (with S unary), where every occurrence of $S(z)$ is replaced by 1; also in these two cases, the resulting doctrine is the same of the one in Example 2.2.4.

Remark 2.2.13. Let $P\colon \mathsf{FinSet} \to \mathsf{BA}$ be a first-order Boolean doctrine. If $i\colon X \hookrightarrow Y$ is an injective function between finite sets, <u>and $X \neq \varnothing$</u>, then $P(i)\colon P(X) \to P(Y)$ is injective (see the footnote for a proof[2]). So, once the context is nonempty, passing to a larger context cannot identify two distinct elements.

Remark 2.2.14. The use of FinSet reflects the fact that, in this chapter, we are only considering <u>one-sorted purely relational</u> first-order logic. If you are curious about how to include function symbols and multiple sorts, read the footnote[3]. If one wants to consider equality, one requires a further condition (sometimes called *elementarity*) that we do not develop here.

The following exercise gives a first calculation with the axioms of a first-order Boolean doctrine. It recovers, on the doctrine side, the phenomenon already seen semantically in Example 2.1.18: although $\exists x\, 1$ need not be true in the empty context, it becomes true in the context $\{y\}$: once y is available as a generic element, one can witness the existential statement by taking x to be y.

Exercise 2.2.15 ($\exists x\, 1$ becomes true in the context $\{y\}$)**.** (We have done this exercise in class.) Let $P\colon \mathsf{FinSet} \to \mathsf{BA}$ be a first-order Boolean doctrine, and let x and y be distinct variables. Show that, in $P(\{y\})$, we have

$$(\exists x)_{\{y\}}\, 1 = 1. \tag{2.2}$$

Consequently, in $P(\varnothing)$, we have

$$(\forall y)_{\varnothing}\big((\exists x)_{\{y\}}\, 1\big) = 1. \tag{2.3}$$

Solution. Since 1 is the greatest element of $P(\{y\})$, to prove the first claim (eq. (2.2)) it is enough to prove

$$1 \leq (\exists x)_{\{y\}}\, 1. \tag{2.4}$$

Let $\sigma\colon \varnothing \to \{y\}$ be the unique function, and let $\widetilde{\sigma}\colon \{y\} \to \{x, y\}$ be the extension of σ that maps y to x. We apply the Beck–Chevalley property to σ, with y as the quantified variable on the top row and x as the quantified variable

[2]Since i is injective, there is a map

$$g\colon Y \to X$$

such that the composite $X \xrightarrow{i} Y \xrightarrow{g} X$ is the identity: indeed, pick any $x_0 \in X$, which exists since $X \neq \varnothing$, and define $g\colon Y \to X$ as the function that is the inverse of i on $i[X]$ and that sends every element of $Y \setminus i[X]$ to x_0. Applying the functor P, we obtain

$$P(g) \circ P(i) = P(g \circ i) = P(\mathrm{id}_X) = \mathrm{id}_{P(X)}.$$

Hence, $P(i)$ has a left inverse, and is therefore injective.

[3]To simultaneously allow function symbols and multiple sorts, an elegant way is to use *algebraic theories*, i.e. categories with finite products (no need for them to be small, here). A *first-order Boolean doctrine over a category* C *with finite products* is a functor $\mathsf{C} \to \mathsf{BA}$ satisfying certain conditions. The definition of first-order Boolean doctrine that we saw in Definition 2.2.1 is the special case $\mathsf{C} = \mathsf{FinSet}^{\mathrm{op}}$. Roughly speaking, the category C plays the role of the algebraic theory of the language under consideration.

on the bottom row.

$$\begin{tikzcd} P(\{y\}) \arrow[r, "(\exists y)_\varnothing"] \arrow[d, "P(\widetilde{\sigma})"'] & P(\varnothing) \arrow[d, "P(\sigma)"] \\ P(\{x,y\}) \arrow[r, "(\exists x)_{\{y\}}"'] & P(\{y\}). \end{tikzcd}$$

Applying the commutativity of the diagram to the element $1 \in P(\{y\})$, we obtain

$$P(\sigma)\big((\exists y)_\varnothing\, 1\big) = (\exists x)_{\{y\}}\big(P(\widetilde{\sigma})(1)\big).$$

Since $P(\widetilde{\sigma})$ is a Boolean homomorphism, it preserves 1, and so $P(\widetilde{\sigma})(1) = 1$. Therefore,

$$P(\sigma)\big((\exists y)_\varnothing\, 1\big) = (\exists x)_{\{y\}}\, 1.$$

So, to prove (2.4), it remains to show that

$$1 \leq P(\sigma)\big((\exists y)_\varnothing\, 1\big).$$

By the adjunction $(\exists y)_\varnothing \dashv P(\sigma)$, this is equivalent to $(\exists y)_\varnothing\, 1 \leq (\exists y)_\varnothing\, 1$, which is tautologically true. This proves the first claim.

For the second claim (eq. (2.3)), by (2.2), it is enough to prove that

$$(\forall y)_\varnothing\, 1 = 1.$$

Since 1 is the greatest element of $P(\varnothing)$, it is enough to prove

$$1 \leq (\forall y)_\varnothing\, 1.$$

By the adjunction $P(\sigma) \dashv (\forall y)_\varnothing$, the inequality $1 \leq (\forall y)_\varnothing\, 1$ is equivalent to $P(\sigma)(1) \leq 1$, which holds because $P(\sigma)(1) = 1$. Hence, $(\forall y)_\varnothing\, 1 = 1$, and therefore

$$(\forall y)_\varnothing\big((\exists x)_{\{y\}}\, 1\big) = (\forall y)_\varnothing\, 1 = 1.$$ ■

2.3 Models and morphisms of first-order Boolean doctrines

2.3.1 Models

In the doctrine setting, the elements of $P(X)$ are abstract formulas in the context X: one no longer distinguishes atomic formulas from the formulas built from them. Accordingly, a model of P must interpret each $\varphi \in P(X)$ as an X-ary relation on an underlying set M, that is, as a subset of M^X.

Definition 2.3.1 (Model of a first-order Boolean doctrine)**.** Let $P\colon \mathsf{FinSet} \to \mathsf{BA}$ be a first-order Boolean doctrine. A ***model of*** P consists of a set M together with, for every finite set X and every $\varphi \in P(X)$, a subset, denoted by φ^M, of M^X (= the set of functions from X to M) with the following properties.

1. (Boolean connectives) For every finite set X,
   (a) for every $\varphi, \psi \in P(X)$, $(\varphi \vee \psi)^M = \varphi^M \cup \psi^M$ and $(\varphi \wedge \psi)^M = \varphi^M \cap \psi^M$,
   (b) for every $\varphi \in P(X)$, $(\neg\varphi)^M = M^X \setminus \varphi^M$,
   (c) $(0)^M = \varnothing$, $(1)^M = M^X$.
2. (Substitutions) For every morphism $\sigma\colon X \to Y$ in FinSet and every $\varphi \in P(X)$,

$$(P(\sigma)(\varphi))^M = \{\, \nu\colon Y \to M \mid \nu \circ \sigma \in \varphi^M \,\}.$$

3. (Quantifiers) For every finite set X, every $y \notin X$, and every $\varphi \in P(X \cup \{y\})$,

$$((\exists y)_X \varphi)^M = \{\, \nu \colon X \to M \mid \text{there is } a \in M \text{ such that } \nu_a \in \varphi^M \,\},$$
$$((\forall y)_X \varphi)^M = \{\, \nu \colon X \to M \mid \text{for every } a \in M,\ \nu_a \in \varphi^M \,\},$$

where $\nu_a \colon X \cup \{y\} \to M$ is the function given by $\nu_a|_X = \nu$ and $\nu_a(y) = a$.

Example 2.3.2 (Ordinary models induce models of the Lindenbaum–Tarski doctrine)**.** Let $\mathcal{T}$ be a theory in a relational first-order language $\mathcal{L}$, and let A be a model of $\mathcal{T}$ in the ordinary sense. Then A defines a model of the doctrine $\mathrm{LT}_{\mathcal{T}}$ by setting, for every finite set X and every class $[\varphi] \in \mathrm{LT}_{\mathcal{T}}(X)$,

$$[\varphi]^A = \{\, \nu \colon X \to A \mid A, \nu \models \varphi \,\}.$$

This is well-defined: if $[\varphi] = [\psi]$ in $\mathrm{LT}_{\mathcal{T}}(X)$, then φ and ψ are semantically equivalent modulo $\mathcal{T}$ in the context X, and hence

$$A, \nu \models \varphi \qquad \Longleftrightarrow \qquad A, \nu \models \psi$$

for every assignment $\nu : X \to A$. The conditions in Definition 2.3.1 are then exactly the usual semantic clauses for Boolean connectives, substitutions, and quantifiers.

Notation 2.3.3 (Satisfaction of sentences in doctrine models)**.** Let P be a first-order Boolean doctrine, let M be a model of P, and let $\varphi \in P(\varnothing)$. Then φ^M is a subset of $M^{\varnothing}$. But $M^{\varnothing}$ is a singleton, since there is exactly one function $\varnothing \to M$. So φ^M is either empty or the whole singleton $M^{\varnothing}$. Accordingly, we write

$$M \models \varphi$$

if $\varphi^M = M^{\varnothing}$.

2.3.2 Morphisms

The previous definition of model admits a useful categorical reformulation. To do so, we first define morphisms between first-order Boolean doctrines.

Definition 2.3.4 (Morphism of first-order Boolean doctrines)**.** Let $P_1, P_2 \colon \mathsf{FinSet} \to \mathsf{BA}$ be first-order Boolean doctrines. A ***morphism of first-order Boolean doctrines*** from P_1 to P_2 is a natural transformation [a] $\mathfrak{m} \colon P_1 \to P_2$ such that, for every finite set X and every $y \notin X$, writing the quantifiers in P_i as

$$(\exists y)^i_X, (\forall y)^i_X \colon P_i(X \cup \{y\}) \to P_i(X) \qquad (i = 1, 2),$$

the following diagrams commute[b]:

$$\begin{tikzcd} P_1(X \cup \{y\}) \arrow[r, "(\exists y)^1_X"] \arrow[d, "\mathfrak{m}_{X \cup \{y\}}"'] & P_1(X) \arrow[d, "\mathfrak{m}_X"] \\ P_2(X \cup \{y\}) \arrow[r, "(\exists y)^2_X"'] & P_2(X) \end{tikzcd} \qquad \begin{tikzcd} P_1(X \cup \{y\}) \arrow[r, "(\forall y)^1_X"] \arrow[d, "\mathfrak{m}_{X \cup \{y\}}"'] & P_1(X) \arrow[d, "\mathfrak{m}_X"] \\ P_2(X \cup \{y\}) \arrow[r, "(\forall y)^2_X"'] & P_2(X) \end{tikzcd}$$

[a] Recall that this consists of a family of Boolean homomorphisms

$$\mathfrak{m}_X \colon P_1(X) \to P_2(X) \qquad (X \in \mathsf{FinSet})$$

such that, for every function $\sigma \colon X \to Y$ in FinSet, we have $P_2(\sigma) \circ \mathfrak{m}_X = \mathfrak{m}_Y \circ P_1(\sigma)$.

[b] The commutativity of one diagram implies the commutativity of the other one. This follows from the facts that $\forall = \neg\exists\neg$ and $\exists = \neg\forall\neg$.

In other words, a morphism of first-order Boolean doctrines is a natural transformation that commutes with existential and universal quantification.

2.3.3 Models as morphisms to the powerset doctrine

Definition 2.3.5 (The powerset doctrine on a set)**.** Let M be a set. The ***powerset doctrine on*** M is the functor

$$\mathcal{P}_M \colon \mathsf{FinSet} \to \mathsf{BA}$$

defined as follows. For every finite set X, let

$$\mathcal{P}_M(X) \coloneqq \mathcal{P}(M^X).$$

For every function $\sigma \colon X \to Y$, define

$$\mathcal{P}_M(\sigma) \colon \mathcal{P}(M^X) \to \mathcal{P}(M^Y)$$

by

$$\mathcal{P}_M(\sigma)(A) \coloneqq \{\nu \colon Y \to M \mid \nu \circ \sigma \in A\}.$$

Finally, for every finite set X and every $y \notin X$, define

$$(\exists y)_X, (\forall y)_X \colon \mathcal{P}(M^{X \cup \{y\}}) \to \mathcal{P}(M^X)$$

by

$$(\exists y)_X(A) = \{\nu \colon X \to M \mid \text{there is } a \in M \text{ such that } \nu_a \in A\},$$
$$(\forall y)_X(A) = \{\nu \colon X \to M \mid \text{for every } a \in M,\ \nu_a \in A\},$$

where

$$\nu_a \colon X \cup \{y\} \to M$$

is the function given by

$$\nu_a{\restriction}_X = \nu \qquad \text{and} \qquad \nu_a(y) = a.$$

One checks that $\mathcal{P}_M$ is a first-order Boolean doctrine.[a]

[a] As a functor, $\mathcal{P}_M$ is the composite of the representable functor $\hom(-, M) \colon \mathsf{FinSet} \to \mathsf{Set}^{\mathrm{op}}$ and the powerset functor $\mathcal{P} \colon \mathsf{Set}^{\mathrm{op}} \to \mathsf{BA}$.

Remark 2.3.6 (Models as morphisms)**.** Let $P \colon \mathsf{FinSet} \to \mathsf{BA}$ be a first-order Boolean doctrine and let M be a set. Giving a model of P with underlying set M in the sense of Definition 2.3.1 is equivalent to giving a morphism of first-order Boolean doctrines from P to $\mathcal{P}_M$.[4]

Remark 2.3.7 (First-order Boolean doctrines as a variety of many-sorted algebras)**.** First-order Boolean doctrines can be presented as a variety of many-sorted algebras. We have one sort for each finite set X, together with the Boolean operations in each sort, the reindexings along functions $\sigma \colon X \to Y$ as unary function symbols from the sort X to the sort Y, and the quantifiers $(\exists y)_X$ and $(\forall y)_X$ as unary function symbols from the sort $X \cup \{y\}$ to the sort X. All axioms can be expressed equationally (remember that the adjunctions can be expressed by the triangle identities, which are equations). The notion of morphism in Definition 2.3.4 is the expected notion of homomorphism between such structures.

[4] We sketch why this is true. Given such a morphism $\mathfrak{m}$, one defines

$$\varphi^M \coloneqq \mathfrak{m}_X(\varphi) \subseteq M^X.$$

Conversely, given a model of P in the sense of Definition 2.3.1, one defines

$$\mathfrak{m}_X(\varphi) \coloneqq \varphi^M.$$

The condition that $\mathfrak{m}_X$ is a Boolean homomorphism corresponds to the clause (1) of Definition 2.3.1 holds. The naturality of $\mathfrak{m}$ in Definition 2.3.4 corresponds to the clause (2) of Definition 2.3.1. The commutative of $\mathfrak{m}$ with the quantifiers in Definition 2.3.4 corresponds to the clause (3) of Definition 2.3.1.

2.4 Gödel's completeness theorem

We shall use the following fact (Gödel's completeness theorem for first-order Boolean doctrines) as a black box, meaning that we skip its (nontrivial!) proof (which uses the axiom of choice). In the logical context, where one refers to a logical calculus (which we have not seen), it is presented in one of the following (easily seen to be equivalent) forms:

1. Every theory that does not prove falsum has a model.
2. If every model of a theory $\mathcal{T}$ satisfies a sentence φ, then φ is provable from $\mathcal{T}$.

These statements are important because they say that the proof system is exactly strong enough to capture semantic consequence: whenever falsum is not derivable, there is a model witnessing this, and whenever a sentence holds in every model, it is already derivable.

For a doctrine P, the condition $0 \neq 1$ in $P(\varnothing)$ is precisely the algebraic form of saying that $\mathcal{T}$ does not prove falsum; Thus, Theorem 2.4.1 below is the doctrine-theoretic version of (1). Likewise, in that same setting, the equality $\varphi = 1$ in $P(\varnothing)$ means that φ is provable from $\mathcal{T}$, so Corollary 2.4.8 below is the doctrine-theoretic version of (2).

Theorem 2.4.1 (Gödel's completeness theorem, for first-order Boolean doctrines)**.** *If $P\colon \mathsf{FinSet} \to \mathsf{BA}$ is a first-order Boolean doctrine and $0 \neq 1$ in $P(\varnothing)$, then P has a model.*

Remark 2.4.2. The condition $0 \neq 1$ in $P(\varnothing)$ is necessary. Indeed, if M is a model of P, then $0^M = \varnothing$ and $1^M = M^{\varnothing}$, and $M^{\varnothing}$ is a singleton.

We now explain a construction that allows one to derive from it a first-order analogue of the Boolean Prime Ideal Theorem.

Definition 2.4.3 (Filter quotient construction)**.** Let $P\colon \mathsf{FinSet} \to \mathsf{BA}$ be a first-order Boolean doctrine, and let G be a filter of $P(\varnothing)$. For every finite set X, let

$$!_X \colon \varnothing \to X$$

be the unique function. Define a relation $\sim_G$ on $P(X)$ by

$$\varphi \sim_G \psi \quad :\Longleftrightarrow \quad \text{there is } \theta \in G \text{ such that } P(!_X)(\theta) \leq (\varphi \leftrightarrow \psi),$$

where

$$\varphi \leftrightarrow \psi \;\coloneqq\; (\neg\varphi \vee \psi) \wedge (\neg\psi \vee \varphi).$$

For every finite set X, write

$$(P/G)(X) \coloneqq P(X)/{\sim_G}.$$

Remark 2.4.4. The reason why we require G to live in $P(\varnothing)$ is that only sentences can be reindexed canonically into every context, namely along the maps

$$!_X \colon \varnothing \to X.$$

Moreover, every $\theta \in G$ becomes equal to 1 in $(P/G)(\varnothing)$, since

$$\theta \leq \theta \leftrightarrow 1.$$

So the quotient P/G may be thought of as the doctrine obtained from P by forcing all sentences in G to hold.

Proposition 2.4.5 (The filter quotient construction defines a first-order Boolean doctrine)**.** *For every function $\sigma\colon X \to Y$, the assignment*

$$(P/G)(\sigma)([\varphi]) \coloneqq [P(\sigma)(\varphi)]$$

is well-defined; and for every finite set X and every $y \notin X$, the assignments

$$(\exists y)_X([\psi]) \coloneqq [(\exists y)_X \psi], \qquad (\forall y)_X([\psi]) \coloneqq [(\forall y)_X \psi]$$

are well-defined.
Together with the quotient Boolean algebra structure on each $(P/G)(X)$, these operations make P/G into a first-order Boolean doctrine.
Moreover, the quotient maps

$$q_X\colon P(X) \to (P/G)(X), \qquad q_X(\varphi) \coloneqq [\varphi],$$

define a morphism of first-order Boolean doctrines

$$q\colon P \to P/G.$$

Proof. We omit the proof. ■

Corollary 2.4.6 (Filter/ideal form of Gödel's completeness theorem)**.** *Let $P\colon \mathsf{FinSet} \to \mathsf{BA}$ be a first-order Boolean doctrine. Let F be a filter of $P(\varnothing)$ and let I be an ideal of $P(\varnothing)$. If*

$$F \cap I = \varnothing,$$

then there is a model M of P such that

$$M \models \varphi \qquad \text{for every } \varphi \in F,$$

and

$$M \not\models \psi \qquad \text{for every } \psi \in I.$$

Proof. Let G be the filter generated by $F \cup \{\, \neg\psi \mid \psi \in I \,\}$; i.e.,[a]

$$G = \uparrow\{\, \varphi \wedge \neg\psi \mid \varphi \in F,\ \psi \in I \,\}.$$

We claim that $0 \notin G$. So, let us assume $0 \in G$, and let us reach a contradiction. Since $0 \in G$, there is $\varphi \in F$ and $\psi \in I$ such that

$$\varphi \wedge \neg\psi \leq 0.$$

Equivalently,

$$\varphi \leq \psi.$$

Since $\varphi \in F$ and F is upward closed, the element ψ belongs to F; since it also belongs to I, we reach a contradiction with $F \cap I = \varnothing$. So, G is proper.

By Proposition 2.4.5, the quotient doctrine P/G is a first-order Boolean doctrine. Since G is proper, we have

$$0 \neq 1 \qquad \text{in } (P/G)(\varnothing).$$

Indeed, if $[0] = [1]$, then there would be $\theta \in G$ such that

$$\theta \leq 0 \leftrightarrow 1 = 0,$$

hence $\theta = 0$, contradicting the fact that G is proper.

By Theorem 2.4.1 (Gödel's completeness theorem, for first-order Boolean doctrines), P/G has a model. Let

$$q\colon P \to P/G$$

be the quotient morphism, given by

$$q_X(\varphi) \coloneqq [\varphi].$$

Viewing models as morphisms, we may compose q with a model of P/G, and thus obtain a model M of P.

Every element of G holds in M. Indeed, if $\theta \in G$, then $[\theta] = [1]$ in $(P/G)(\varnothing)$, so every model of P/G satisfies $[\theta]$, and hence the induced model M of P satisfies θ. In particular,

$$M \models \varphi \qquad \text{for every } \varphi \in F,$$

and

$$M \not\models \psi \qquad \text{for every } \psi \in I.$$

■

[a] In fact, by distributivity, the upward closure "↑" is not necessary in the following display.

Remark 2.4.7. The previous corollary can be seen as a first-order analogue of the Boolean Prime Ideal Theorem (Theorem 1.3.17).

Corollary 2.4.8 (True in all models implies provable)**.** *Let $P\colon \mathsf{FinSet} \to \mathsf{BA}$ be a first-order Boolean doctrine, and let $\varphi \in P(\varnothing)$. If for every model M of P we have $M \models \varphi$, then $\varphi = 1$.*

Proof. The contrapositive is obtained by applying the filter/ideal form of Gödel's completeness theorem (Corollary 2.4.6) to the ideal $\downarrow\varphi$ and the filter $\{1\}$. ■

2.5 Elementary equivalence and types

Definition 2.5.1 (Elementary equivalence and types)**.** Let $P\colon \mathsf{FinSet} \to \mathsf{BA}$ be a first-order Boolean doctrine, and let M, N be models of P. We say that M and N are ***elementarily equivalent***, and write

$$M \equiv N,$$

if, for every $\varphi \in P(\varnothing)$,

$$M \models \varphi \qquad \Longleftrightarrow \qquad N \models \varphi.$$

A ***type*** of P is an equivalence class of models of P under this relation; the type of a model M is denoted by $[M]$. We write

$$\mathrm{Mod}(P)$$

for the class of models of P, and

$$\mathrm{Typ}(P) \coloneqq \mathrm{Mod}(P)/{\equiv}$$

for the quotient of $\mathrm{Mod}(P)$ by elementary equivalence.

Remark 2.5.2. A type records exactly the information about a model that can be expressed by sentences, i.e., by elements of $P(\varnothing)$. In particular, two models define the same type if and only if they are indistinguishable by any sentence of the doctrine.

Although $\mathrm{Mod}(P)$ is in general a proper class, the quotient $\mathrm{Typ}(P)$ is a set, since each type is completely determined by the subset of the set $P(\varnothing)$ consisting of the sentences it satisfies.

Theorem 2.5.3 (Types $\overset{1:1}{\leftrightarrow}$ ultrafilters)**.** *Let $P\colon \mathsf{FinSet} \to \mathsf{BA}$ be a first-order Boolean doctrine. There is a bijection*

$$\begin{aligned} \Gamma\colon \mathrm{Typ}(P) &\longrightarrow \mathrm{Ult}(P(\varnothing)) \\ [M] &\longmapsto \{\, \varphi \in P(\varnothing) \mid M \models \varphi \,\}. \end{aligned}$$

Proof. First, Γ is well-defined. If $[M] = [N]$, then $M \equiv N$, and hence

$$M \models \varphi \qquad \Longleftrightarrow \qquad N \models \varphi$$

for every $\varphi \in P(\varnothing)$. So $\Gamma([M]) = \Gamma([N])$.

For every model M of P, $\Gamma([M])$ is indeed an ultrafilter by the definition of $\models$ (Notation 2.3.3) and the defining properties of models relatively to Boolean operations (Definition 2.3.1). So Γ indeed lands in $\mathrm{Ult}(P(\varnothing))$.

Injectivity is immediate from the definition of elementary equivalence.

For surjectivity, let $U \in \mathrm{Ult}(P(\varnothing))$. Since U is a filter and $P(\varnothing) \setminus U$ is an ideal, the filter/ideal form of Gödel's completeness theorem (Corollary 2.4.6) yields a model M of P such that

$$M \models \varphi \qquad \text{for every } \varphi \in U,$$

and

$$M \not\models \psi \qquad \text{for every } \psi \in P(\varnothing) \setminus U.$$

Hence,

$$\Gamma([M]) = U.$$

■

Via this bijection, $\mathrm{Typ}(P)$ inherits a Stone topology from $\mathrm{Ult}(P(\varnothing))$. This can be defined directly as follows.

Definition 2.5.4 (The space of types)**.** Let $P\colon \mathsf{FinSet} \to \mathsf{BA}$ be a first-order Boolean doctrine. We equip $\mathrm{Typ}(P)$ with the topology that has as a basis the sets

$$[\![\varphi]\!] \;\coloneqq\; \{\, [M] \in \mathrm{Typ}(P) \mid M \models \varphi \,\}, \qquad\qquad \varphi \in P(\varnothing).$$

These subsets are well-defined, since satisfaction of sentences depends only on the type of the model, and they do form a basis, since they are closed under finite intersections[a].

[a] Indeed, $[\![\varphi]\!] \cap [\![\psi]\!] = [\![\varphi \wedge \psi]\!]$ and $[\![1]\!] = \mathrm{Typ}(P)$.

Proposition 2.5.5 (The space of types is the Stone dual of the algebra of sentences)**.** *Let $P\colon \mathsf{FinSet} \to \mathsf{BA}$ be a first-order Boolean doctrine. The bijection Γ of Theorem 2.5.3 is a homeomorphism, where $\mathrm{Ult}(P(\varnothing))$ carries the Stone topology. Consequently, $\mathrm{Typ}(P)$ is a Stone space, whose clopens are precisely the sets of the form*

$$[\![\varphi]\!] \;=\; \{\, [M] \in \mathrm{Typ}(P) \mid M \models \varphi \,\}, \qquad\qquad \varphi \in P(\varnothing).$$

Proof. For every $\varphi \in P(\varnothing)$,

$$\begin{aligned} \Gamma^{-1}\big[\eta_{P(\varnothing)}(\varphi)\big] &= \{\, [M] \in \mathrm{Typ}(P) \mid \Gamma([M]) \in \eta_{P(\varnothing)}(\varphi) \,\} \\ &= \{\, [M] \in \mathrm{Typ}(P) \mid \varphi \in \Gamma([M]) \,\} \\ &= \{\, [M] \in \mathrm{Typ}(P) \mid M \models \varphi \,\} \\ &= [\![\varphi]\!]. \end{aligned}$$

So, along the bijection Γ, Γ maps the basis

$$\{\, [\![\varphi]\!] \mid \varphi \in P(\varnothing) \,\}$$

of $\mathrm{Typ}(P)$ bijectively onto the basis

$$\{\, \eta_{P(\varnothing)}(\varphi) \mid \varphi \in P(\varnothing) \,\}$$

of $\mathrm{Ult}(P(\varnothing))$, and is therefore a homeomorphism. The final claim follows from the fact that the space of ultrafilters is a Stone space (Theorem 1.5.7) and the characterization of clopens in the space of ultrafilters (Lemma 1.5.8). ■

Remark 2.5.6. In other words, the space of types of P is the Stone dual of the Boolean algebra of sentences $P(\varnothing)$. The nontrivial point is surjectivity of Γ, which is exactly where the filter/ideal form of Gödel's completeness theorem (Corollary 2.4.6) is used.

Example 2.5.7 (Types in the language with a unary relation symbol)**.** Let $\mathcal{L} = \{R\}$, where R is a unary relation symbol, and let $\mathcal{T} = \varnothing$. Set

$$P \coloneqq \mathrm{LT}_{\mathcal{T}}.$$

Concretely, the four types of P may be represented, for instance, by the following models of $\mathcal{T}$:

1. the empty model, corresponding to $\neg\exists x\, R(x) \wedge \neg\exists x\, \neg R(x)$;
2. a singleton model in which R holds, corresponding to $\exists x\, R(x) \wedge \neg\exists x\, \neg R(x)$;
3. a singleton model in which R does not hold, corresponding to $\neg\exists x\, R(x) \wedge \exists x\, \neg R(x)$;
4. a two-element model in which one element satisfies R and one does not, corresponding to $\exists x\, R(x) \wedge \exists x\, \neg R(x)$.

These four correspond bijectively to the elements of $\mathrm{Typ}(P)$.[5] Its power set (i.e., the set of clopens) corresponds bijectively to the elements $\mathrm{LT}_{\mathcal{T}}(\varnothing)$.

2.6 Pointed models and types in context

We now generalize the previous discussion from the empty context to an arbitrary finite context. Since an element of $P(X)$ should be thought of as a formula in the variables X, the relevant semantic object is not just a model, but a model together with an assignment of the variables in X.

Definition 2.6.1 (Pointed models and types in a context)**.** Let $P\colon \mathsf{FinSet} \to \mathsf{BA}$ be a first-order Boolean doctrine, and let X be a finite set. An ***X-pointed model of*** P is a pair (M, ν) consisting of a model M of P and a function $\nu\colon X \to M$. If (M, ν) is an X-pointed model of P, and $\varphi \in P(X)$, we write

$$M, \nu \models \varphi$$

if

$$\nu \in \varphi^M.$$

Two X-pointed models (M, ν) and (N, μ) are ***elementarily equivalent*** (in the context X), written $(M, \nu) \equiv (N, \mu)$, if, for every $\varphi \in P(X)$,

$$M, \nu \models \varphi \qquad \Longleftrightarrow \qquad N, \mu \models \varphi.$$

[5] For instance, any model with five elements, two of which satisfy R and three of which do not, has the same type as the fourth representative. In the language $\{R\}$ without equality, a sentence can only detect whether there are elements satisfying R and whether there are elements satisfying $\neg R$; in particular, it cannot distinguish between one such element and several (because we do not have equality).

An ***X-type of*** P is an equivalence class of X-pointed models under this relation. We write

$$\mathrm{Mod}_X(P)$$

for the class of X-pointed models of P, and

$$\mathrm{Typ}_X(P) \coloneqq \mathrm{Mod}_X(P)/\equiv$$

for the quotient of $\mathrm{Mod}_X(P)$ by elementary equivalence in the context X.

Remark 2.6.2. For fixed X, every X-type is determined by the subset

$$\{\, \varphi \in P(X) \mid M, \nu \models \varphi \,\} \subseteq P(X),$$

so $\mathrm{Typ}_X(P)$ is a set. Moreover, when $X = \varnothing$, the unique map $\varnothing \to M$ identifies a $\varnothing$-pointed model with a model of P, and the notation $M, \nu \models \varphi$ agrees with Notation 2.3.3. Thus, the present definition extends the empty-context case.

Example 2.6.3 (The assignment matters)**.** Let $\mathcal{L} = \{R\}$, where R is a unary relation symbol, let $\mathcal{T} = \varnothing$, and let

$$P \coloneqq \mathrm{LT}_{\mathcal{T}}.$$

Let A be the $\mathcal{L}$-structure with two elements a, b, such that $a \in R^A$ and $b \notin R^A$. By Example 2.3.2, we may regard A as a model of P. Set

$$X \coloneqq \{x\},$$

and let $\nu, \mu \colon X \to A$ be given by

$$\nu(x) = a, \qquad \mu(x) = b.$$

Then the underlying models of (A, ν) and (A, μ) are equal, hence certainly elementarily equivalent as models. However, (A, ν) and (A, μ) are not elementarily equivalent as X-pointed models, because

$$A, \nu \models R(x) \qquad \text{but} \qquad A, \mu \not\models R(x).$$

Thus, elementary equivalence of pointed models is genuinely finer than elementary equivalence of the underlying models.

As in the empty context, to identify the Stone dual of $P(X)$ we need a filter/ideal form of Gödel's completeness theorem in the context X.

Theorem 2.6.4 (Filter/ideal form of Gödel's completeness theorem in a context)**.** *Let $P \colon \mathsf{FinSet} \to \mathsf{BA}$ be a first-order Boolean doctrine, and let X be a finite set. Let F be a filter of $P(X)$ and I an ideal of $P(X)$ such that $F \cap I = \varnothing$. There is an X-pointed model (M, ν) of P such that*

$$M, \nu \models \varphi \qquad \textit{for every } \varphi \in F,$$

and

$$M, \nu \not\models \psi \qquad \textit{for every } \psi \in I.$$

Proof. We omit the proof.[a] ■

[a] One usually deduces this theorem from the version in the empty context by adding a constant symbol for each element of X. Since our setting does not allow for the usage of constants, we shall simply use this result as a black box.

Theorem 2.6.5 (X-types $\overset{1:1}{\leftrightarrow}$ ultrafilters of $P(X)$)**.** *Let $P\colon \mathsf{FinSet} \to \mathsf{BA}$ be a first-order Boolean doctrine, and let X be a finite set. The assignment*

$$\begin{aligned} \Gamma_X \colon \mathrm{Typ}_X(P) &\longrightarrow \mathrm{Ult}(P(X)) \\ [(M,\nu)] &\longmapsto \{\, \varphi \in P(X) \mid M,\nu \models \varphi \,\}, \end{aligned}$$

defines a bijection.

Proof. First, Γ_X is well-defined. If

$$[(M,\nu)] = [(N,\mu)]\,,$$

then $(M,\nu) \equiv (N,\mu)$, and hence

$$M,\nu \models \varphi \qquad \Longleftrightarrow \qquad N,\mu \models \varphi$$

for every $\varphi \in P(X)$. So

$$\Gamma_X([(M,\nu)]) = \Gamma_X([(N,\mu)]).$$

For every X-pointed model (M,ν) of P, the set $\Gamma_X([(M,\nu)])$ is indeed an ultrafilter by the definition of $M,\nu \models \varphi$ and the defining properties of models relatively to Boolean operations (Definition 2.3.1). So Γ_X indeed lands in $\mathrm{Ult}(P(X))$.

Injectivity is immediate from the definition of elementary equivalence in the context X.

For surjectivity, let $U \in \mathrm{Ult}(P(X))$. Since U is a filter and $P(X) \setminus U$ is an ideal, the filter/ideal form of Gödel's completeness theorem in context (Theorem 2.6.4) yields an X-pointed model (M,ν) of P such that

$$M,\nu \models \varphi \qquad \text{for every } \varphi \in U,$$

and

$$M,\nu \not\models \psi \qquad \text{for every } \psi \in P(X) \setminus U.$$

Hence

$$\Gamma_X([(M,\nu)]) = U.$$

■

Via this bijection, $\mathrm{Typ}_X(P)$ inherits a Stone topology from $\mathrm{Ult}(P(X))$. This can be defined directly as follows.

Definition 2.6.6 (The space of X-types)**.** Let $P\colon \mathsf{FinSet} \to \mathsf{BA}$ be a first-order Boolean doctrine, and let X be a finite set. We equip $\mathrm{Typ}_X(P)$ with the topology that has as a basis the sets

$$[\![\varphi]\!] \;\coloneqq\; \{\, [(M,\nu)] \in \mathrm{Typ}_X(P) \mid M,\nu \models \varphi \,\}, \qquad\qquad \varphi \in P(X).$$

These subsets are well-defined, since satisfaction of formulas in the context X depends only on the X-type of the pointed model, and they do form a basis, since they are closed under finite intersections[a].

[a] Indeed, $[\![\varphi]\!] \cap [\![\psi]\!] = [\![\varphi \wedge \psi]\!]$ and $[\![1]\!] = \mathrm{Typ}_X(P)$.

Theorem 2.6.7 (The space of X-types is the Stone dual of $P(X)$)**.** *Let $P\colon \mathsf{FinSet} \to \mathsf{BA}$ be a first-order Boolean doctrine, and X a finite set. The bijection $\Gamma_X\colon \mathrm{Typ}_X(P) \to \mathrm{Ult}(P(X))$ of Theorem 2.6.5 is a homeomorphism, where $\mathrm{Ult}(P(X))$ carries the Stone topology. Consequently, $\mathrm{Typ}_X(P)$ is a Stone space, whose clopens are precisely the sets of the form*

$$[\![\varphi]\!] \;=\; \{\, [(M,\nu)] \in \mathrm{Typ}_X(P) \mid M,\nu \models \varphi \,\}, \qquad\qquad \varphi \in P(X).$$

Proof. For every $\varphi \in P(X)$,

$$\begin{aligned}\Gamma_X^{-1}\big[\eta_{P(X)}(\varphi)\big] &= \{\, [(M,\nu)] \in \mathrm{Typ}_X(P) \mid \Gamma_X([(M,\nu)]) \in \eta_{P(X)}(\varphi) \,\} \\ &= \{\, [(M,\nu)] \in \mathrm{Typ}_X(P) \mid \varphi \in \Gamma_X([(M,\nu)]) \,\} \\ &= \{\, [(M,\nu)] \in \mathrm{Typ}_X(P) \mid M,\nu \models \varphi \,\} \\ &= [\![\varphi]\!].\end{aligned}$$

So, along the bijection Γ_X, the basis

$$\{\, [\![\varphi]\!] \mid \varphi \in P(X) \,\}$$

of $\mathrm{Typ}_X(P)$ corresponds exactly to the basis

$$\{\, \eta_{P(X)}(\varphi) \mid \varphi \in P(X) \,\}$$

of $\mathrm{Ult}(P(X))$. Hence Γ_X is a homeomorphism. The final claim follows from the fact that the space of ultrafilters is a Stone space (Theorem 1.5.7) and the characterization of clopens in the space of ultrafilters (Lemma 1.5.8). ■

Remark 2.6.8. In other words, for every finite set X, the space of X-types of P is the Stone dual of the Boolean algebra $P(X)$. As in the empty context, the compactness of $\mathrm{Typ}_X(P)$ is not evident from the definition of the topology; it ultimately comes from Gödel's completeness theorem through the surjectivity part of Theorem 2.6.5.

Example 2.6.9 (The space of $\{x\}$-types for a unary relation symbol)**.** Let $\mathcal{L} = \{R\}$, where R is a unary relation symbol, let $\mathcal{T} = \varnothing$, and set

$$P \coloneqq \mathrm{LT}_{\mathcal{T}}, \qquad X \coloneqq \{x\}.$$

By Example 2.2.5, the Boolean algebra

$$P(X) = \mathrm{LT}_{\mathcal{T}}(\{x\})$$

has exactly the following four atoms:

$$\begin{gathered} R(x) \wedge \exists y\, \neg R(y), \\ R(x) \wedge \neg\exists y\, \neg R(y), \\ \neg R(x) \wedge \exists y\, R(y), \\ \neg R(x) \wedge \neg\exists y\, R(y). \end{gathered}$$

Hence, by Proposition 1.3.11 and Theorem 2.6.5, the set $\mathrm{Typ}_X(P)$ has exactly four elements.

These four X-types may be represented by the following X-pointed models:

1. a two-element model in which one element satisfies R and one does not, with x assigned to an element satisfying R;
2. a singleton model whose unique element satisfies R, with x assigned to that element;
3. a two-element model in which one element satisfies R and one does not, with x assigned to an element not satisfying R;
4. a singleton model whose unique element does not satisfy R, with x assigned to that element.

These pointed models are pairwise non-elementarily equivalent, since each satisfies exactly one of the four displayed atoms. For instance, the second representative satisfies $\forall y\, R(y)$, whereas the first does not.

Conversely, every X-pointed model (A,ν) of P is elementarily equivalent to exactly one of the four representatives above. Indeed, in the language $\{R\}$ without equality, a formula with free variable contained in $\{x\}$ can only detect whether $\nu(x)$ belongs to R^A and whether the opposite region is empty or nonempty.

Therefore the four atoms above define the four singleton clopens of $\mathrm{Typ}_X(P)$, so this space is discrete. In particular,

$$P(X) \cong \mathrm{Clop}(\mathrm{Typ}_X(P)) = \mathcal{P}(\mathrm{Typ}_X(P)).$$

2.7 The type space functor

The doctrine P does not only assign to each finite set X the Boolean algebra $P(X)$: it also assigns to each function $f\colon X \to Y$ a Boolean homomorphism

$$P(f)\colon P(X) \to P(Y).$$

Since, under Stone duality, Boolean homomorphisms correspond to continuous maps in the opposite direction, it is natural to ask whether the spaces of types assemble into a contravariant functor. They do.

Definition 2.7.1 (The maps induced on type spaces)**.** Let $P\colon \mathsf{FinSet} \to \mathsf{BA}$ be a first-order Boolean doctrine, let X, Y be finite sets, and let $f\colon X \to Y$. We define

$$\mathrm{Typ}_f(P)\colon \mathrm{Typ}_Y(P) \longrightarrow \mathrm{Typ}_X(P), \qquad [(M,\nu)] \longmapsto [(M, \nu \circ f)]\,.$$

Remark 2.7.2. If $f\colon \varnothing \to X$ is the unique map, then

$$\mathrm{Typ}_f(P)\colon \mathrm{Typ}_X(P) \to \mathrm{Typ}(P)$$

simply forgets the pointing.

Example 2.7.3. Let $\mathcal{L} = \{R\}$, $\mathcal{T} = \varnothing$, $P \coloneqq \mathrm{LT}_{\mathcal{T}}$, $X \coloneqq \{x\}$, and let $f\colon \varnothing \to X$ be the unique map. Then

$$\mathrm{Typ}_f(P)\colon \mathrm{Typ}_X(P) \to \mathrm{Typ}(P)$$

simply forgets the pointing.

With the representatives of Examples 2.5.7 and 2.6.9, the first and third X-types have the same image, while the type of the empty model is not in the image. Thus, $\mathrm{Typ}_f(P)$ need not be either injective or surjective.

Proposition 2.7.4 (Well-definedness and continuity)**.** *Let $P\colon \mathsf{FinSet} \to \mathsf{BA}$ be a first-order Boolean doctrine, let $f\colon X \to Y$, and let*

$$\mathrm{Typ}_f(P)\colon \mathrm{Typ}_Y(P) \to \mathrm{Typ}_X(P)$$

be as in Definition 2.7.1. Then $\mathrm{Typ}_f(P)$ is well-defined and continuous. More precisely, for every $\varphi \in P(X)$,

$$\mathrm{Typ}_f(P)^{-1}\big[[\![\varphi]\!]\big] = [\![P(f)(\varphi)]\!],$$

where on the left $[\![\varphi]\!] \subseteq \mathrm{Typ}_X(P)$ and on the right $[\![P(f)(\varphi)]\!] \subseteq \mathrm{Typ}_Y(P)$ denote the basic clopens of Definition 2.6.6.

Proof. To prove that $\mathrm{Typ}_f(P)$ is well-defined, suppose that

$$[(M,\nu)] = [(N,\mu)]$$

in $\mathrm{Typ}_Y(P)$. Let $\varphi \in P(X)$. Then

$$\begin{aligned} M, \nu \circ f \models \varphi &\iff \nu \circ f \in \varphi^M \\ &\iff \nu \in (P(f)(\varphi))^M \\ &\iff M, \nu \models P(f)(\varphi) \\ &\iff N, \mu \models P(f)(\varphi) \\ &\iff \mu \in (P(f)(\varphi))^N \\ &\iff \mu \circ f \in \varphi^N \\ &\iff N, \mu \circ f \models \varphi, \end{aligned}$$

where the second and sixth equivalences use the substitution clause in Definition 2.3.1. Therefore $(M, \nu \circ f)$ and $(N, \mu \circ f)$ are elementarily equivalent in the context X, as required.

To prove continuity, let $\varphi \in P(X)$ and let $[(M, \nu)] \in \mathrm{Typ}_Y(P)$. Then

$$\begin{aligned} [(M, \nu)] \in \mathrm{Typ}_f(P)^{-1}[[\![\varphi]\!]] &\iff \mathrm{Typ}_f(P)([(M, \nu)]) \in [\![\varphi]\!] \\ &\iff M, \nu \circ f \models \varphi \\ &\iff M, \nu \models P(f)(\varphi) \\ &\iff [(M, \nu)] \in [\![P(f)(\varphi)]\!]. \end{aligned}$$

So the displayed equality holds, and this proves continuity. ■

Proposition 2.7.5 (The type space functor)**.** *Let $P\colon \mathsf{FinSet} \to \mathsf{BA}$ be a first-order Boolean doctrine. The assignments*

$$X \longmapsto \mathrm{Typ}_X(P), \qquad (f\colon X \to Y) \longmapsto \mathrm{Typ}_f(P)\colon \mathrm{Typ}_Y(P) \to \mathrm{Typ}_X(P)$$

define a functor

$$\mathrm{Typ}_{_}(P)\colon \mathsf{FinSet}^{\mathrm{op}} \longrightarrow \mathsf{Stone}.$$

Proof. By Theorem 2.6.7, for every finite set X the space $\mathrm{Typ}_X(P)$ is a Stone space, and by Proposition 2.7.4 every map $\mathrm{Typ}_f(P)$ is continuous. Moreover,

$$\mathrm{Typ}_{\mathrm{id}_X}(P)([(M, \nu)]) = [(M, \nu \circ \mathrm{id}_X)] = [(M, \nu)]$$

for every $[(M, \nu)] \in \mathrm{Typ}_X(P)$, so identities are preserved. Finally, if

$$f\colon X \to Y, \qquad g\colon Y \to Z,$$

then for every $[(M, \nu)] \in \mathrm{Typ}_Z(P)$ we have

$$\begin{aligned} (\mathrm{Typ}_f(P) \circ \mathrm{Typ}_g(P))([(M, \nu)]) &= \mathrm{Typ}_f(P)([(M, \nu \circ g)]) \\ &= [(M, \nu \circ g \circ f)] \\ &= \mathrm{Typ}_{g \circ f}(P)([(M, \nu)]). \end{aligned}$$

So compositions are preserved as well. ■

Proposition 2.7.6 (Naturality of the homeomorphisms Γ_X)**.** *Let $P\colon \mathsf{FinSet} \to \mathsf{BA}$ be a first-order*

Boolean doctrine, and let $f\colon X \to Y$. Then the diagram

$$\begin{tikzcd} \mathrm{Typ}_Y(P) \arrow[r, "\Gamma_Y"] \arrow[d, "\mathrm{Typ}_f(P)"'] & \mathrm{Ult}(P(Y)) \arrow[d, "\mathrm{Ult}(P(f))"] \\ \mathrm{Typ}_X(P) \arrow[r, "\Gamma_X"'] & \mathrm{Ult}(P(X)) \end{tikzcd}$$

commutes. In particular, the family $(\Gamma_X)_{X \in \mathsf{FinSet}}$ is a natural isomorphism between $\mathrm{Typ}_{_}(P)$ and the functor obtained from the composite

$$\mathsf{FinSet} \xrightarrow{P} \mathsf{BA} \xrightarrow{\mathrm{Ult}} \mathsf{Stone}^{\mathrm{op}}$$

by passing to the opposite categories.

Proof. Recall from Definition 2.8.11 that $\mathrm{Ult}(P(f))$ is the inverse image map

$$P(f)^{-1}[-]\colon \mathrm{Ult}(P(Y)) \to \mathrm{Ult}(P(X)).$$

Let $[(M,\nu)] \in \mathrm{Typ}_Y(P)$. Then

$$\begin{aligned} (\Gamma_X \circ \mathrm{Typ}_f(P))([(M,\nu)]) &= \Gamma_X([(M,\nu\circ f)]) \\ &= \{\, \varphi \in P(X) \mid M,\nu\circ f \models \varphi \,\} \\ &= \{\, \varphi \in P(X) \mid M,\nu \models P(f)(\varphi) \,\} \\ &= P(f)^{-1}\big[\{\, \psi \in P(Y) \mid M,\nu \models \psi \,\}\big] \\ &= \mathrm{Ult}(P(f))(\Gamma_Y([(M,\nu)])). \end{aligned}$$

Therefore the diagram commutes. Since each Γ_X is a homeomorphism by Theorem 2.6.7, the last claim follows. ∎

Remark 2.7.7. The commuting square in Proposition 2.7.6 says exactly that, under the homeomorphisms Γ_X, the Boolean homomorphism $P(f)$ corresponds under Stone duality to the continuous map $\mathrm{Typ}_f(P)$.

Remark 2.7.8 (Towards polyadic spaces)**.** By Propositions 2.7.5 and 2.7.6, every first-order Boolean doctrine P gives rise to a functor

$$\mathrm{Typ}_{_}(P)\colon \mathsf{FinSet}^{\mathrm{op}} \longrightarrow \mathsf{Stone}$$

which is naturally isomorphic to the opposite of the composite

$$\mathsf{FinSet} \xrightarrow{P} \mathsf{BA} \xrightarrow{\mathrm{Ult}} \mathsf{Stone}^{\mathrm{op}}.$$

Thus, after Stone duality, a first-order Boolean doctrine may be encoded by a Stone-valued functor on $\mathsf{FinSet}^{\mathrm{op}}$.

The next step will be to characterize which functors

$$F\colon \mathsf{FinSet}^{\mathrm{op}} \longrightarrow \mathsf{Stone}$$

arise in this way. This will lead to a duality between first-order Boolean doctrines and a suitable category of Stone-valued functors, which we shall later call polyadic spaces.

2.8 Polyadic spaces

Every first-order Boolean doctrine $\mathsf{FinSet} \to \mathsf{BA}$ gives rise, by composition with Stone duality, to a functor

$$\mathsf{FinSet}^{\mathrm{op}} \longrightarrow \mathsf{Stone}.$$

Our next question is to characterize which functors arise in this way. The functors satisfying the characterization will be called *polyadic spaces*. This will yield a duality between first-order Boolean doctrines and polyadic spaces.

To give the correct definition of polyadic spaces, we need to find the correct dual formulation of (i) the existence of quantifiers and (ii) the Beck–Chevalley condition.

2.8.1 Existence of adjoints is dual to openness

We want to dualize the first axiom for a first-order Boolean doctrine $P\colon \mathsf{FinSet} \to \mathsf{BA}$, which was:

For every finite set X and every $y \notin X$, the Boolean homomorphism $P(X \hookrightarrow X \cup \{y\})\colon P(X) \to P(X \cup \{y\})$ has a left adjoint (or, equivalently, a right adjoint).

Therefore, we answer the following question: given a continuous map

$$f\colon S \to T$$

between Stone spaces, when does the inverse-image function

$$f^{-1}[-]\colon \mathrm{Clop}(T) \to \mathrm{Clop}(S)$$

have a left adjoint? As shown in the next result, the answer is: precisely when f is open.

Proposition 2.8.1 (Existence of adjoints is dual to openness)**.** *Let*

$$f\colon S \to T$$

be a continuous map of Stone spaces. The following are equivalent.

1. *(Existence of adjoints) The function $f^{-1}[-]\colon \mathrm{Clop}(T) \to \mathrm{Clop}(S)$ has a left adjoint (equivalently, a right adjoint).*[a]
2. *(Clopen map) For every clopen A of S, the image $f[A]$ is clopen in T.*
3. *(Open map) The map f is open, i.e., for every open A of S, the image $f[A]$ is open in T.*

When these conditions hold, the left, resp. right adjoint, are

$$\begin{aligned} \mathrm{Clop}(S) &\longrightarrow \mathrm{Clop}(T) \\ A &\longmapsto f[A] \end{aligned} \qquad\qquad \begin{aligned} \mathrm{Clop}(S) &\longrightarrow \mathrm{Clop}(T) \\ A &\longmapsto T \setminus f[S \setminus A]. \end{aligned}$$

[a] Recall that a Boolean homomorphism $g\colon A \to B$ between Boolean algebras has a left adjoint if and only if it has a right adjoint: given a left adjoint l, one obtains a right adjoint r as $r(x) := \neg l(\neg x)$, and, given a right adjoint r, one obtains a left adjoint l as $l(x) := \neg r(\neg x)$.

Proof. (2) $\Rightarrow$ (1). The map

$$\begin{aligned} \mathrm{Clop}(S) &\longrightarrow \mathrm{Clop}(T) \\ A &\longmapsto f[A] \end{aligned}$$

is well-defined by (2). Now let $A \in \mathrm{Clop}(S)$ and $B \in \mathrm{Clop}(T)$. Then

$$f[A] \subseteq B \iff A \subseteq f^{-1}[B].$$

So $f[-] \dashv f^{-1}[-]$.

(1) $\Rightarrow$ (2). Suppose that

$$L \dashv f^{-1}[-]$$

for some function

$$L\colon \mathrm{Clop}(S) \to \mathrm{Clop}(T).$$

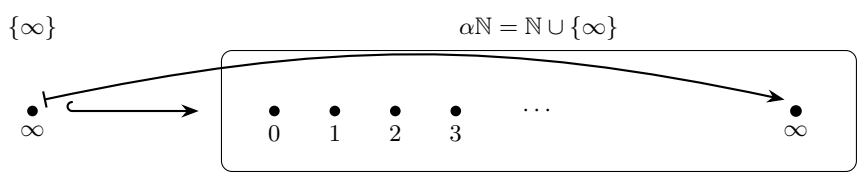
To prove that the function f is clopen, it is enough to prove that, for every $A \in \mathrm{Clop}(S)$, $f[A] = L(A)$. Fix $A \in \mathrm{Clop}(S)$. For every $B \in \mathrm{Clop}(T)$,

$$f[A] \subseteq B \iff A \subseteq f^{-1}[B] \iff L(A) \subseteq B,$$

where the second equivalence is again the adjunction. Since $f[A]$ and $L(A)$ are closed, and a closed set in a Stone space is the intersection of the clopens in which it is contained (Lemma 1.7.11), we have

$$f[A] = \bigcap\{\, B \in \mathrm{Clop}(T) \mid f[A] \subseteq B \,\} = \bigcap\{\, B \in \mathrm{Clop}(T) \mid L(A) \subseteq B \,\} = L(A). \qquad \blacksquare$$

A continuous function between Stone spaces is always closed but need not be open: consider, for example, the inclusion of the singleton $\{\infty\}$ in the one-point compactification $\alpha\mathbb{N}$ of the set of natural numbers $\mathbb{N}$ (see Example 1.5.4(2) for the definition of this space). Indeed, the clopen $\{\infty\}$ of $\{\infty\}$ is mapped to the subset $\{\infty\}$ of $\alpha\mathbb{N}$, which is closed but not open in $\alpha\mathbb{N}$.



In conclusion, the first axiom for a polyadic space $E\colon \mathsf{FinSet}^{\mathrm{op}} \to \mathsf{Stone}$ will be

For every finite set X and every $y \notin X$, the map $E(X \hookrightarrow X \cup \{y\})\colon E(X \cup \{y\}) \to E(X)$ is open.

2.8.2 Beck–Chevalley is dual to quasi-pullback

We want to dualize the second axiom for a first-order Boolean doctrine $P\colon \mathsf{FinSet} \to \mathsf{BA}$, which was:

For every function $\sigma\colon X \to X'$ between finite sets, every $y \notin X$ and every $y' \notin X'$, calling $\widetilde{\sigma}\colon X \cup \{y\} \to X' \cup \{y'\}$ the extension of σ that maps y to y', the following diagram commutes.

$$\begin{tikzcd} P(X \cup \{y\}) \arrow[r, "(\exists y)_X"] \arrow[d, "P(\widetilde{\sigma})"'] & P(X) \arrow[d, "P(\sigma)"] \\ P(X' \cup \{y'\}) \arrow[r, "(\exists y')_{X'}"'] & P(X') \end{tikzcd} \qquad \begin{tikzcd} P(X \cup \{y\}) \arrow[r, "(\forall y)_X"] \arrow[d, "P(\widetilde{\sigma})"'] & P(X) \arrow[d, "P(\sigma)"] \\ P(X' \cup \{y'\}) \arrow[r, "(\forall y')_{X'}"'] & P(X') \end{tikzcd}$$

The square appearing in the diagram is made up of two parallel Boolean homomorphisms and two parallel left adjoints of Boolean homomorphisms. Moreover, the four Boolean homomorphisms at play form a commutative square. This leads us to consider the following.

Proposition 2.8.2 (Beck–Chevalley is dual to quasi-pullback)**.** *Let*

$$\begin{tikzcd} X \arrow[r, "g"] \arrow[d, "f"'] & Z \arrow[d, "v"] \\ Y \arrow[r, "u"'] & W \end{tikzcd} \tag{2.5}$$

be a commutative square of continuous maps between Stone spaces, with g and u open. The following are equivalent.

1. *(Beck–Chevalley) The following square of functions between Boolean algebras commutes.*

$$\begin{tikzcd} \mathrm{Clop}(X) \arrow[r, "g[-]"] & \mathrm{Clop}(Z) \\ \mathrm{Clop}(Y) \arrow[u, "f^{-1}[-]"] \arrow[r, "u[-]"'] & \mathrm{Clop}(W) \arrow[u, "v^{-1}[-]"'] \end{tikzcd}$$

2. *(Quasi-pullback) For all $y \in Y$ and $z \in Z$ such that $u(y) = v(z)$, there is $x \in X$ such that $f(x) = y$ and $g(x) = z$.*

$$\begin{tikzcd} x \arrow[r, mapsto, dashed, "g"] \arrow[d, mapsto, dashed, "f"'] & z \arrow[d, mapsto, "v"] \\ y \arrow[r, mapsto, "u"'] & u(y) = v(z) \end{tikzcd}$$

In other words, the square in Stone *in* (2.5) *is a* quasi-pullback*, in the sense of Definition 2.8.3 below.*

Proof. (2) $\Rightarrow$ (1). Let $B \in \mathrm{Clop}(Y)$, and let us prove $g[f^{-1}[B]] = v^{-1}[u[B]]$. The inclusion $g[f^{-1}[B]] \subseteq v^{-1}[u[B]]$ is immediate, in the sense that it is a straightforward consequence of the commutativity of the square (2.5) of Stone spaces (no need to use the hypothesis (2)); see the footnote for the details[a].

We are left to prove the remaining inclusion: $v^{-1}[u[B]] \subseteq g[f^{-1}[B]]$. Here is where we use the hypothesis (2). Let $z \in v^{-1}[u[B]]$; i.e, let $z \in Z$ be such that there is $y \in B$ such that $u(y) = v(z)$. We shall prove $z \in g[f^{-1}[B]]$. By the hypothesis (2), there is $x \in X$ such that $f(x) = y$ and $g(x) = z$. Thus, $z \in g[f^{-1}[B]]$.

(1) $\Rightarrow$ (2). Fix $y \in Y$ and $z \in Z$ such that $u(y) = v(z)$. We shall prove that there is x in X such that $f(x) = y$ and $g(x) = z$, i.e., we shall prove $g^{-1}[\{z\}] \cap f^{-1}[\{y\}] \neq \varnothing$.

For every $B \in \mathrm{Clop}(Y)$ with $y \in B$, we have $v(z) = u(y) \in u[B]$ and hence $z \in v^{-1}[u[B]]$ and hence, by (1), $z \in g[f^{-1}[B]]$. In other words, for every $B \in \mathrm{Clop}(Y)$ containing y, the set

$$E_B \coloneqq g^{-1}[\{z\}] \cap f^{-1}[B]$$

is nonempty.

Since Stone spaces are Hausdorff, the singleton $\{y\}$ is closed in Y. Therefore, using the fact that any closed set in a Stone space is the intersection of the clopens in which it is contained (Lemma 1.7.11),

$$\{y\} = \bigcap \{\, B \in \mathrm{Clop}(Y) \mid y \in B \,\}.$$

Hence,

$$g^{-1}[\{z\}] \cap f^{-1}[\{y\}] = g^{-1}[\{z\}] \cap f^{-1}\Bigg[\bigcap_{B \in \mathrm{Clop}(Y) : y \in B} B\Bigg] = \bigcap_{B \in \mathrm{Clop}(Y) : y \in B} \big(g^{-1}[\{z\}] \cap f^{-1}[B]\big). \tag{2.6}$$

So, to prove that this set is nonempty, it is enough to prove that the family of closed sets

$$\big(E_B\big)_{B\in\mathrm{Clop}(Y),\, y\in B}$$

has the finite intersection property.

Let $B_1, \dots, B_n \in \mathrm{Clop}(Y)$ all contain y. Then

$$B \coloneqq B_1 \cap \cdots \cap B_n$$

is clopen and contains y. By the previous paragraph,

$$E_B = g^{-1}[\{z\}] \cap f^{-1}[B] \neq \varnothing.$$

Moreover,

$$E_B \subseteq E_{B_1} \cap \cdots \cap E_{B_n}.$$

So every finite intersection of the family (E_B) is nonempty. Therefore, since every E_B is closed, the intersection $\bigcap_{B\in Clop(Y):y\in B} E_B$ is nonempty, i.e., by (2.6),

$$g^{-1}[\{z\}] \cap f^{-1}[\{y\}] \neq \varnothing,$$

as desired. ■

[a] Let $z \in g[f^{-1}[B]]$. Then there is $x \in X$ such that $g(x) = z$ and $f(x) \in B$. Using the commutativity of the original square, we obtain $v(z) = v(g(x)) = u(f(x))$. Since $f(x) \in B$, we have $u(f(x)) \in u[B]$, hence $v(z) \in u[B]$. So, $z \in v^{-1}[u[B]]$. Therefore, $g[f^{-1}[B]] \subseteq v^{-1}[u[B]]$.

The condition (2) in Proposition 2.8.2, represented pictorially as

$$\begin{tikzcd} x \arrow[r, mapsto, dashed, "g"] \arrow[d, mapsto, dashed, "f"'] & z \arrow[d, mapsto, "v"] \\ y \arrow[r, mapsto, "u"'] & u(y) = v(z) \end{tikzcd}$$

, can be expressed succinctly by saying that the square

$$\begin{tikzcd} X \arrow[r, "g"] \arrow[d, "f"'] & Z \arrow[d, "v"] \\ Y \arrow[r, "u"'] & W \end{tikzcd}$$

is a *quasi-pullback*, in the following sense.

Definition 2.8.3 (Quasi-pullback)**.** Let $\mathcal{C}$ be a category with pullbacks. A commutative square

$$\begin{tikzcd} X \arrow[r, "g"] \arrow[d, "f"'] & Z \arrow[d, "v"] \\ Y \arrow[r, "u"'] & W. \end{tikzcd}$$

is said to be a ***quasi-pullback***[a] if the morphism χ from X to the pullback $Y \times_W Z$ of u and v induced

by the universal property of the pullback is an epimorphism.

$$\begin{tikzcd}
X \arrow[rr, "g"] \arrow[dr, dashed, "\chi"] \arrow[dd, "f"'] & & Z \arrow[dd, "v"] \\
& Y \times_W Z \arrow[ur, "\pi_2"'] \arrow[dl, "\pi_1"] & \\
Y \arrow[rr, "u"'] & & W
\end{tikzcd}$$

[a]Quasi-pullbacks are called *epi-pullbacks* by some authors.

Remark 2.8.4. Recall that pullbacks in Stone are computed as in Set (Proposition 1.7.8), and that epimorphisms in Stone are precisely the surjective continuous maps (Theorem 1.7.16). Hence, a commutative square of Stone spaces

$$\begin{tikzcd}
X \arrow[r, "g"] \arrow[d, "f"'] & Z \arrow[d, "v"] \\
Y \arrow[r, "u"'] & W
\end{tikzcd}$$

is a quasi-pullback if and only if the canonical map $X \to Y \times_W Z$ is surjective if and only if for every $y \in Y$ and every $z \in Z$ such that $u(y) = v(z)$ there is $x \in X$ such that $f(x) = y$ and $g(x) = z$.

In conclusion, the second axiom for a polyadic space $E\colon \mathsf{FinSet}^{\mathrm{op}} \to \mathsf{Stone}$ will be

For every morphism $\sigma\colon X \to X'$ in FinSet, every $y \notin X$ and every $y' \notin X'$, letting $\widetilde{\sigma}\colon X \cup \{y\} \longrightarrow X' \cup \{y'\}$ denote the extension of σ mapping y to y', the following diagram is a quasi-pullback in Stone (equivalently, in Set).

$$\begin{tikzcd}
E(X' \cup \{y'\}) \arrow[r, "E(X' \hookrightarrow X' \cup \{y'\})"] \arrow[d, "E(\widetilde{\sigma})"'] & E(X') \arrow[d, "E(\sigma)"] \\
E(X \cup \{y\}) \arrow[r, "E(X \hookrightarrow X \cup \{y\})"'] & E(X)
\end{tikzcd}$$

2.8.3 Polyadic spaces

Definition 2.8.5 (Polyadic space)**.** A ***polyadic space***[a] is a functor

$$E\colon \mathsf{FinSet}^{\mathrm{op}} \longrightarrow \mathsf{Stone}$$

with the following properties.

1. For every finite set X and every $y \notin X$, the map

$$E(X \hookrightarrow X \cup \{y\})\colon E(X \cup \{y\}) \longrightarrow E(X)$$

is open.

2. For every function $\sigma\colon X \to X'$ between finite sets, every $y \notin X$ and every $y' \notin X'$, letting

$$\widetilde{\sigma}\colon X \cup \{y\} \longrightarrow X' \cup \{y'\}$$

denote the extension of σ that maps y to y', the diagram

$$\begin{tikzcd} E(X' \cup \{y'\}) \arrow[r, "E(X' \hookrightarrow X' \cup \{y'\})"] \arrow[d, "E(\widetilde{\sigma})"'] & E(X') \arrow[d, "E(\sigma)"] \\ E(X \cup \{y\}) \arrow[r, "E(X \hookrightarrow X \cup \{y\})"'] & E(X) \end{tikzcd}$$

is a quasi-pullback in Stone.

[a]The name "polyadic space" is motivated by the term "polyadic algebra", which is another algebraic approach to first-order logic.

We now discuss some basic examples of polyadic spaces.

Example 2.8.6 (Representable polyadic spaces)**.** Let M be a finite set. Define a functor

$$E_M \colon \mathsf{FinSet}^{\mathrm{op}} \longrightarrow \mathsf{Stone}$$

as follows: for every $X \in \mathsf{FinSet}$,

$$E_M(X) \coloneqq \hom_{\mathsf{FinSet}}(X, M) = M^X,$$

equipped with the discrete topology, and for every morphism $f \colon X \to Y$ in FinSet,

$$\begin{aligned} E_M(f) \colon M^Y &\longrightarrow M^X \\ \nu &\longmapsto \nu \circ f. \end{aligned}$$

Then, E_M is a polyadic space.[6]

[6]We prove that it is indeed a polyadic space. For every finite set X, the set M^X is finite, and hence with the discrete topology it is a Stone space. So $E_M \colon \mathsf{FinSet}^{\mathrm{op}} \to \mathsf{Stone}$ is a well-defined functor.

Let $X \in \mathsf{FinSet}$ and $y \notin X$. Then, $E_M(X \hookrightarrow X \cup \{y\}) \colon M^{X \cup \{y\}} \longrightarrow M^X$ is the restriction map $\nu \mapsto \nu\restriction_X$. Since both the domain and the codomain are discrete, this map is open. So axiom (1) of the definition of polyadic spaces holds.

Now let $\sigma \colon X \to X'$ be a map between finite sets, let $y \notin X$ and $y' \notin X'$, and let $\widetilde{\sigma} \colon X \cup \{y\} \to X' \cup \{y'\}$ be the extension of σ sending y to y'. We claim that the square

$$\begin{tikzcd} M^{X' \cup \{y'\}} \arrow[r, "E_M(X' \hookrightarrow X' \cup \{y'\})"] \arrow[d, "E_M(\widetilde{\sigma})"'] & M^{X'} \arrow[d, "E_M(\sigma)"] \\ M^{X \cup \{y\}} \arrow[r, "E_M(X \hookrightarrow X \cup \{y\})"'] & M^X \end{tikzcd}$$

is a quasi-pullback (in fact, a pullback) in Set, and hence also in Stone. This follows from the fact that the square

$$\begin{tikzcd} X \arrow[r, hook] \arrow[d, "\sigma"'] & X \cup \{y\} \arrow[d, "\widetilde{\sigma}"] \\ X' \arrow[r, hook] & X' \cup \{y'\} \end{tikzcd}$$

is a pushout, and hence it is mapped by any contravariant representable functor $E \colon \mathsf{FinSet}^{\mathrm{op}} \to \mathsf{Set}$ into a pullback. (Recall that a contravariant representable functor $\mathsf{C}^{\mathrm{op}} \to \mathsf{Set}$ maps colimits of C to limits of Set.)

If you don't like this argument for quasi-pullback, here is an alternative hands-on one. Let $\alpha \in M^{X \cup \{y\}}$ and $\beta \in M^{X'}$ be such that $E_M(X \hookrightarrow X \cup \{y\})(\alpha) = E_M(\sigma)(\beta)$, i.e. $\alpha\restriction_X = \beta \circ \sigma$. Define $\gamma \colon X' \cup \{y'\} \to M$ by

$$\gamma(x') \coloneqq \beta(x') \qquad (x' \in X'), \qquad \gamma(y') \coloneqq \alpha(y).$$

Then,

$$E_M(X' \hookrightarrow X' \cup \{y'\})(\gamma) = \gamma\restriction_{X'} = \beta.$$

Moreover, for every $x \in X$ we have

$$(E_M(\widetilde{\sigma})(\gamma))(x) = \gamma(\sigma(x)) = \beta(\sigma(x)) = \alpha(x),$$

Example 2.8.7 (Two representable examples)**.** We consider a couple of special cases of representable polyadic spaces (Example 2.8.6).

1. If $M = 1$, then for every finite set X,
$$E_1(X) = 1^X \cong 1.$$
So E_1 is the constant functor at a singleton.

2. If $M = \varnothing$, then
$$E_\varnothing(X) = \hom_{\mathsf{FinSet}}(X, \varnothing) = \begin{cases} 1, & X = \varnothing, \\ \varnothing, & X \neq \varnothing. \end{cases}$$

Example 2.8.8 (Constant polyadic spaces)**.** Let S be a Stone space. The constant functor
$$\Delta_S \colon \mathsf{FinSet}^{\mathrm{op}} \longrightarrow \mathsf{Stone}, \qquad X \longmapsto S,$$
which sends every morphism to id_S, is a polyadic space.[7]

2.8.4 Morphisms of polyadic spaces

Recall that a morphism of first-order Boolean doctrines is a natural transformation commuting with existential quantification. By Proposition 2.8.2, the Stone-side translation of this condition is that a certain square be a quasi-pullback. This motivates the following definition.

Definition 2.8.9 (Morphism of polyadic spaces)**.** Let
$$E_1, E_2 \colon \mathsf{FinSet}^{\mathrm{op}} \longrightarrow \mathsf{Stone}$$
be polyadic spaces. A ***morphism of polyadic spaces***
$$\alpha \colon E_1 \longrightarrow E_2$$
is a natural transformation such that, for every finite set X and every $y \notin X$, the diagram
$$\begin{tikzcd} E_1(X \cup \{y\}) \arrow[r, "E_1(X \hookrightarrow X \cup \{y\})"] \arrow[d, "\alpha_{X \cup \{y\}}"'] & E_1(X) \arrow[d, "\alpha_X"] \\ E_2(X \cup \{y\}) \arrow[r, "E_2(X \hookrightarrow X \cup \{y\})"'] & E_2(X) \end{tikzcd}$$
is a quasi-pullback.

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while
$$(E_M(\widetilde{\sigma})(\gamma))(y) = \gamma(y') = \alpha(y);$$
hence
$$E_M(\widetilde{\sigma})(\gamma) = \alpha.$$
So the square is a quasi-pullback.

[7] We prove it. For every finite set X and every $y \notin X$, the map
$$\Delta_S(X \hookrightarrow X \cup \{y\}) \colon S \to S$$
is the identity, and hence open. Similarly, the square below is a quasi-pullback (in fact, a pullback).
$$\begin{tikzcd} S \arrow[r, "\mathrm{id}_S"] \arrow[d, "\mathrm{id}_S"'] & S \arrow[d, "\mathrm{id}_S"] \\ S \arrow[r, "\mathrm{id}_S"'] & S \end{tikzcd}$$
So Δ_S satisfies both axioms and is therefore a polyadic space.

2.8.5 Joyal duality for first-order Boolean doctrines

In this subsection, we arrive at the duality between first-order Boolean doctrines and polyadic spaces.

Definition 2.8.10 (The categories FOBD and PolySp)**.**

1. We let FOBD denote the category of first-order Boolean doctrines (Definition 2.2.1) and morphisms between them (Definition 2.3.4).
2. We let PolySp denote the category of polyadic spaces (Definition 2.8.5) and morphisms between them (Definition 2.8.9).

It is straightforward to see that these are well-defined categories.

Definition 2.8.11 (The functors Clop^* and Ult^* between FOBD and PolySp)**.**

1. ($\mathrm{Clop}^* \coloneqq \mathrm{Clop} \circ (-)^{\mathrm{op}}$) We define a functor
$$\mathrm{Clop}^* \colon \mathsf{PolySp}^{\mathrm{op}} \longrightarrow \mathsf{FOBD}$$
as follows:
   - (On objects:) it maps a polyadic space $E \colon \mathsf{FinSet}^{\mathrm{op}} \to \mathsf{Stone}$ to the first-order Boolean doctrine obtained as the composite
   $$\mathsf{FinSet} \xrightarrow{E^{\mathrm{op}}} \mathsf{Stone}^{\mathrm{op}} \xrightarrow{\mathrm{Clop}} \mathsf{BA}$$
   - (On morphisms:) it maps a morphism of polyadic spaces $\alpha \colon E_1 \to E_2$ to the natural transformation
   $$\big(\mathrm{Clop}(\alpha_X)\big)_{X \in \mathsf{FinSet}}$$
   from $\mathrm{Clop} \circ E_2^{\mathrm{op}}$ to $\mathrm{Clop} \circ E_1^{\mathrm{op}}$; we recall that, for every $X \in \mathsf{FinSet}$,
   $$\mathrm{Clop}(\alpha_X) = \alpha_X^{-1}[-] \colon \mathrm{Clop}(E_2(X)) \to \mathrm{Clop}(E_1(X)).$$
2. ($\mathrm{Ult}^* \coloneqq \mathrm{Ult}^{\mathrm{op}} \circ (-)^{\mathrm{op}}$) We define a functor
$$\mathrm{Ult}^* \colon \mathsf{FOBD} \longrightarrow \mathsf{PolySp}^{\mathrm{op}}$$
as follows:
   - (On objects:) it maps a first-order Boolean doctrine $P \colon \mathsf{FinSet} \to \mathsf{BA}$ to the composite
   $$\mathsf{FinSet}^{\mathrm{op}} \xrightarrow{P^{\mathrm{op}}} \mathsf{BA}^{\mathrm{op}} \xrightarrow{\mathrm{Ult}^{\mathrm{op}}} \mathsf{Stone}.$$
   - (On morphisms:) it maps a morphism of first-order Boolean doctrines $\mathfrak{m} \colon P_1 \to P_2$ to the natural transformation
   $$\big(\mathrm{Ult}(\mathfrak{m}_X)\big)_{X \in \mathsf{FinSet}}$$
   from $\mathrm{Ult}^{\mathrm{op}} \circ P_2^{\mathrm{op}}$ to $\mathrm{Ult}^{\mathrm{op}} \circ P_1^{\mathrm{op}}$; we recall that, for every $X \in \mathsf{FinSet}$,
   $$\mathrm{Ult}(\mathfrak{m}_X) = \mathfrak{m}_X^{-1}[-] \colon \mathrm{Ult}(P_2(X)) \to \mathrm{Ult}(P_1(X)).$$

Remark 2.8.12 (Well-definedness of the two functors)**.** Proposition 2.8.1 ("Existence of adjoints is dual to openness") and Proposition 2.8.2 ("Beck–Chevalley is dual to quasi-pullback") guarantee that these functors are well-defined; see the footnotes for the details of the proofs of the facts that Clop^* is well-defined on

objects[8] and on morphisms[9] and that Ult^* is well-defined on objects[10] and on morphisms[11].

[8]Let $E\colon \mathsf{FinSet}^{\mathrm{op}} \to \mathsf{Stone}$ be a polyadic space, and let us show that the composite $\mathsf{FinSet} \xrightarrow{E^{\mathrm{op}}} \mathsf{Stone}^{\mathrm{op}} \xrightarrow{\mathrm{Clop}} \mathsf{BA}$ is a first-order Boolean doctrine. Let $X \in \mathsf{FinSet}$ and $y \notin X$. By axiom (1) in the definition of polyadic spaces (Definition 2.8.5), the map $E(X \hookrightarrow X \cup \{y\})\colon E(X \cup \{y\}) \to E(X)$ is open. Therefore, by Proposition 2.8.1 ("Existence of adjoints is dual to openness"), the inverse-image Boolean homomorphism $E(X \hookrightarrow X \cup \{y\})^{-1}[-]\colon \mathrm{Clop}(E(X)) \to \mathrm{Clop}(E(X \cup \{y\}))$ has a left adjoint—namely, the direct image function. For the Beck–Chevalley condition, let $\sigma\colon X \to X'$ be a morphism in FinSet, let $y \notin X$ and $y' \notin X'$, and let $\widetilde{\sigma}\colon X \cup \{y\} \to X' \cup \{y'\}$ be the extension of σ sending y to y'. By axiom (2) in the definition of polyadic spaces, the square

$$\begin{tikzcd} E(X' \cup \{y'\}) \arrow[r, "E(X' \hookrightarrow X' \cup \{y'\})"] \arrow[d, "E(\widetilde{\sigma})"'] & E(X') \arrow[d, "E(\sigma)"] \\ E(X \cup \{y\}) \arrow[r, "E(X \hookrightarrow X \cup \{y\})"'] & E(X) \end{tikzcd}$$

is a quasi-pullback. Since the horizontal maps are open and this is a quasi-pullback, Proposition 2.8.2 ("Beck–Chevalley is dual to quasi-pullback") yields the commutativity of the corresponding square on clopens with the left adjoints:

$$\begin{tikzcd} \mathrm{Clop}(E(X \cup \{y\})) \arrow[r, "E(\widetilde{\sigma})^{-1}[-]"] \arrow[d, "(\exists y)^E_X"'] & \mathrm{Clop}(E(X' \cup \{y'\})) \arrow[d, "(\exists y')^E_{X'}"] \\ \mathrm{Clop}(E(X)) \arrow[r, "E(\sigma)^{-1}[-]"'] & \mathrm{Clop}(E(X')). \end{tikzcd}$$

This is the Beck–Chevalley condition for $\mathrm{Clop}^*(E)$.

[9]Let $\alpha\colon E_1 \to E_2$ be a morphism of polyadic spaces and let us prove that $\mathrm{Clop}^*(\alpha)$ is a morphism of first-order Boolean doctrines from $\mathrm{Clop}^*(E_2)$ to $\mathrm{Clop}^*(E_1)$. Each map $\alpha_X^{-1}[-]\colon \mathrm{Clop}(E_2(X)) \to \mathrm{Clop}(E_1(X))$ is clearly a Boolean homomorphism. To prove naturality, let $f\colon X \to Y$ be a morphism of FinSet. Since α is a natural transformation, the square

$$\begin{tikzcd} E_1(Y) \arrow[r, "E_1(f)"] \arrow[d, "\alpha_Y"'] & E_1(X) \arrow[d, "\alpha_X"] \\ E_2(Y) \arrow[r, "E_2(f)"'] & E_2(X) \end{tikzcd}$$

commutes. Taking the inverse image on clopens, we obtain $E_1(f)^{-1}[-] \circ \alpha_X^{-1}[-] = \alpha_Y^{-1}[-] \circ E_2(f)^{-1}[-]$, which is exactly the naturality of the family $\left(\alpha_X^{-1}[-]\right)_X$. It remains to verify compatibility with existential quantification. Fix $X \in \mathsf{FinSet}$ and $y \notin X$. Since α is a morphism of polyadic spaces, the following is a quasi-pullback.

$$\begin{tikzcd} E_1(X \cup \{y\}) \arrow[r, "E_1(X \hookrightarrow X \cup \{y\})"] \arrow[d, "\alpha_{X \cup \{y\}}"'] & E_1(X) \arrow[d, "\alpha_X"] \\ E_2(X \cup \{y\}) \arrow[r, "E_2(X \hookrightarrow X \cup \{y\})"'] & E_2(X) \end{tikzcd}$$

By Proposition 2.8.2 ("Beck–Chevalley is dual to quasi-pullback"), this is equivalent to the commutativity of the square

$$\begin{tikzcd} \mathrm{Clop}(E_2(X \cup \{y\})) \arrow[r, "\alpha_{X \cup \{y\}}^{-1}[-]"] \arrow[d, "(\exists y)^{E_2}_X"'] & \mathrm{Clop}(E_1(X \cup \{y\})) \arrow[d, "(\exists y)^{E_1}_X"] \\ \mathrm{Clop}(E_2(X)) \arrow[r, "\alpha_X^{-1}[-]"'] & \mathrm{Clop}(E_1(X)). \end{tikzcd}$$

[10]Let $P\colon \mathsf{FinSet} \to \mathsf{BA}$ be a first-order Boolean doctrine, and let us prove that $\mathrm{Ult}^*(P)\colon \mathsf{FinSet}^{\mathrm{op}} \to \mathsf{Stone}$ is a polyadic space. It is clearly a functor. We verify the two axioms of Definition 2.8.5. *Axiom* (1). Let $X \in \mathsf{FinSet}$ and $y \notin X$. Then $P(X \hookrightarrow X \cup \{y\})\colon P(X) \to P(X \cup \{y\})$ has a left adjoint by the definition of first-order Boolean doctrine. Under the Boolean algebra isomorphisms

$$\eta_{P(X)}\colon P(X) \xrightarrow{\sim} \mathrm{Clop}(\mathrm{Ult}(P(X))), \qquad \eta_{P(X \cup \{y\})}\colon P(X \cup \{y\}) \xrightarrow{\sim} \mathrm{Clop}(\mathrm{Ult}(P(X \cup \{y\}))),$$

the homomorphism $P(X \hookrightarrow X \cup \{y\})$ corresponds to inverse image along

$$\mathrm{Ult}(P(X \hookrightarrow X \cup \{y\}))\colon \mathrm{Ult}(P(X \cup \{y\})) \to \mathrm{Ult}(P(X)).$$

Therefore, by Proposition 2.8.1, the map $\mathrm{Ult}(P(X \hookrightarrow X \cup \{y\}))$ is open. So axiom (1) holds. *Axiom* (2). Let $\sigma\colon X \to X'$ be a morphism in FinSet, let $y \notin X$ and $y' \notin X'$, and let $\widetilde{\sigma}\colon X \cup \{y\} \to X' \cup \{y'\}$ be the extension of σ sending y to y'. By the

Theorem 2.8.13 (Joyal duality for first-order Boolean doctrines)**.** *The categories* FOBD *of first-order Boolean doctrines and* PolySp *of polyadic spaces are dually equivalent*

$$\mathsf{FOBD} \simeq \mathsf{PolySp}^{\mathrm{op}},$$

as witnessed by the quasi-inverse functors $\mathrm{Clop}^*\colon \mathsf{PolySp}^{\mathrm{op}} \to \mathsf{FOBD}$ *and* $\mathrm{Ult}^*\colon \mathsf{FOBD} \to \mathsf{PolySp}^{\mathrm{op}}$.

Proof. By Remark 2.8.12, the functors Clop^* and Ult^* are well-defined.

The natural isomorphisms are defined as follows:

1. (On the doctrine side) The component at a first-order Boolean doctrine P is the natural transformation $P \to \mathrm{Clop}^*\mathrm{Ult}^*(P)$ whose component at $X \in \mathsf{FinSet}$ is
$$\widetilde{\eta}^P_X \coloneqq \eta_{P(X)}\colon P(X) \xrightarrow{\sim} \mathrm{Clop}(\mathrm{Ult}(P(X))).$$
2. (On the polyadic-space side) The component at a polyadic space E is the natural transformation $E \to \mathrm{Ult}^*\mathrm{Clop}^*(E)$ whose component at $X \in \mathsf{FinSet}$ is
$$\widetilde{\varepsilon}^E_X \coloneqq \varepsilon_{E(X)}\colon E(X) \xrightarrow{\sim} \mathrm{Ult}(\mathrm{Clop}(E(X))).$$

Beck–Chevalley axiom for P, the square

$$\begin{tikzcd} P(X) \arrow[r, "P(X \hookrightarrow X \cup \{y\})"] \arrow[d, "P(\sigma)"'] & P(X \cup \{y\}) \arrow[d, "P(\widetilde{\sigma})"] \\ P(X') \arrow[r, "P(X' \hookrightarrow X' \cup \{y'\})"'] & P(X' \cup \{y'\}) \end{tikzcd}$$

commutes together with the corresponding left adjoints. Passing to Stone duals, and using the isomorphisms $\eta_{P(-)}$, this says exactly that the square

$$\begin{tikzcd} \mathrm{Ult}(P(X' \cup \{y'\})) \arrow[r, "\mathrm{Ult}(P(X' \hookrightarrow X' \cup \{y'\}))"] \arrow[d, "\mathrm{Ult}(P(\widetilde{\sigma}))"'] & \mathrm{Ult}(P(X')) \arrow[d, "\mathrm{Ult}(P(\sigma))"] \\ \mathrm{Ult}(P(X \cup \{y\})) \arrow[r, "\mathrm{Ult}(P(X \hookrightarrow X \cup \{y\}))"'] & \mathrm{Ult}(P(X)) \end{tikzcd}$$

satisfies the Beck–Chevalley commutativity condition on clopens. Since the horizontal maps are open by the previous paragraph, Proposition 2.8.2 implies that this square is a quasi-pullback. So axiom (2) holds. Hence $\mathrm{Ult}^*(P)$ is a polyadic space.

[11] Let $\mathfrak{m}\colon P_1 \to P_2$ be a morphism of first-order Boolean doctrines, and let us prove that $(\mathrm{Ult}(\mathfrak{m}_X))_{X \in \mathsf{FinSet}}$ is a morphism of polyadic spaces. Since $\mathfrak{m}$ is a natural transformation and Ult is contravariant, the family $\mathrm{Ult}^*(\mathfrak{m}) = \big(\mathrm{Ult}(\mathfrak{m}_X)\big)_{X \in \mathsf{FinSet}}$ is a natural transformation $\mathrm{Ult}^*(P_2) \to \mathrm{Ult}^*(P_1)$. It remains to prove the quasi-pullback condition of Definition 2.8.9. Fix $X \in \mathsf{FinSet}$ and $y \notin X$. Since $\mathfrak{m}$ is a morphism of first-order Boolean doctrines, the square

$$\begin{tikzcd} P_1(X) \arrow[r, "P_1(X \hookrightarrow X \cup \{y\})"] \arrow[d, "\mathfrak{m}_X"'] & P_1(X \cup \{y\}) \arrow[d, "\mathfrak{m}_{X \cup \{y\}}"] \\ P_2(X) \arrow[r, "P_2(X \hookrightarrow X \cup \{y\})"'] & P_2(X \cup \{y\}) \end{tikzcd}$$

commutes together with the corresponding existential quantifiers. Passing to Stone duals, and using the Boolean algebra isomorphisms $\eta_{P_i(-)}$, this says exactly that the associated square on clopens commutes. Since the horizontal maps

$$\mathrm{Ult}(P_i(X \hookrightarrow X \cup \{y\}))\colon \mathrm{Ult}(P_i(X \cup \{y\})) \to \mathrm{Ult}(P_i(X))$$

are open by Proposition 2.8.1, Proposition 2.8.2 implies that the square

$$\begin{tikzcd} \mathrm{Ult}(P_2(X \cup \{y\})) \arrow[r, "\mathrm{Ult}(P_2(X \hookrightarrow X \cup \{y\}))"] \arrow[d, "\mathrm{Ult}(\mathfrak{m}_{X \cup \{y\}})"'] & \mathrm{Ult}(P_2(X)) \arrow[d, "\mathrm{Ult}(\mathfrak{m}_X)"] \\ \mathrm{Ult}(P_1(X \cup \{y\})) \arrow[r, "\mathrm{Ult}(P_1(X \hookrightarrow X \cup \{y\}))"'] & \mathrm{Ult}(P_1(X)) \end{tikzcd}$$

is a quasi-pullback. Therefore, $\mathrm{Ult}^*(\mathfrak{m})$ is a morphism of polyadic spaces.

We omit the verification that $\widetilde{\eta}^P$ and $\widetilde{\varepsilon}^E$ are natural transformation and that they define natural transformations $\widetilde{\eta}$ and $\widetilde{\varepsilon}$. We know from Stone duality that all $\widetilde{\eta}^P_X$ and $\widetilde{\varepsilon}^E_X$ are isomorphisms. ■

Bibliography

[Baez and Dolan, 2001] Baez, J. C. and Dolan, J. (2001). From finite sets to Feynman diagrams. In *Mathematics unlimited—2001 and beyond*, pages 29–50. Springer, Berlin. ↑6

[Bergman, 2015] Bergman, G. M. (2015). *An invitation to general algebra and universal constructions.* Universitext. Cham: Springer, 2nd ed. edition. ↑35, ↑36

[Birkhoff, 1967] Birkhoff, G. (1967). *Lattice theory*, volume 25 of *Colloq. Publ., Am. Math. Soc.* American Mathematical Society (AMS), Providence, RI, 3 edition. ↑11, ↑12

[Burris and Sankappanavar, 2012] Burris, S. N. and Sankappanavar, H. P. (2012). *A Course in Universal Algebra, The Millenium Edition.* Freely available at https://www.math.uwaterloo.ca/~snburris/htdocs/UALG/univ-algebra2012.pdf. ↑8, ↑14, ↑15

[Crama and Hammer, 2011] Crama, Y. and Hammer, P. L. (2011). *Boolean Functions: Theory, Algorithms, and Applications.* Encyclopedia of Mathematics and its Applications. Cambridge University Press. ↑9

[Davey and Priestley, 2002] Davey, B. A. and Priestley, H. A. (2002). *Introduction to Lattices and Order.* Cambridge University Press, Cambridge, 2 edition. ↑11, ↑12

[Enderton, 2001] Enderton, H. B. (2001). *A Mathematical Introduction to Logic.* Academic Press, 2 edition. ↑9, ↑34

[Givant and Halmos, 2009] Givant, S. and Halmos, P. (2009). *Introduction to Boolean algebras.* Undergraduate Texts in Mathematics. Springer, New York. ↑22, ↑47

[Halpern and Lévy, 1971] Halpern, J. D. and Lévy, A. (1971). The Boolean prime ideal theorem does not imply the axiom of choice. In Scott, D. S., editor, *Axiomatic Set Theory*, volume 13 of *Proceedings of Symposia in Pure Mathematics*, pages 83–134. American Mathematical Society, Providence, RI. ↑22

[Jech, 1973] Jech, T. (1973). *The Axiom of Choice*, volume 75 of *Studies in Logic and the Foundations of Mathematics.* North-Holland, Amsterdam. ↑22

[Lawvere, 1969] Lawvere, F. W. (1969). Adjointness in foundations. *Dialectica*, 23(3–4):281–296. ↑53

[Lawvere, 1970] Lawvere, F. W. (1970). Equality in hyperdoctrines and comprehension schema as an adjoint functor. *Applications of Categorical Algebra*, 17:1–14. ↑53

[Mac Lane, 1998] Mac Lane, S. (1998). *Categories for the Working Mathematician*, volume 5 of *Graduate Texts in Mathematics.* Springer, 2 edition. ↑8, ↑31

[Munkres, 2000] Munkres, J. R. (2000). *Topology.* Prentice Hall, Upper Saddle River, NJ, 2 edition. ↑25, ↑27, ↑37, ↑47

[Stone, 1936] Stone, M. H. (1936). The theory of representations for Boolean algebras. *Trans. Amer. Math. Soc.*, 40(1):37–111. ↑22